FIAP GRADUAÇÃO

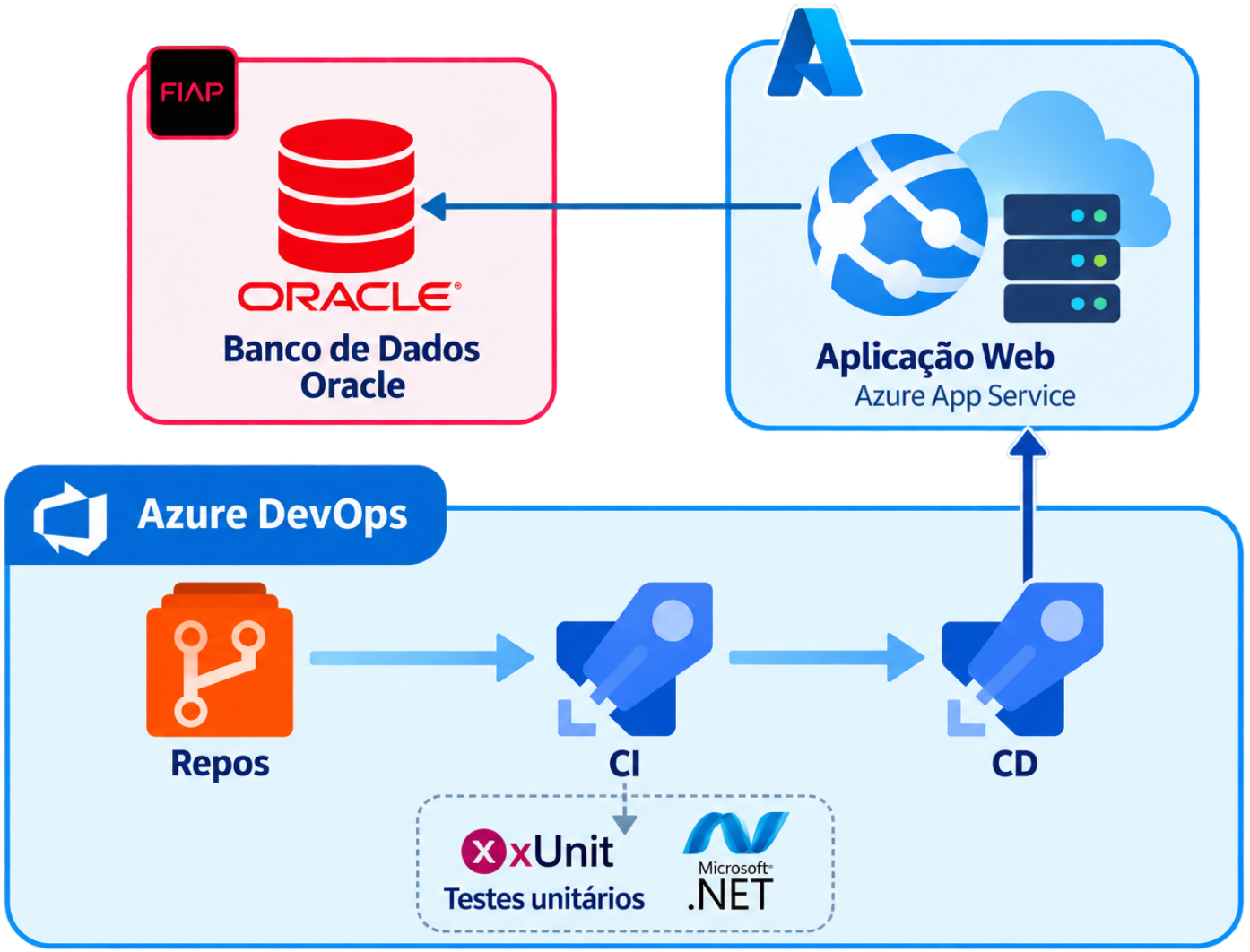
TECNOLOGIA EM DESENVOLVIMENTO DE SISTEMAS

DevOps Tools & Cloud Computing

Azure Pipelines API .NET com Oracle

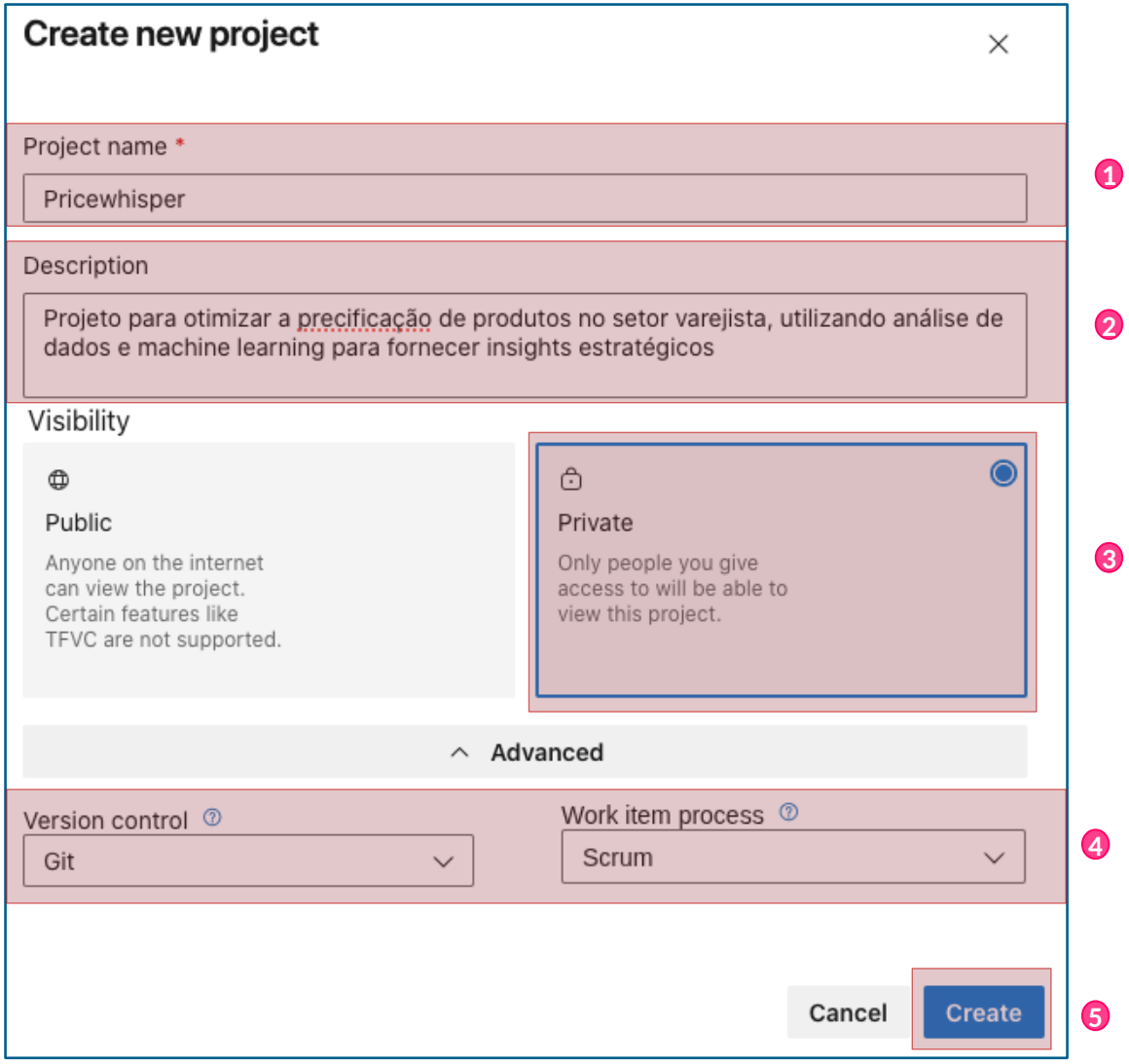
PROF. JOÃO MENK profjoao.menk@fiap.com.br

Azure Pipelines API .NET com Oracle



Azure Pipelines API .NET com Oracle

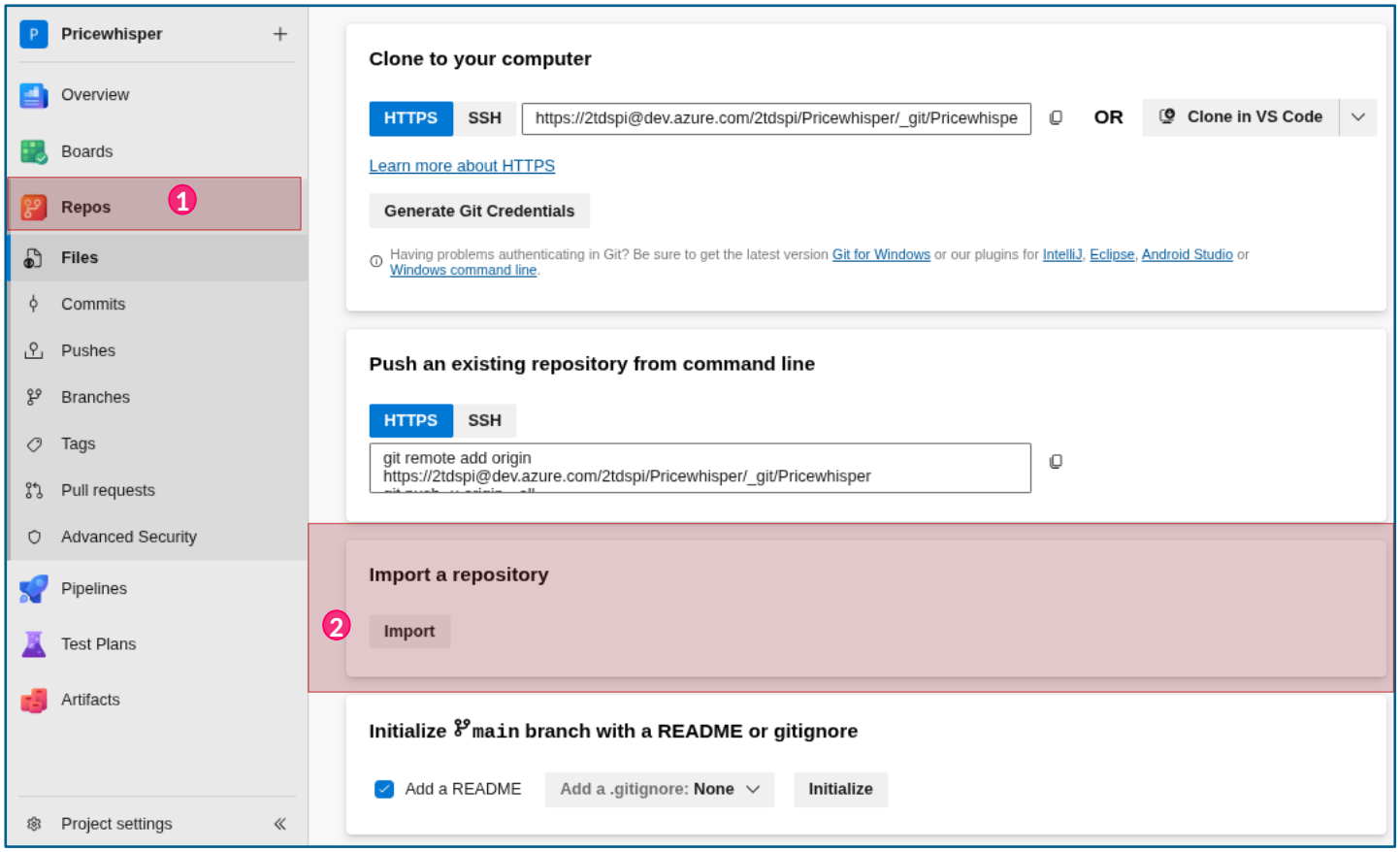
Para iniciar vamos criar um novo projeto no Azure Devops



Azure Pipelines API .NET com Oracle

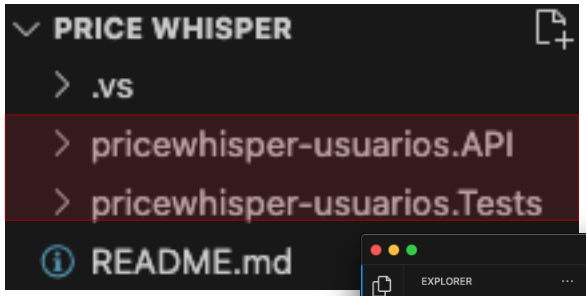
Agora vamos importar o projeto que está no GitHub para o Azure Repos

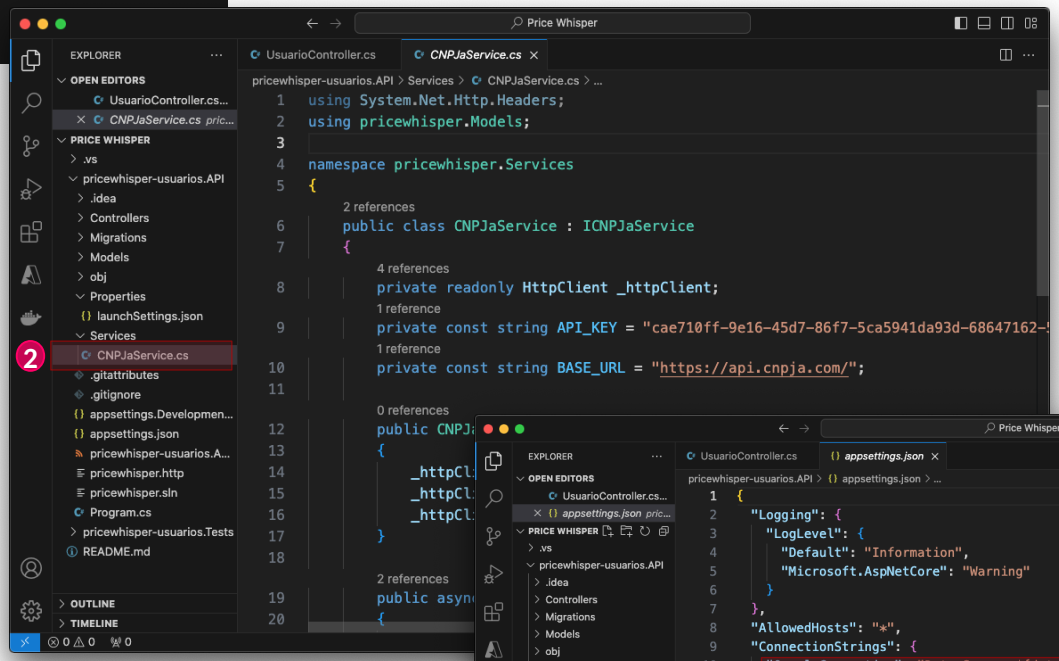
https://github.com/profjoaomenk/pricewhisper-api-dotnet-oracle.git

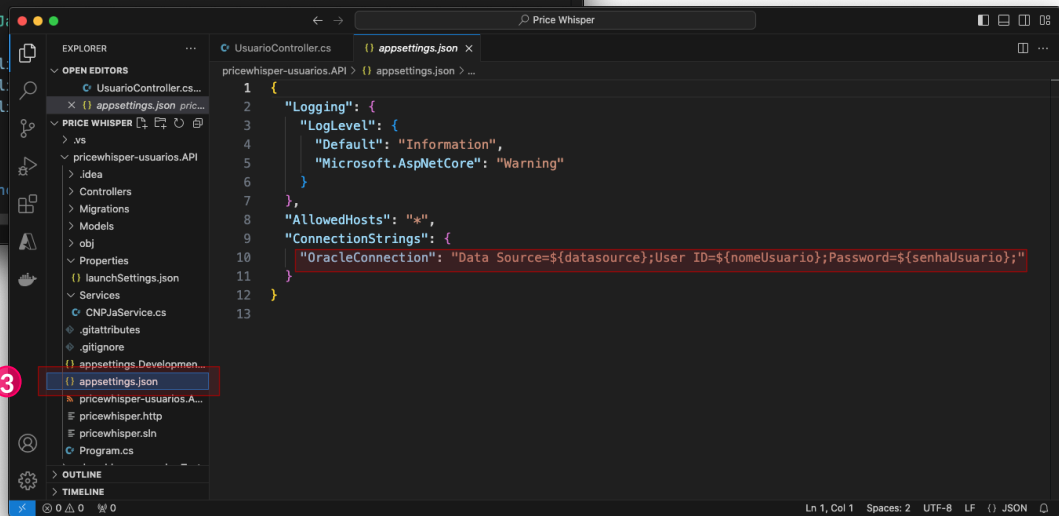


Azure Pipelines API .NET com Oracle

Analise o Código do Projeto juntamente com seu Professor

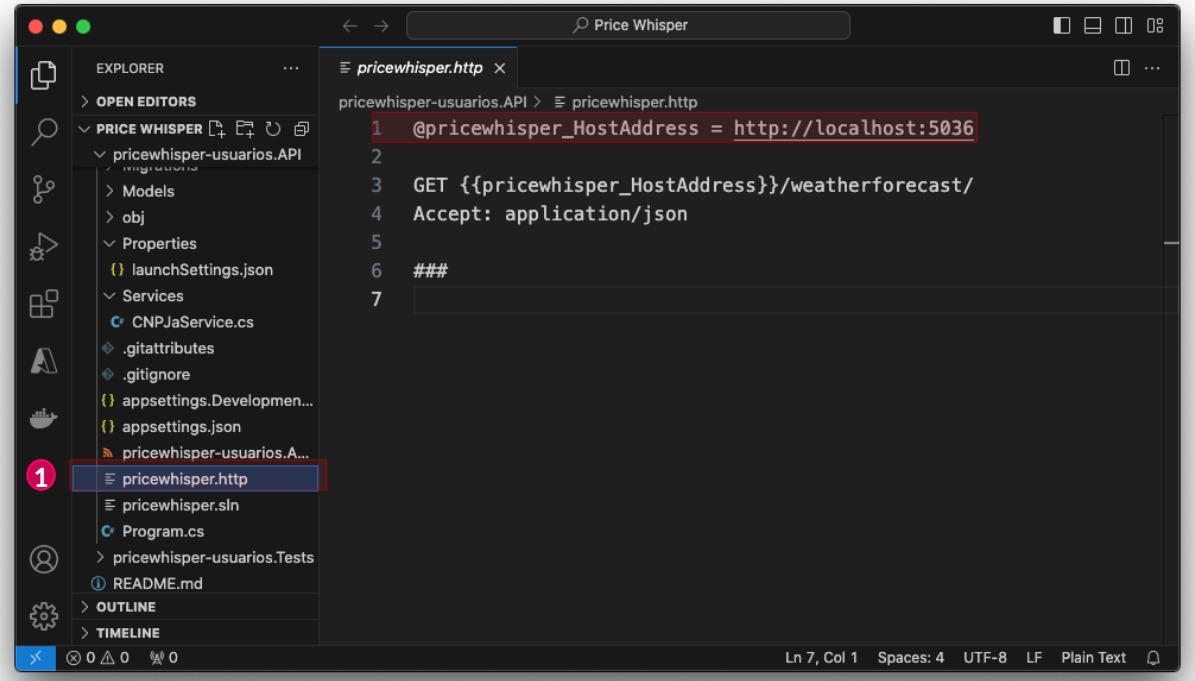






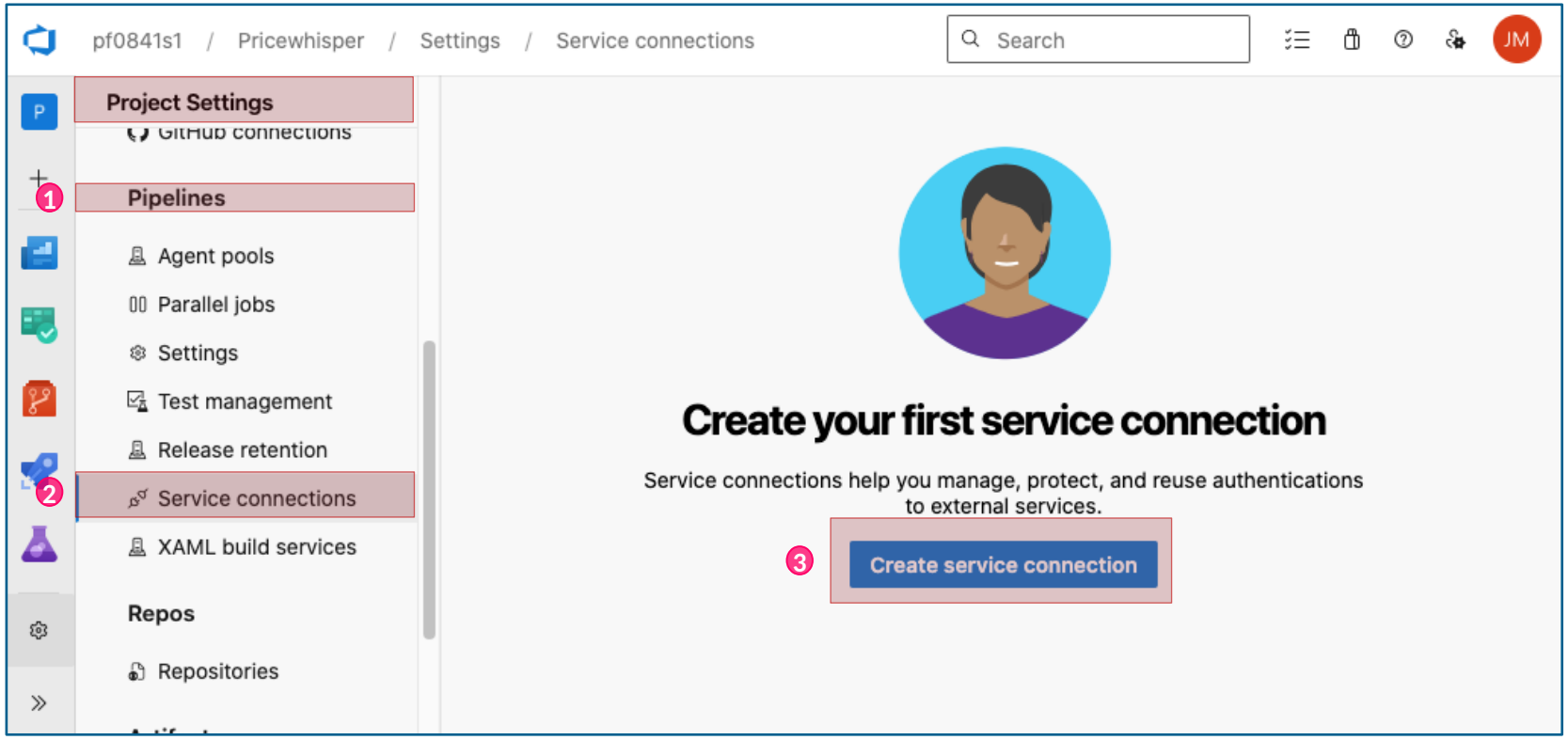
Azure Pipelines API .NET com Oracle

Analise o Código do Projeto juntamente com seu Professor



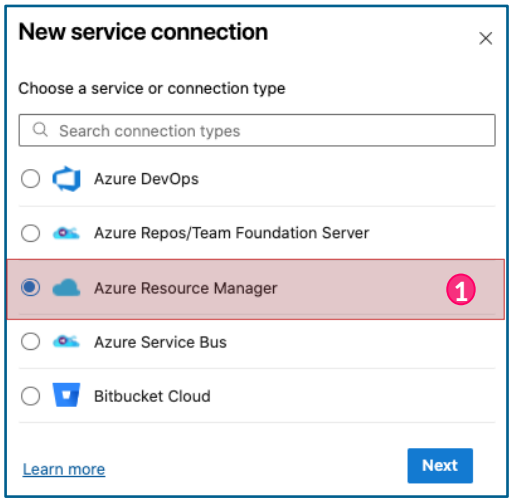
Azure Pipelines API .NET com Oracle

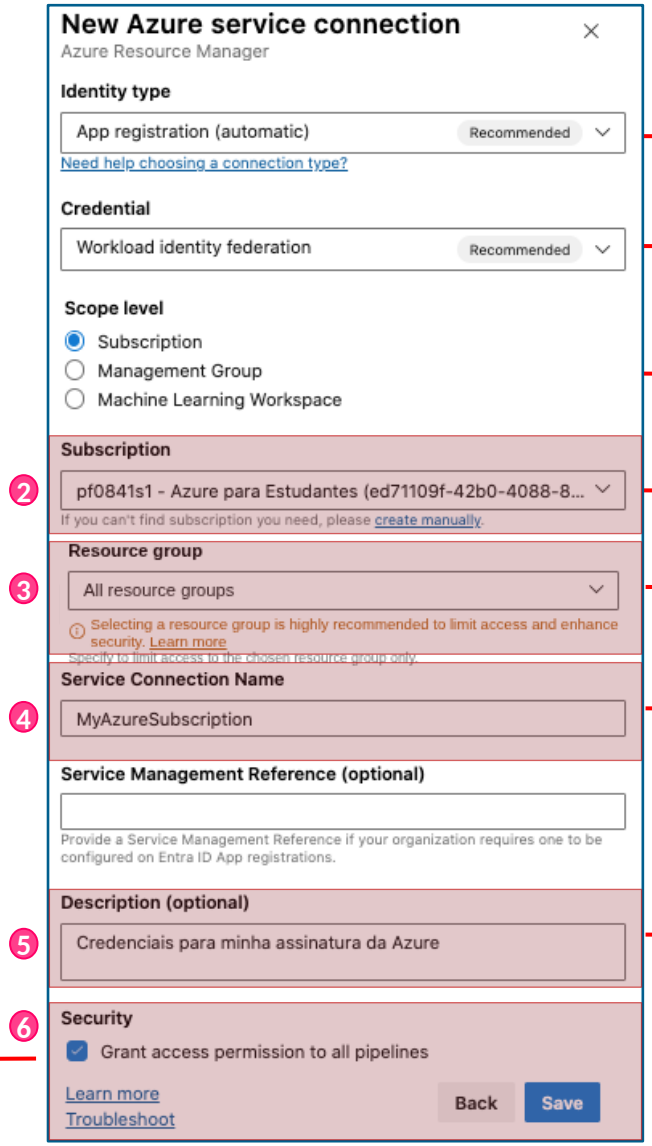
Agora que temos o Projeto criado e seu código fonte vamos criar uma **Conexão de Serviço (Service Connection)** com a nossa assinatura do Azure



Azure Pipelines API .NET com Oracle

Escolha **Azure Resource Manager** como o Service connection e o configure para sua assinatura da Azure





Deixe o padrão

Deixe o padrão

Nível do escopo: Subscription

Sua assinatura

Grupo de Recursos

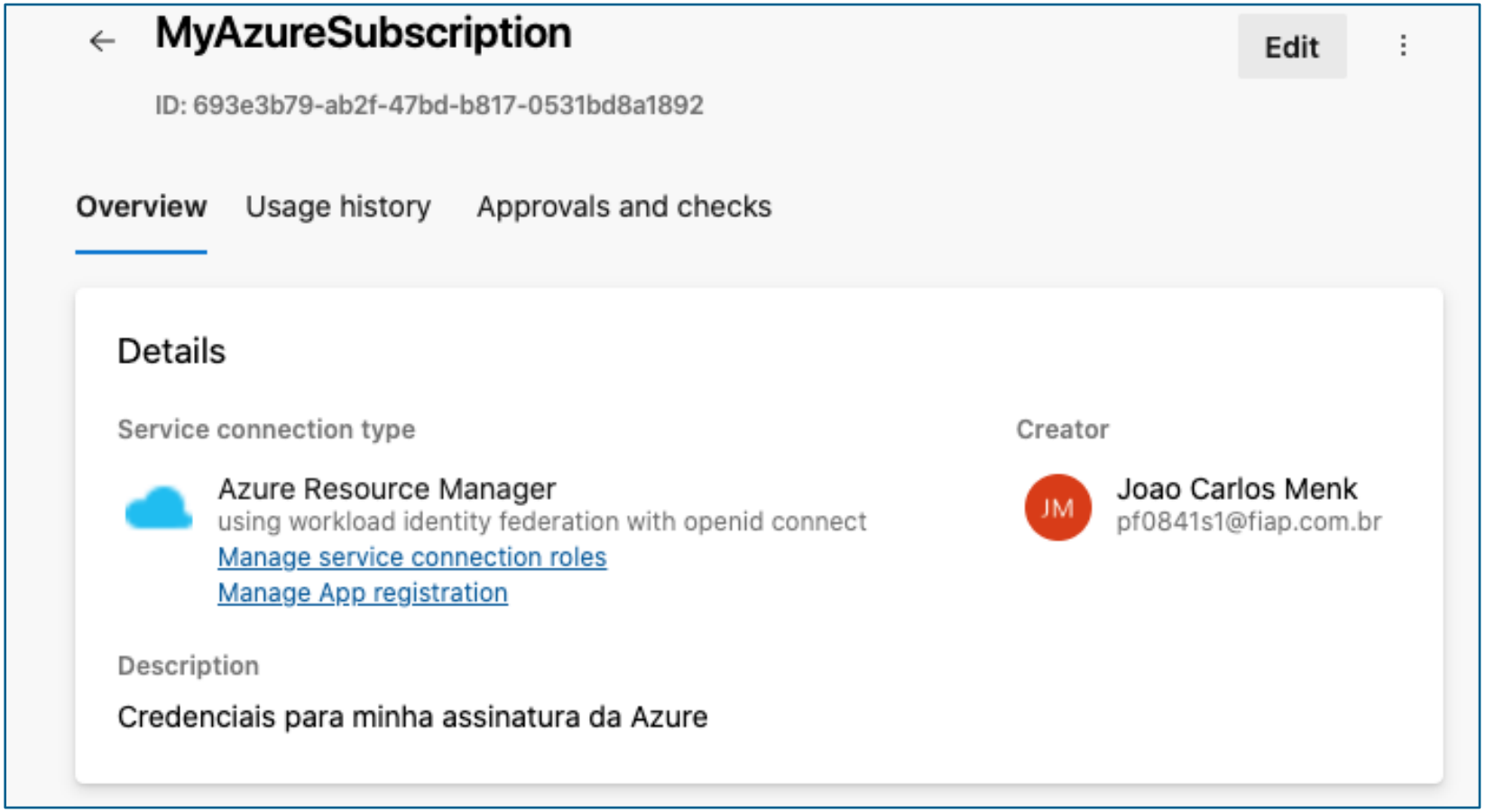
Nome da Conexão de Serviço

Descrição

Todas as Pipelines podem acessar esse Serviço

Azure Pipelines API .NET com Oracle

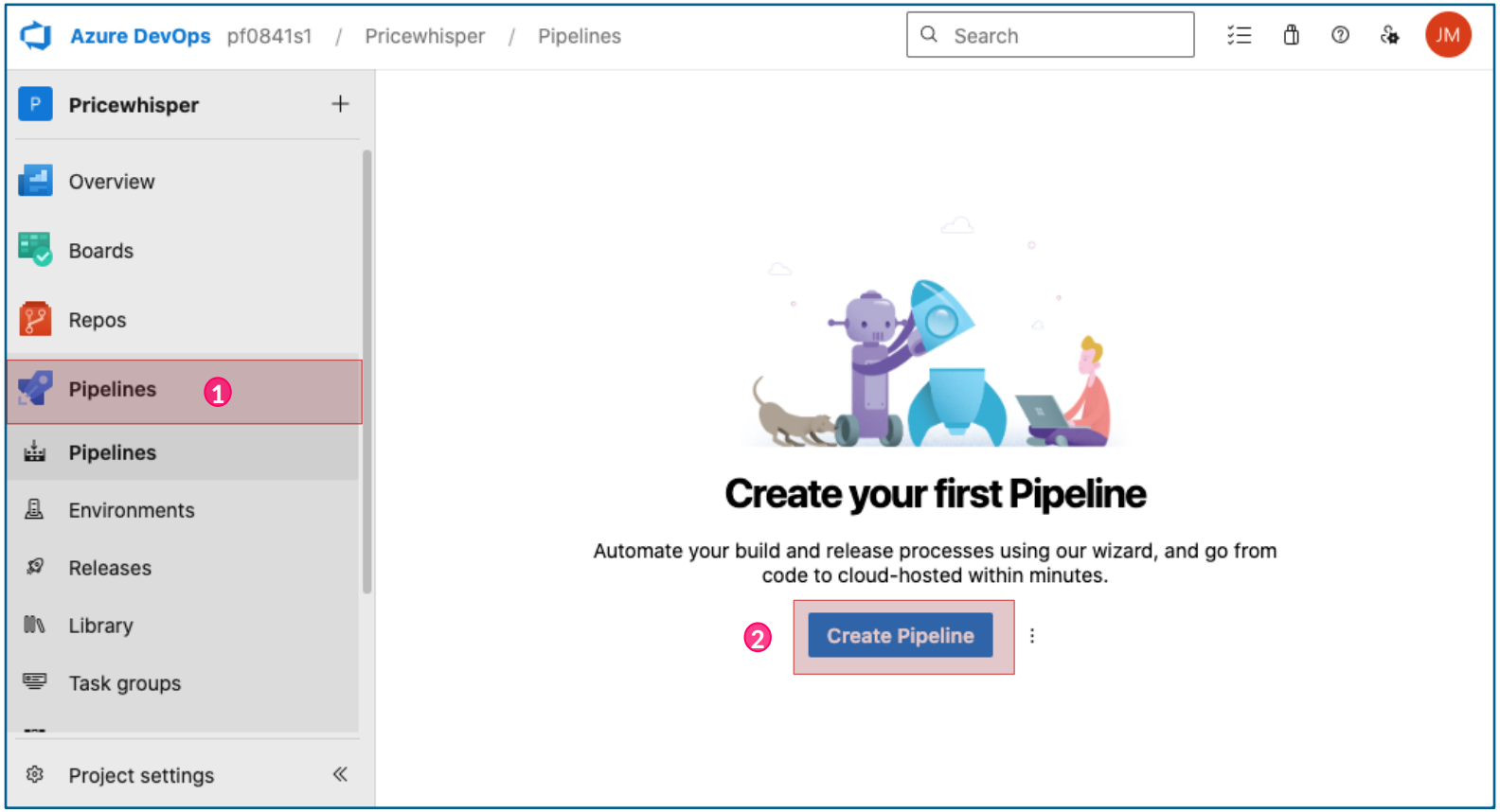
Service connection criado



Azure Pipelines API .NET com Oracle

Agora sim podemos começar a criar nossa Pipeline CI/CD. Vamos começar com a nossa Pipeline de CI (Build)

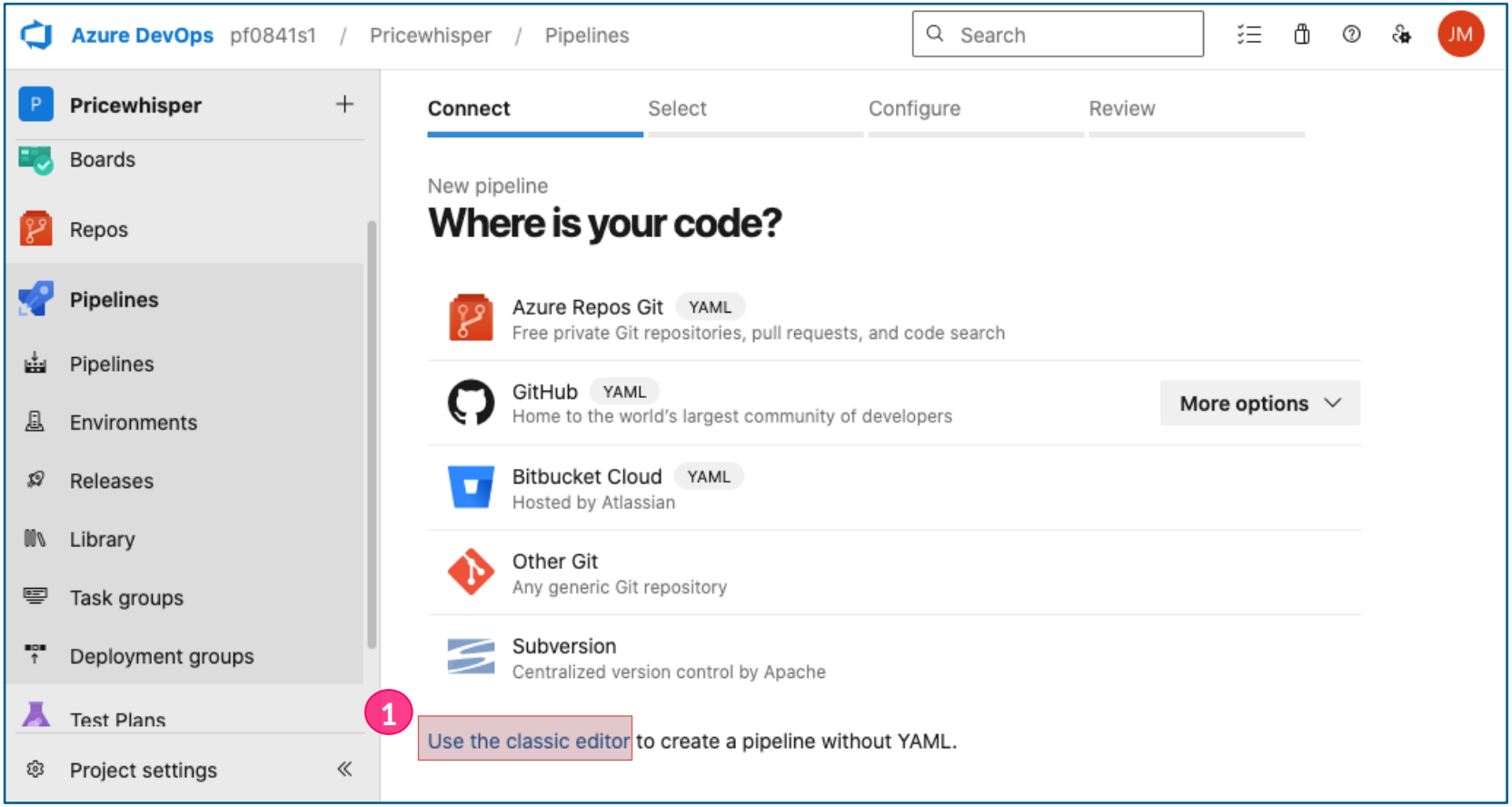
Acesse Pipelines e clique em **Create Pipeline** para começarmos



Azure Pipelines API .NET com Oracle

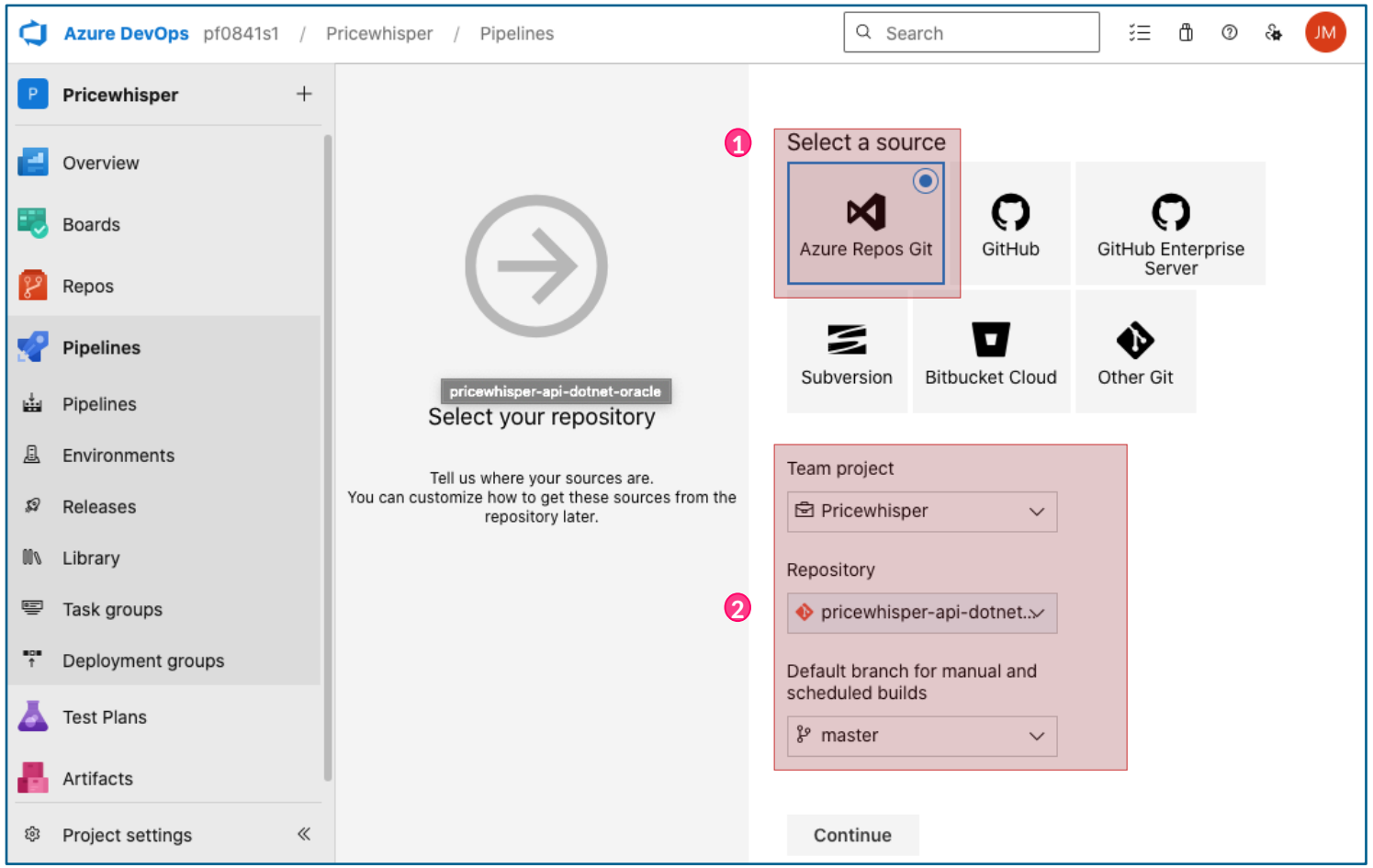
FIAP

Utilize o Classic Editor para essa Pipeline



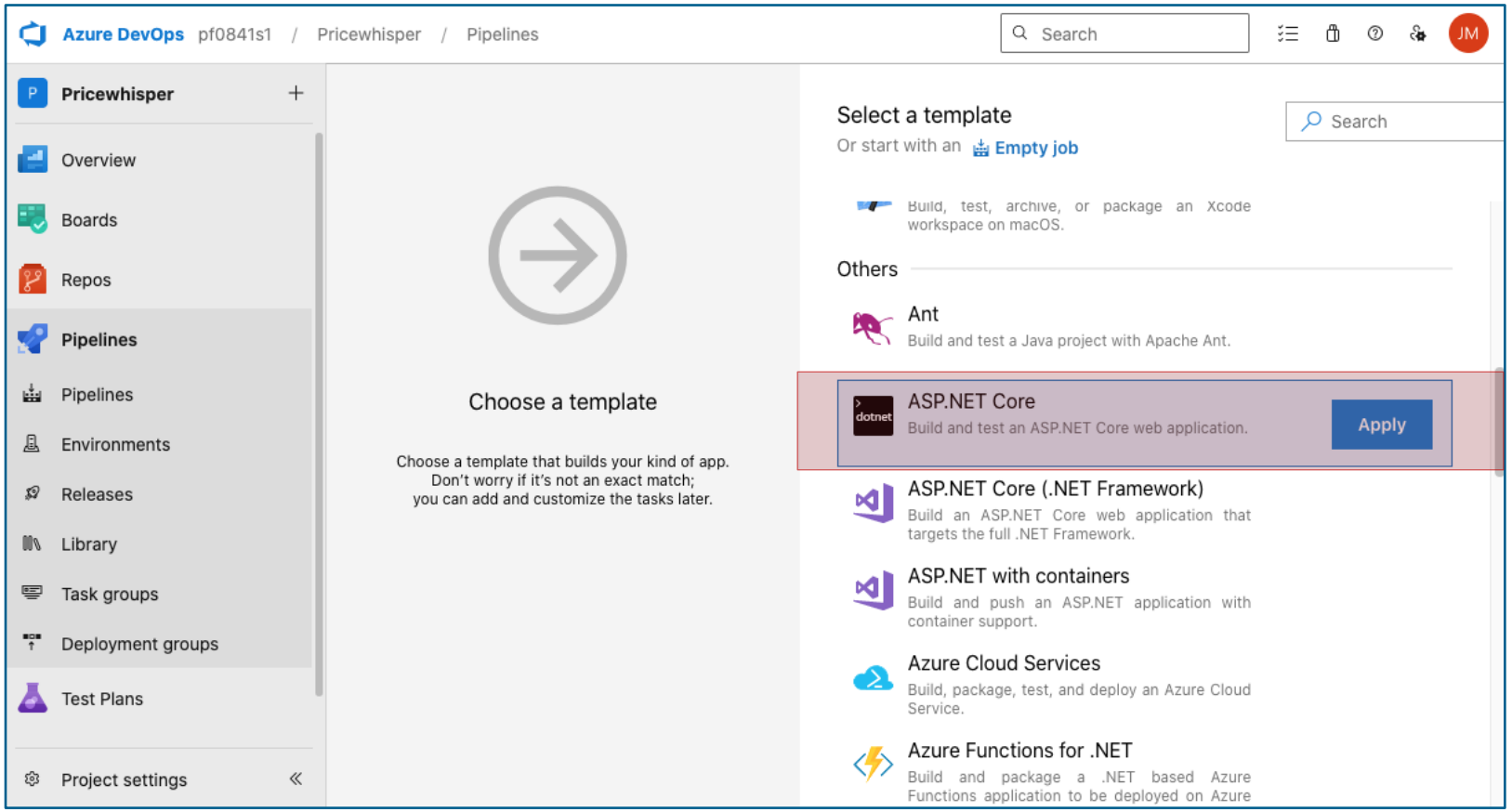
Azure Pipelines API .NET com Oracle

Escolha o Repositório do Projeto que acabamos de importar



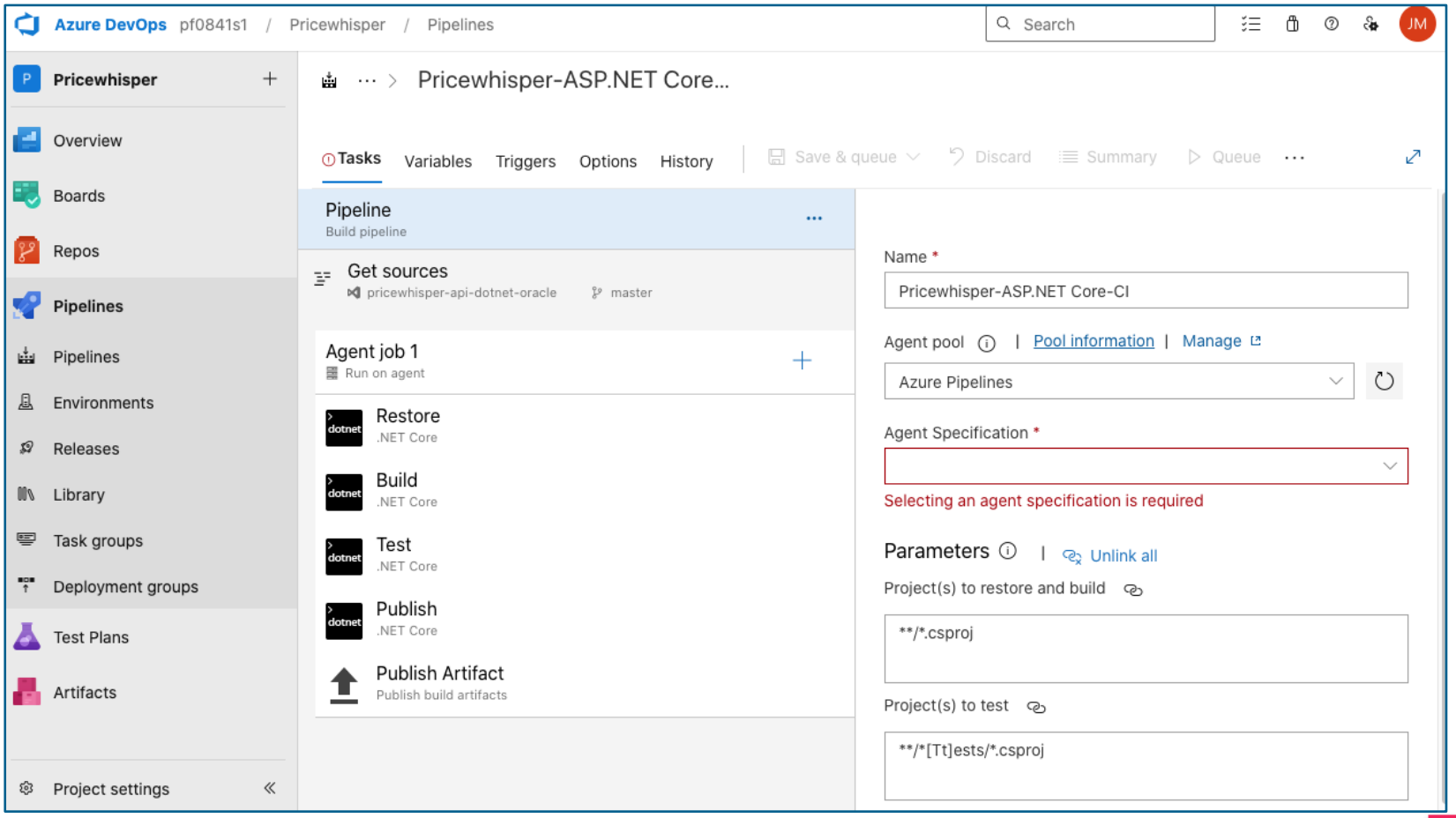
Azure Pipelines API .NET com Oracle

Procure pelo Template: **ASP.NET Core** e selecione



Azure Pipelines API .NET com Oracle

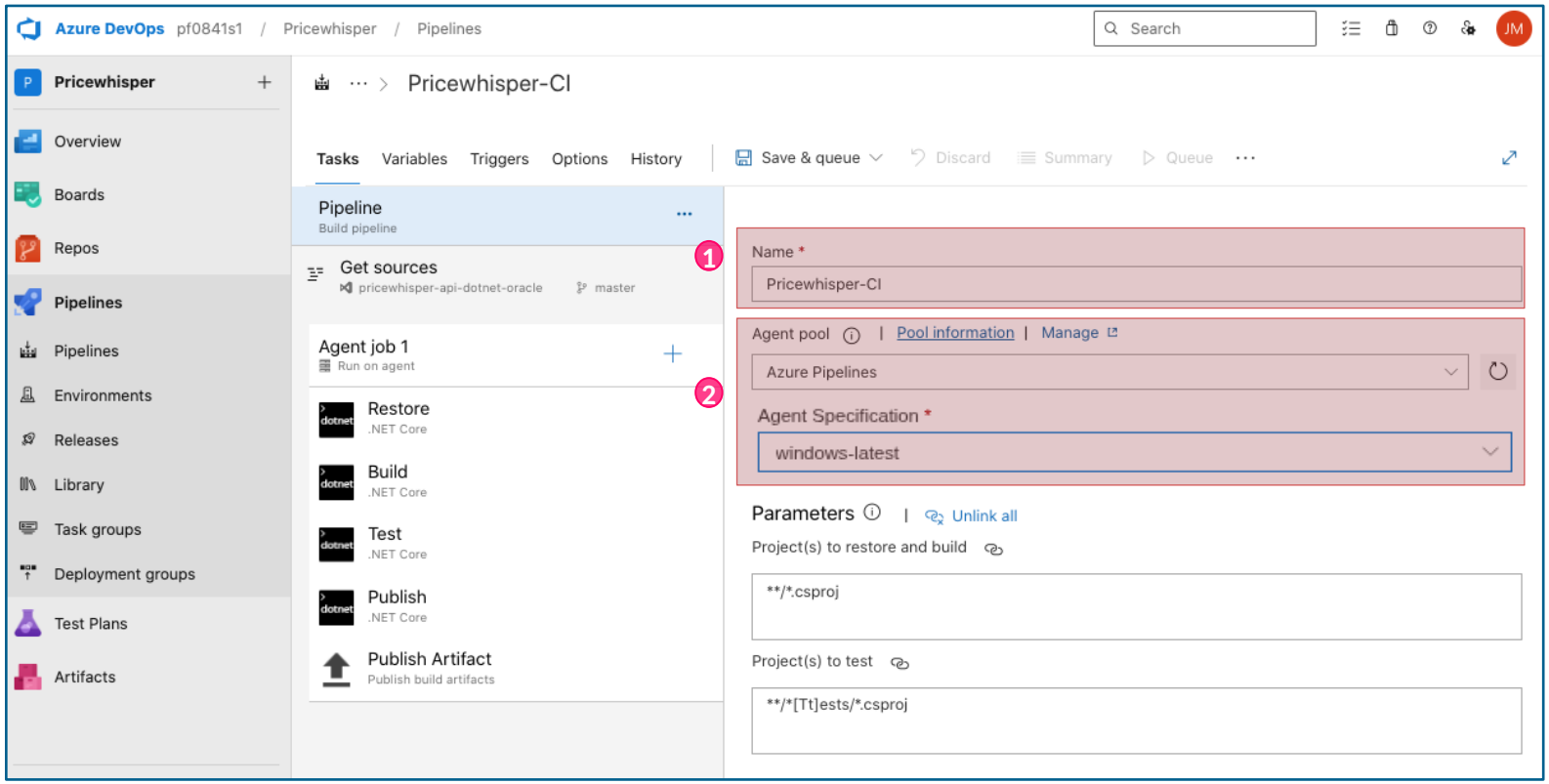
Template criado, agora podemos realizar as adaptações necessárias



Azure Pipelines API .NET com Oracle

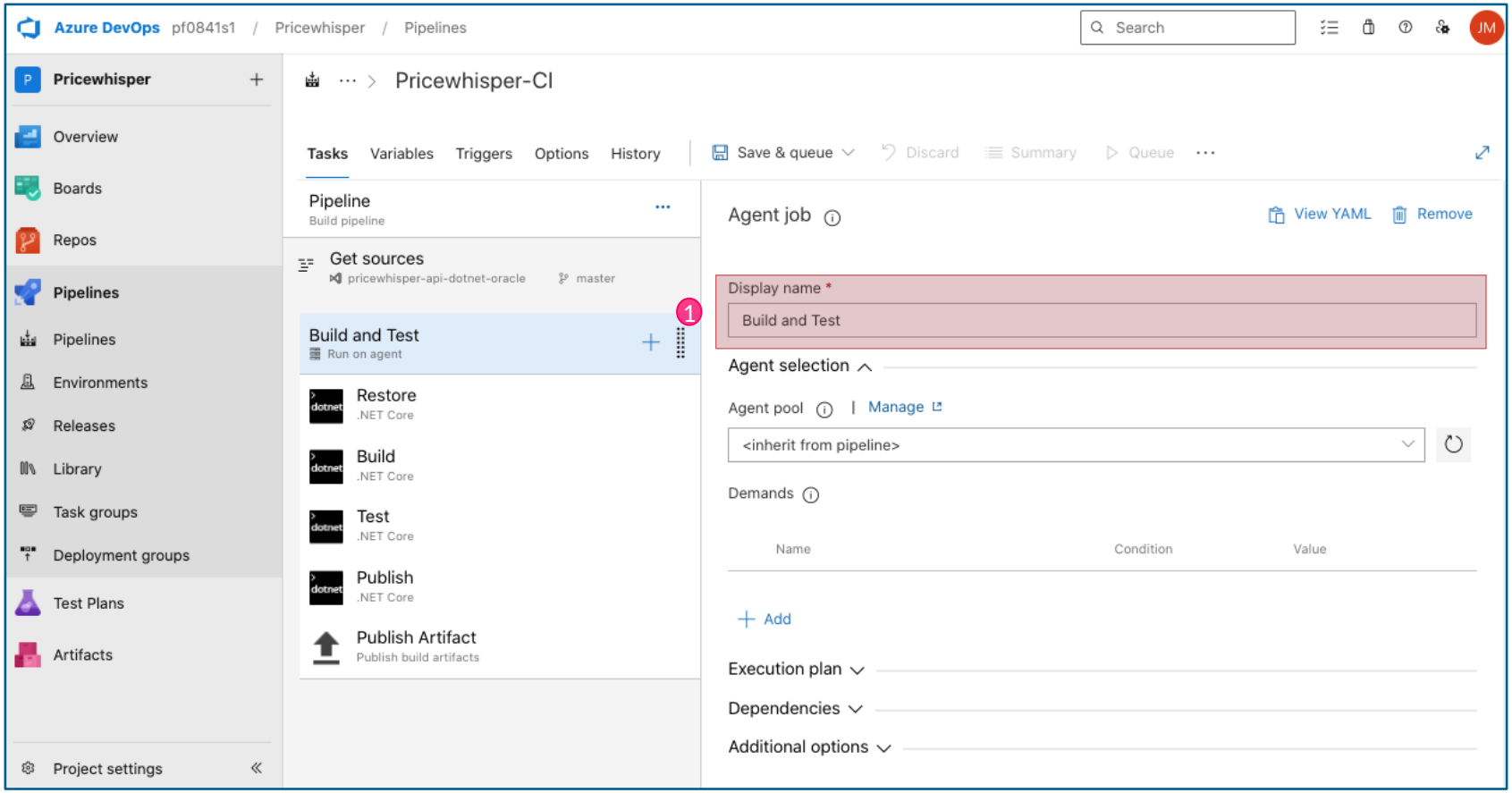
Vamos começar a informar as propriedades da nossa Pipeline de CI

Informe um **novo nome para nossa Pipeline** e como será **o ambiente de execução do CI**



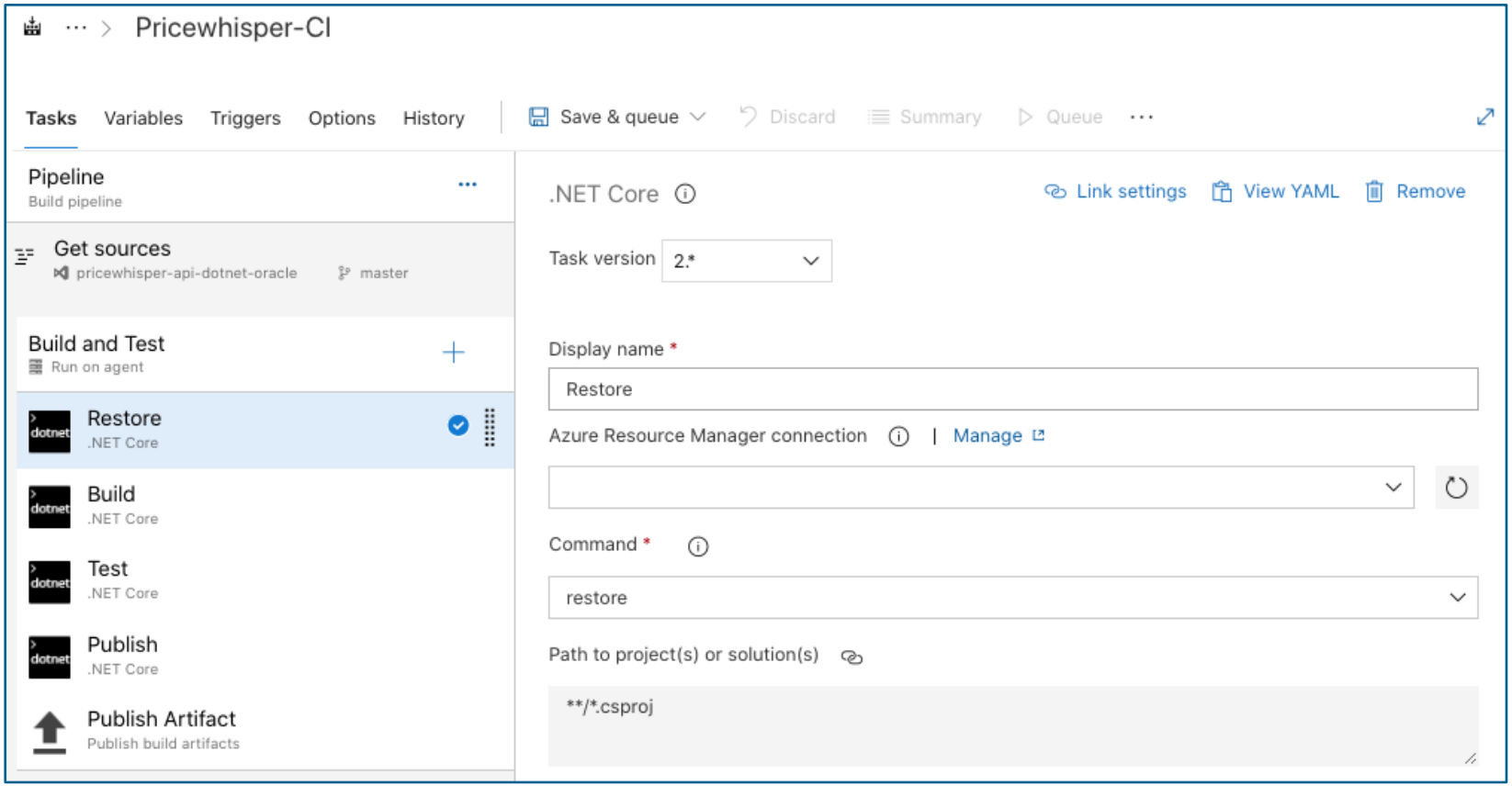
Azure Pipelines API .NET com Oracle

O nome do nosso Job pode ficar como: **Build and Test**



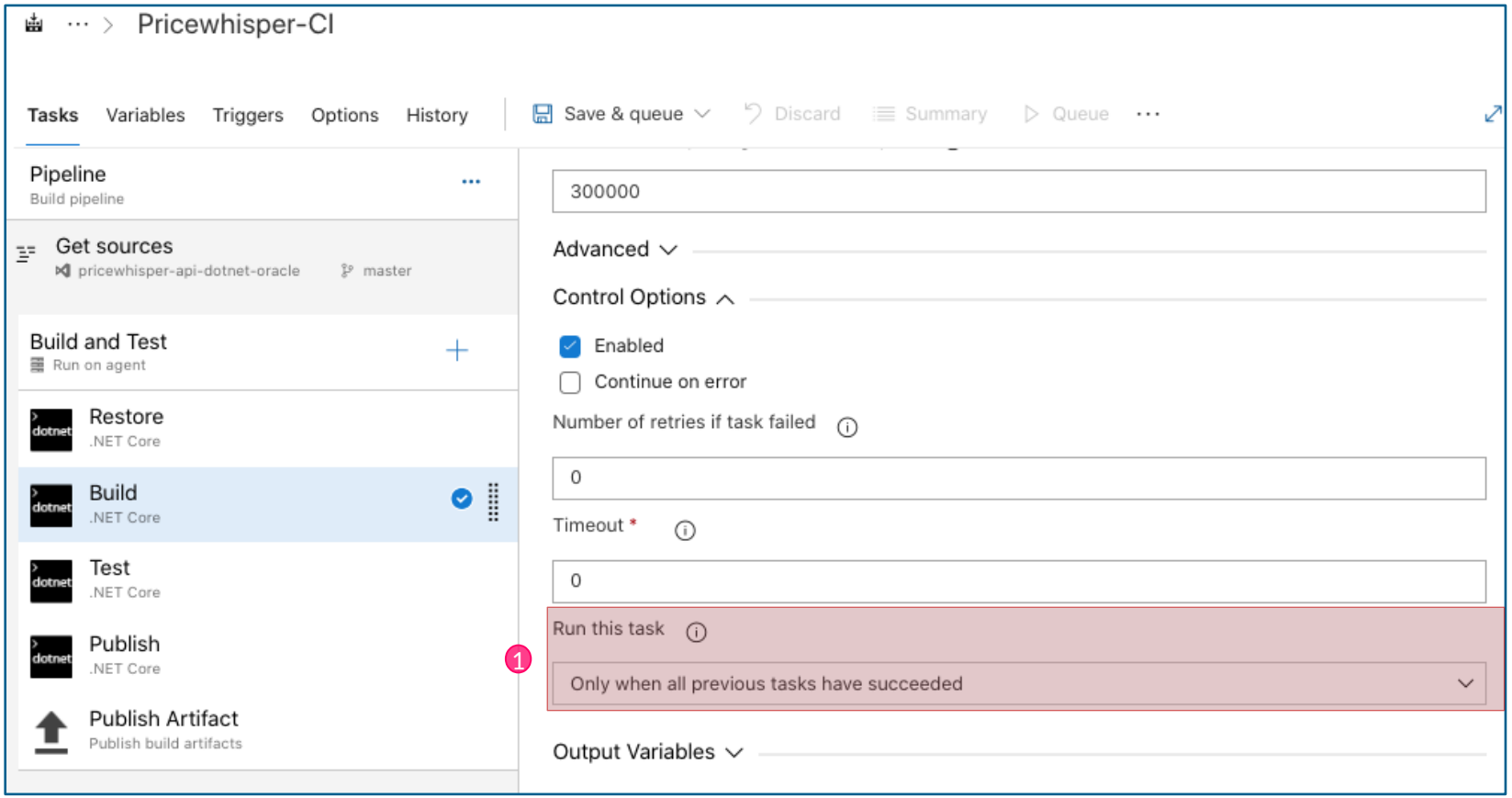
Azure Pipelines API .NET com Oracle

A Task **Restore** não iremos atualizar, aceitaremos os padrões oferecidos



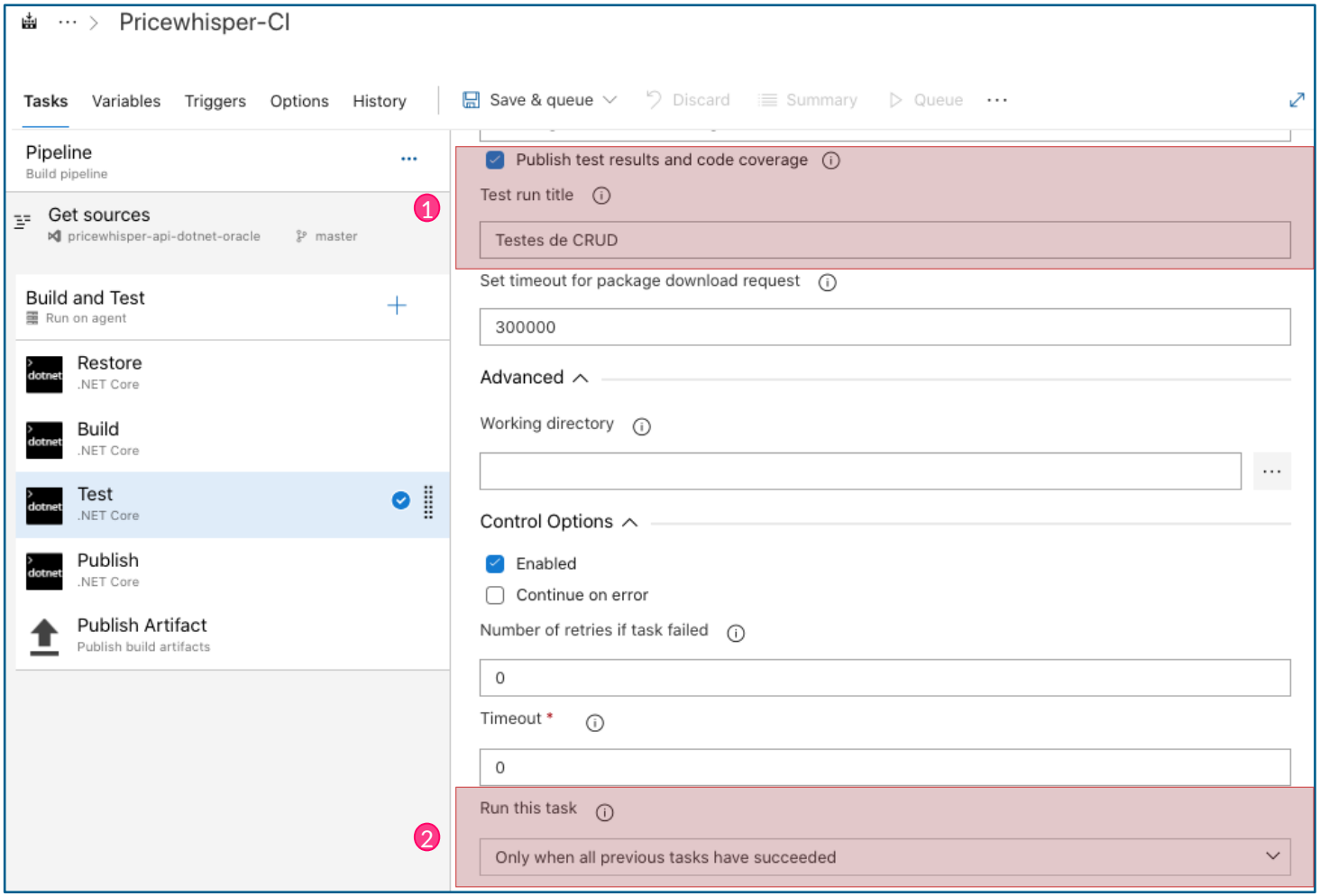
Azure Pipelines API .NET com Oracle

Na Task **Build** só iremos atualizar a precedência de execução



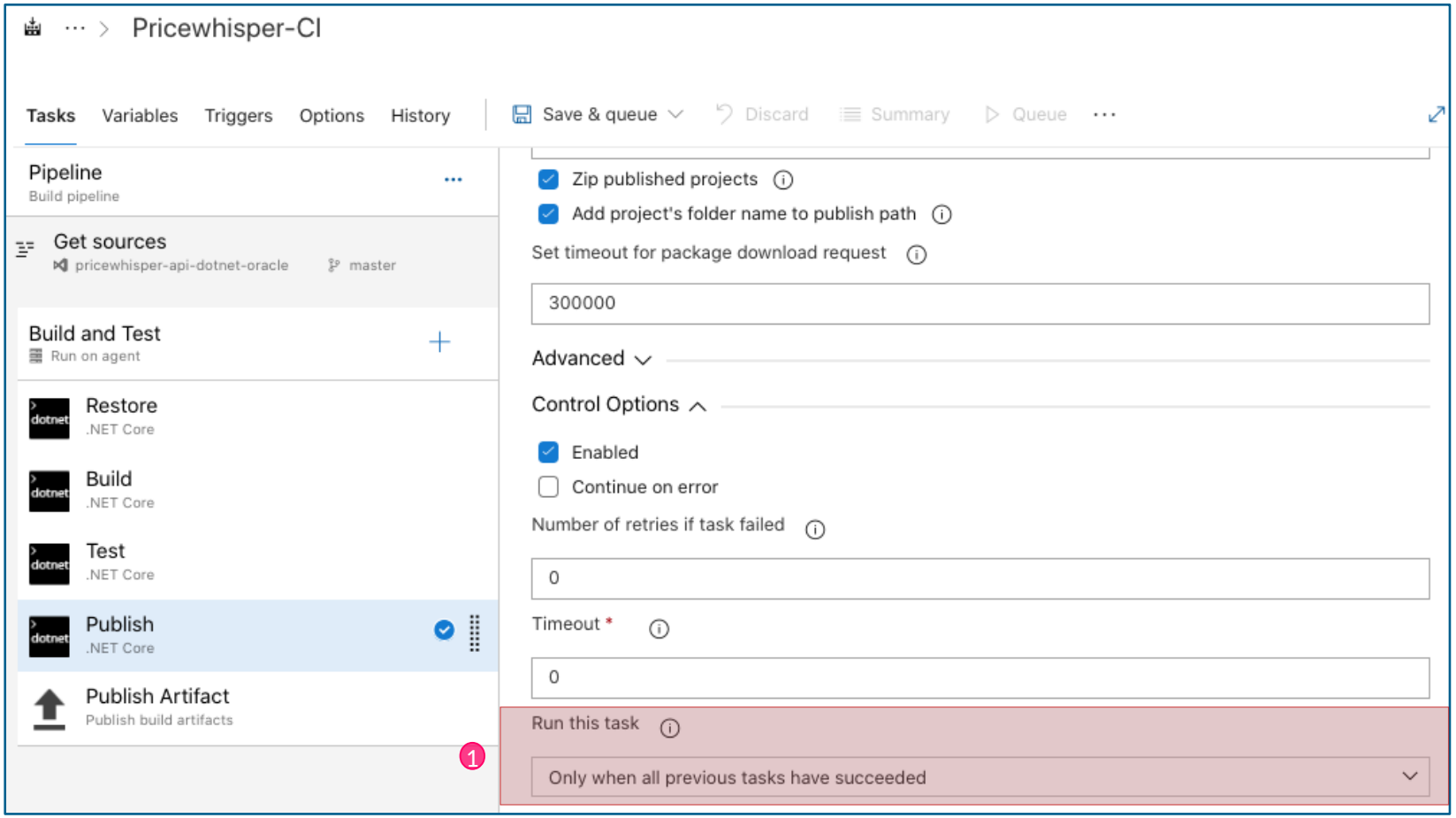
Azure Pipelines API .NET com Oracle

Na Task **Test** iremos apenas adicionar um título para os testes realizados e a precedência de execução



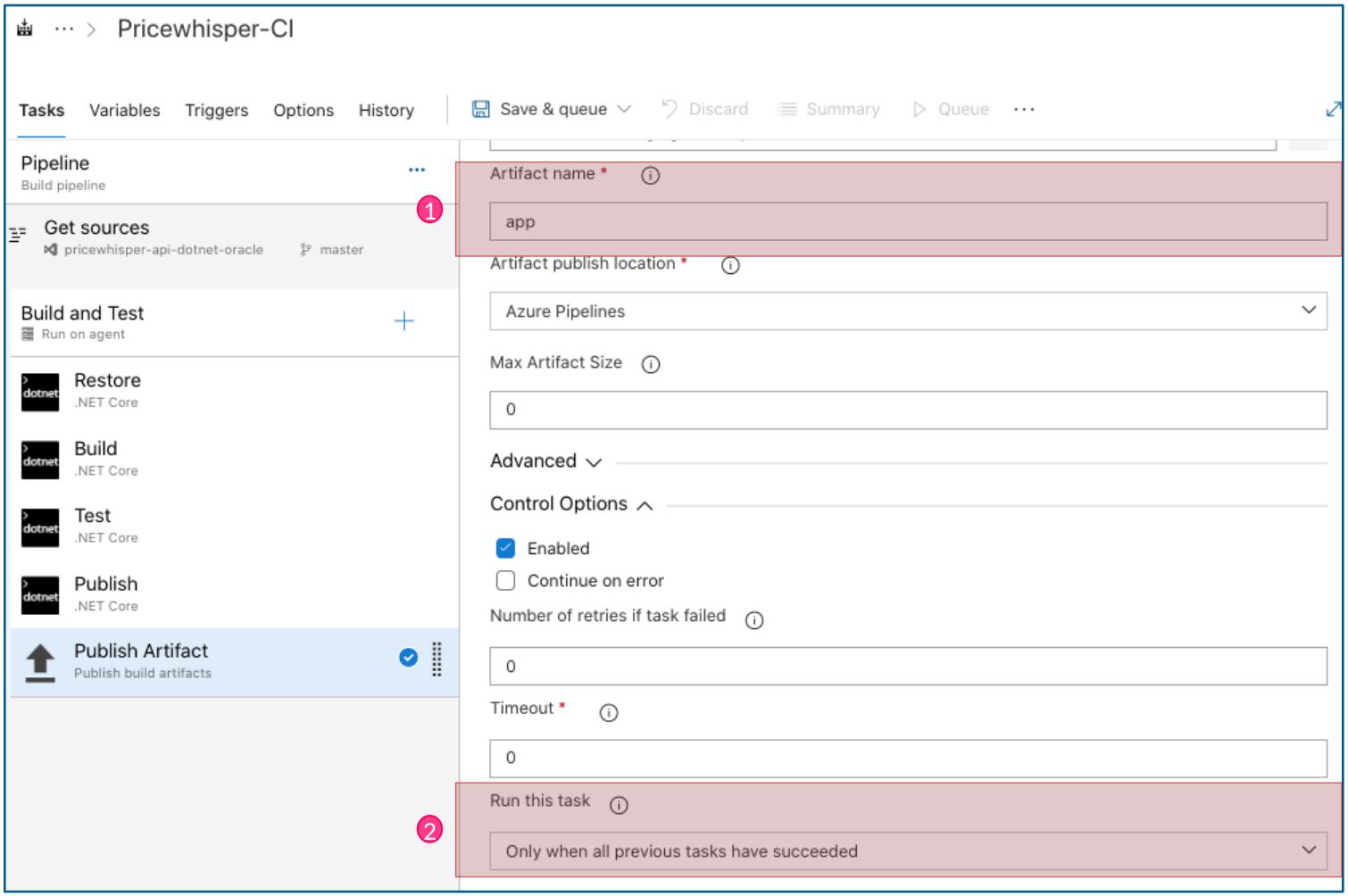
Azure Pipelines API .NET com Oracle

Na Task **Publish** só iremos atualizar a precedência de execução



Azure Pipelines API .NET com Oracle

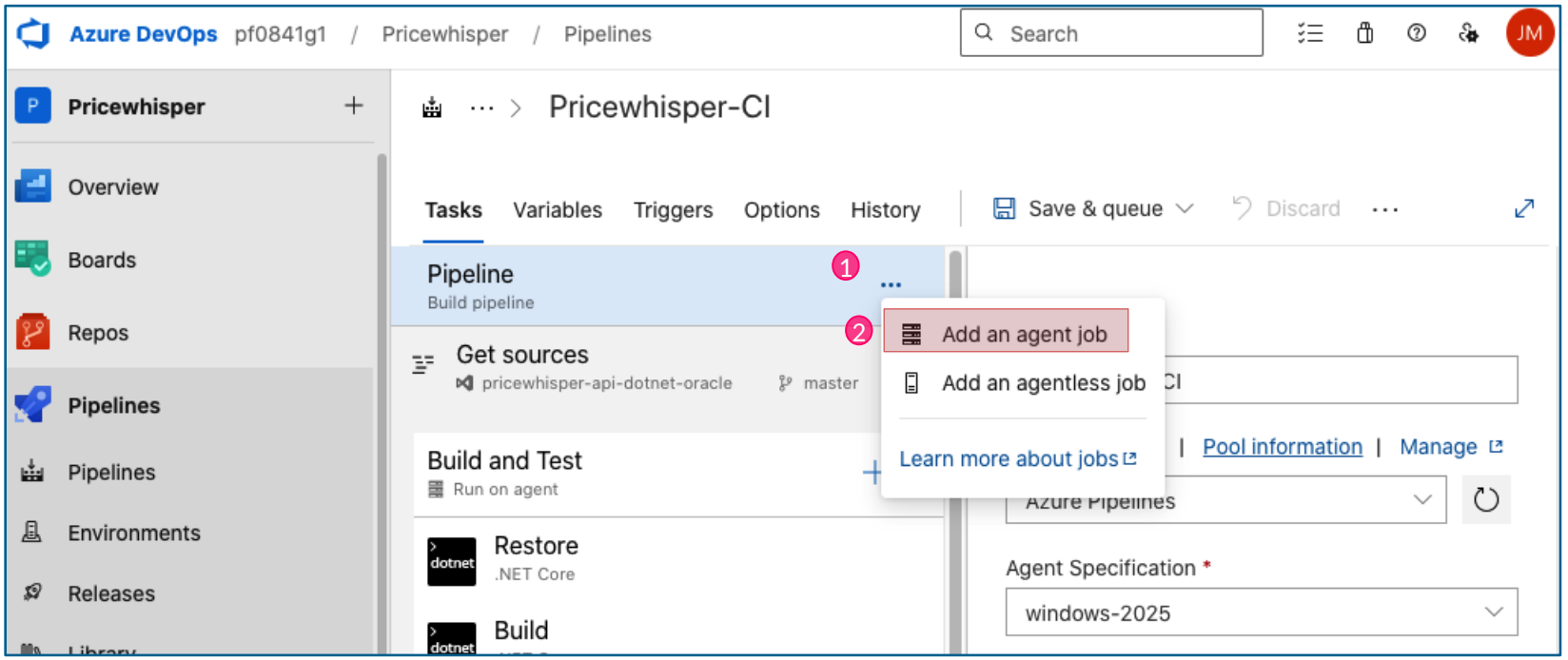
Na Task **Publish Artifact** só iremos alterar o nome do Artefato e atualizar a precedência de execução



Azure Pipelines API .NET com Oracle

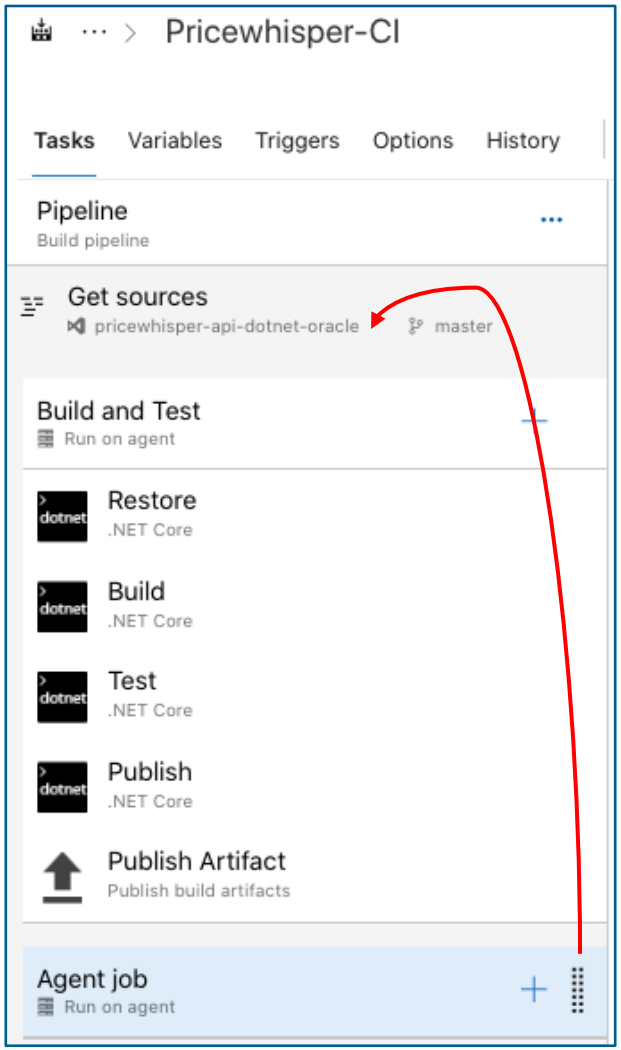
Agora que temos o CI completo e configurado vamos incluir outra execução do Agent, **uma Etapa para criar a Infra necessária do Projeto**, caso seja necessário

Clique na Pipeline e adicione um Agent job

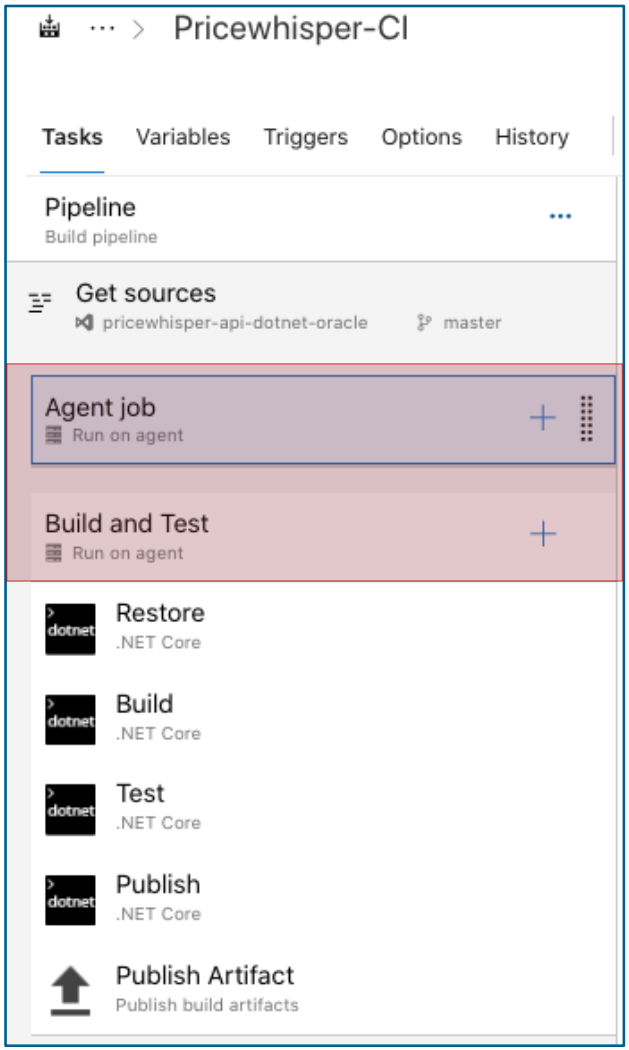


Azure Pipelines API .NET com Oracle

Arraste esse Job para ser o primeiro na lista

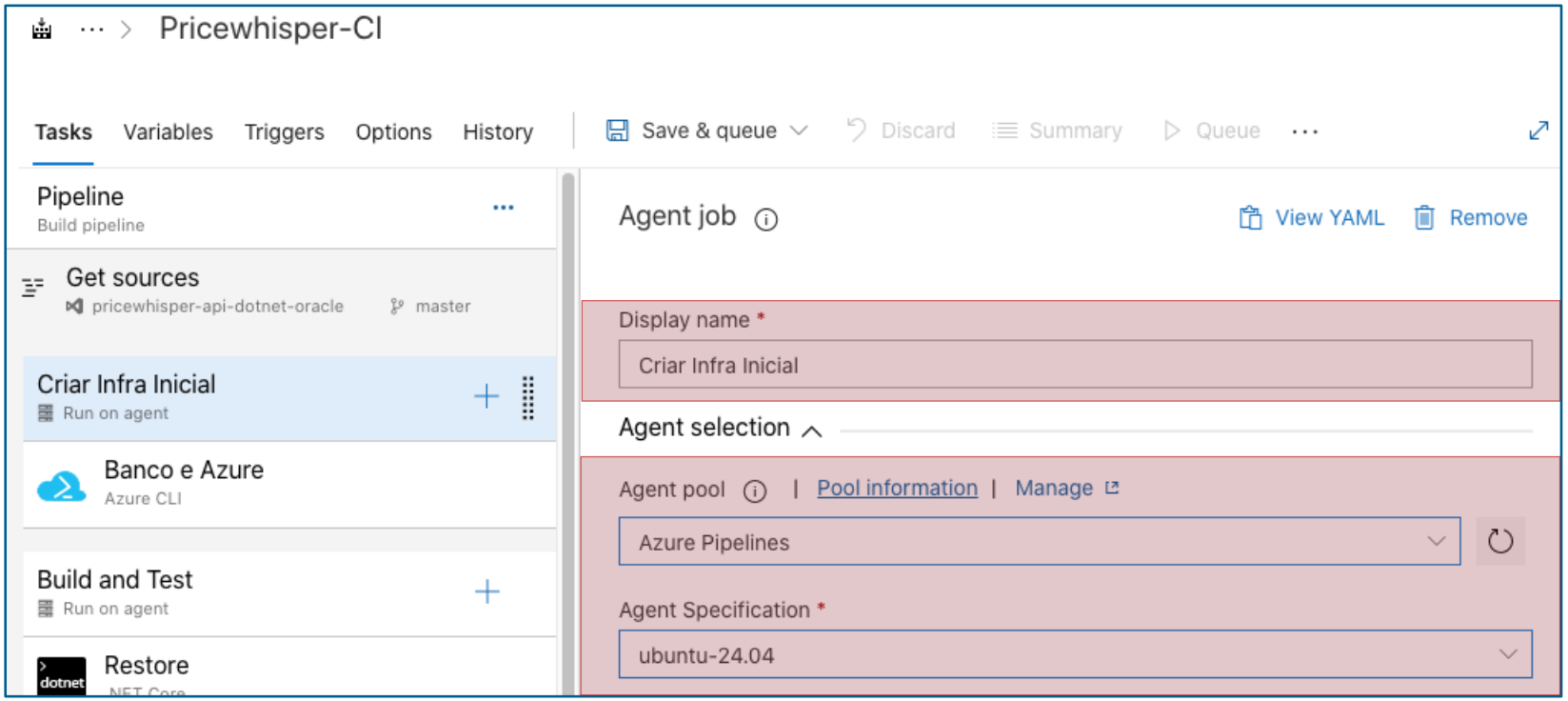






Azure Pipelines API .NET com Oracle

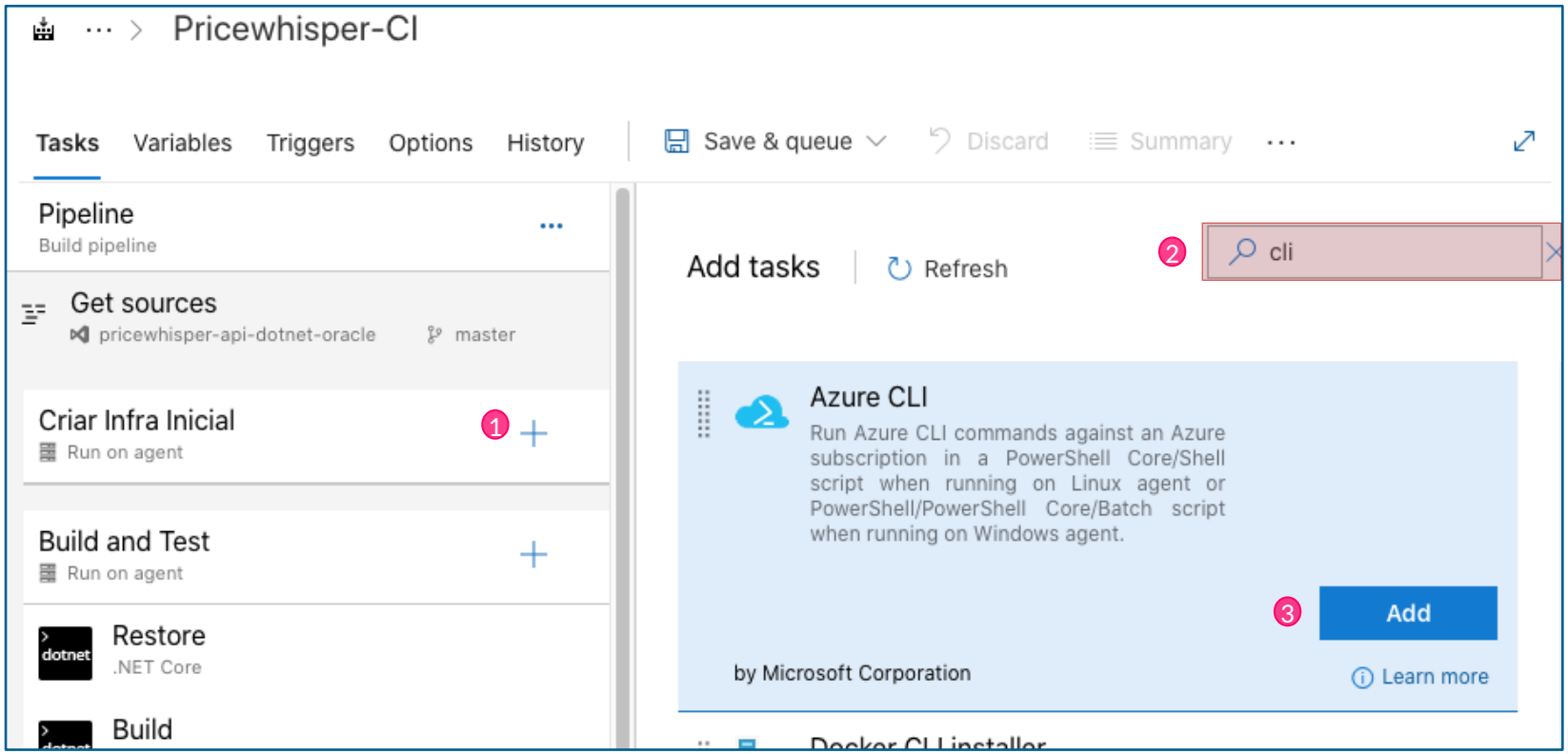
Clique no Agent job criado e altere seu **nome** e a **execução do Job**



Azure Pipelines API .NET com Oracle

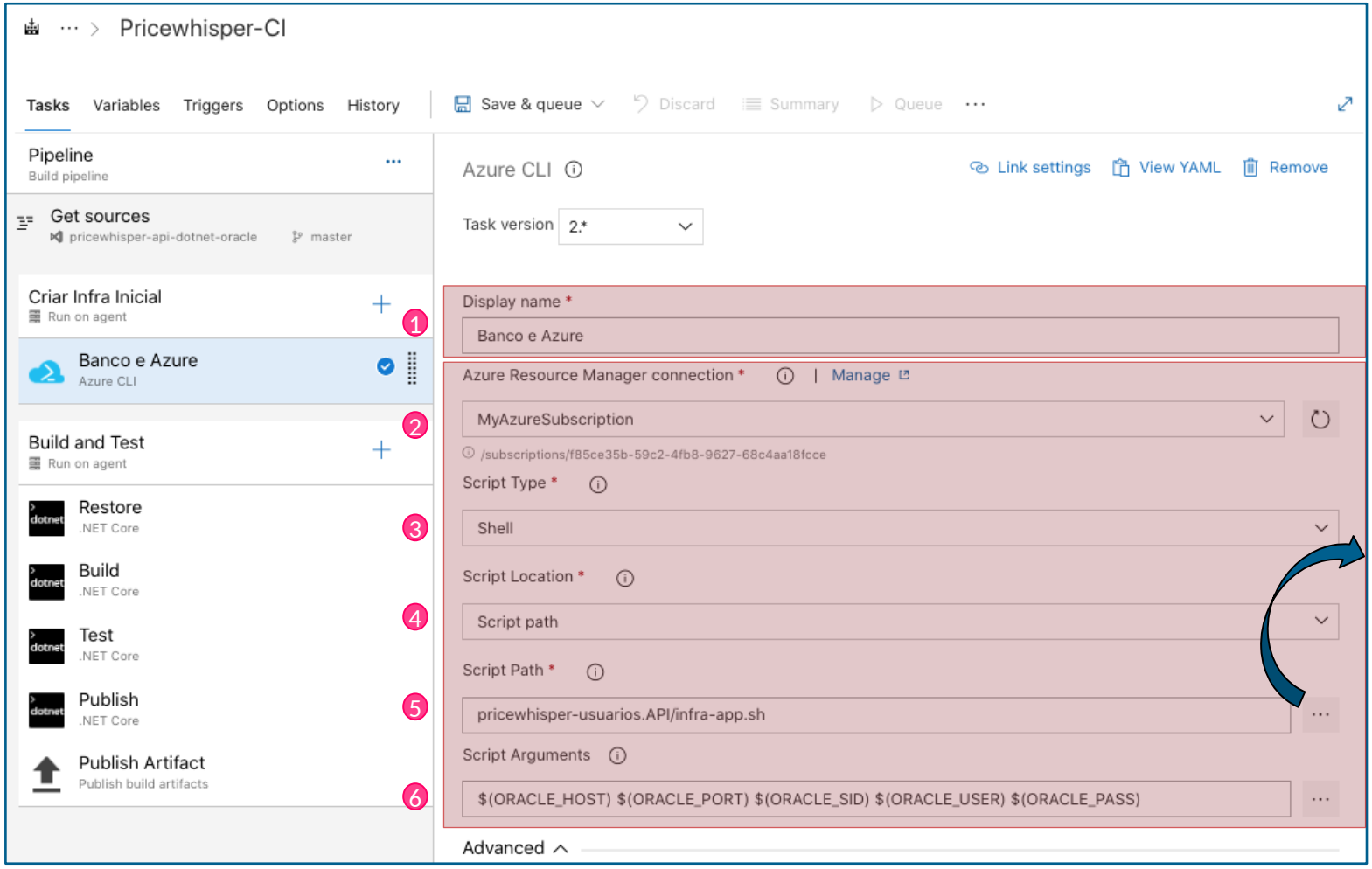
Agora vamos incluir uma nova tarefa para essa Etapa, a tarefa do Azure CLI

Clique no sinal de + no Agent Job criado, procure por CLI e clique no botão Add



Azure Pipelines API .NET com Oracle

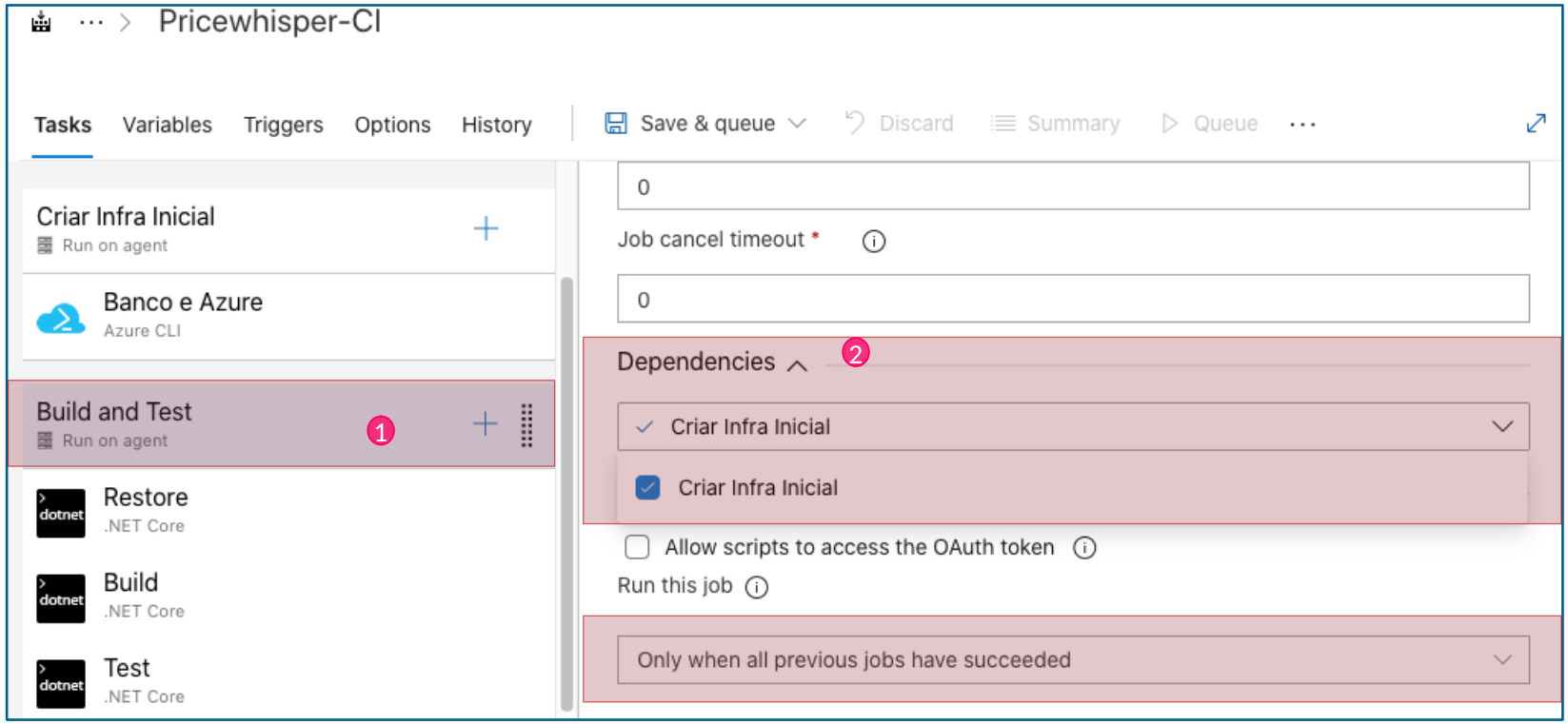
Pronto, incluímos uma nova Tarefa em uma Etapa. Selecione a nova Tarefa e **altere as propriedades conforme abaixo**



Procure o Script no Repos

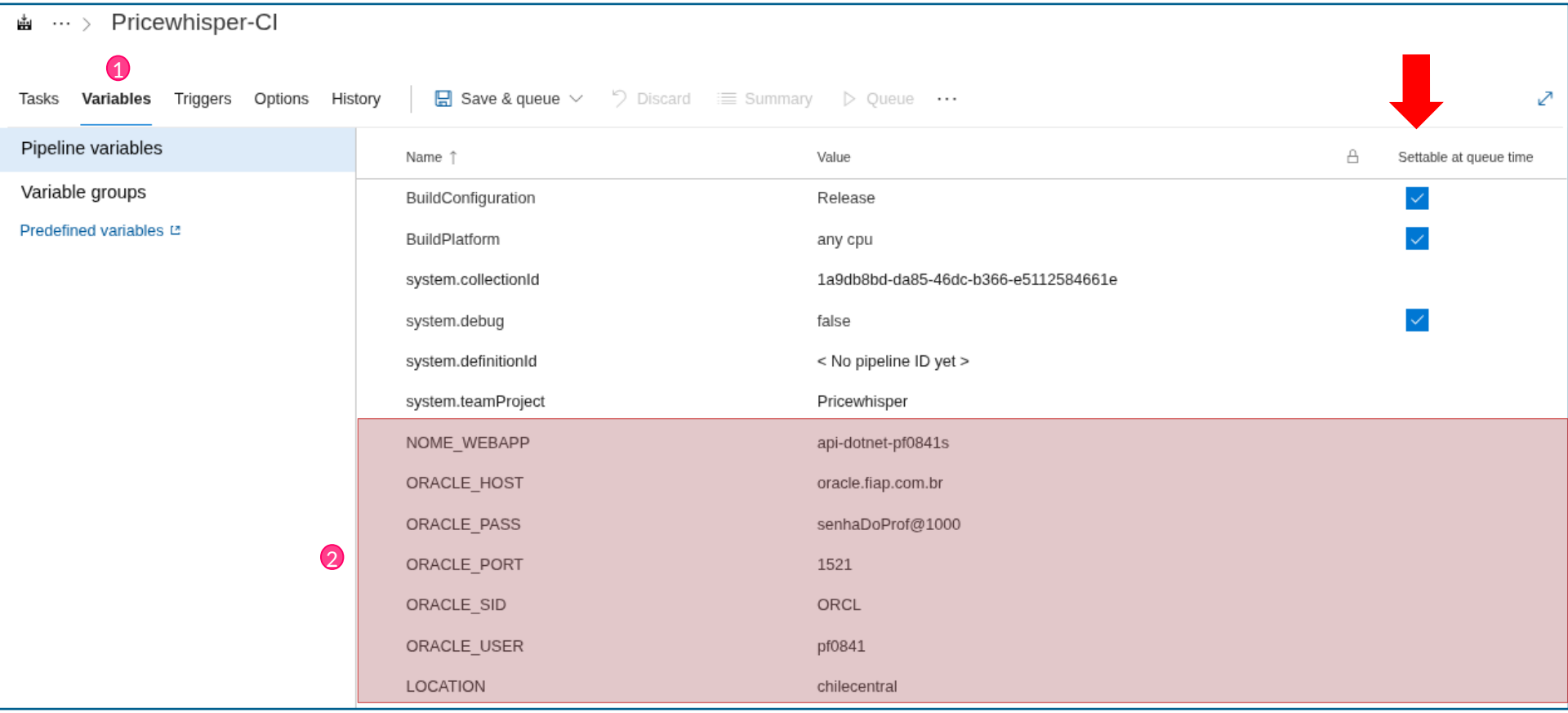
Azure Pipelines API .NET com Oracle

Agora só falta incluir a dependência entre as Etapas, **primeiro a Infra inicial**



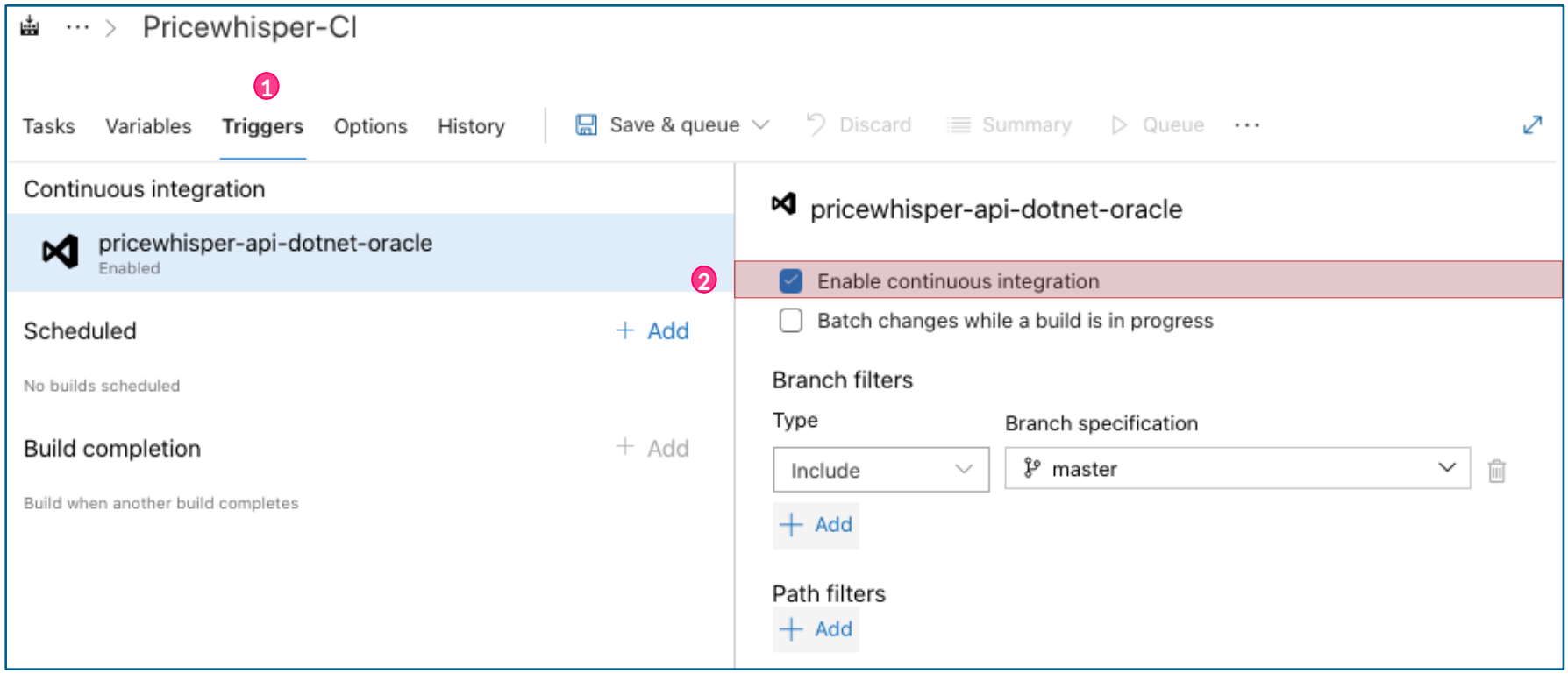
Azure Pipelines API .NET com Oracle

Vamos definir nossas Variáveis de Ambiente na sessão **Variables**



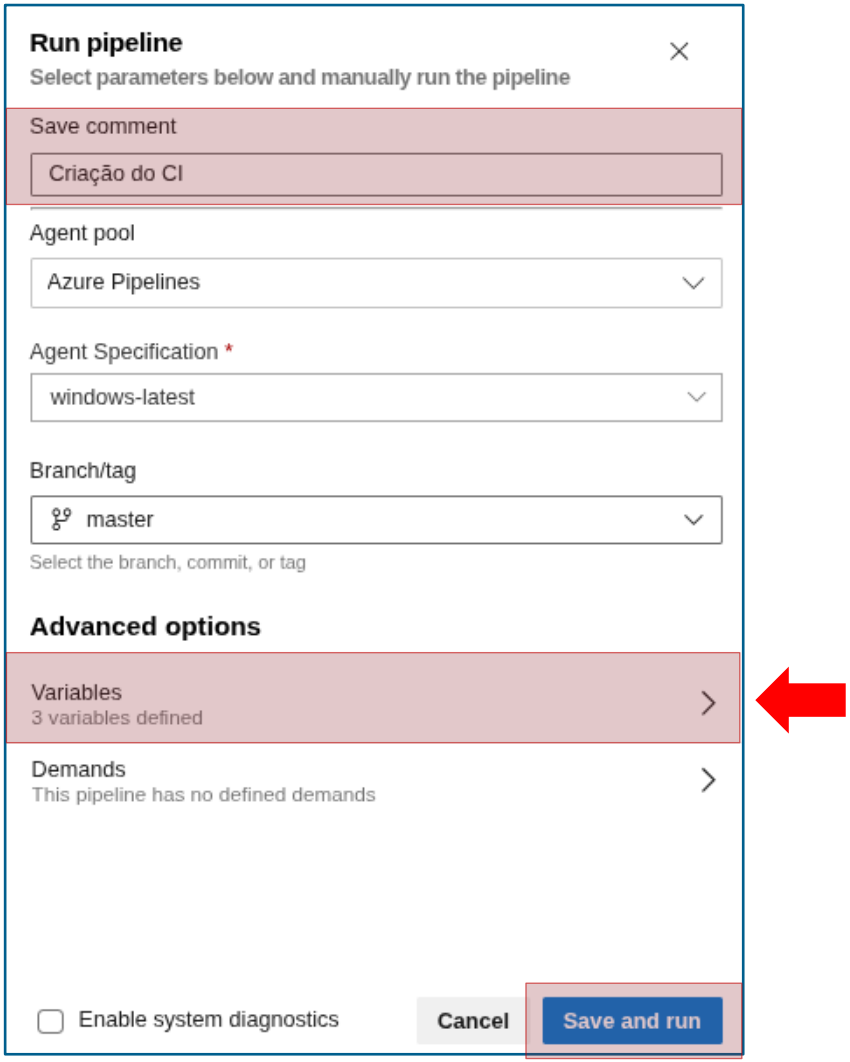
Azure Pipelines API .NET com Oracle

Na aba Triggers habilite a Integração Contínua



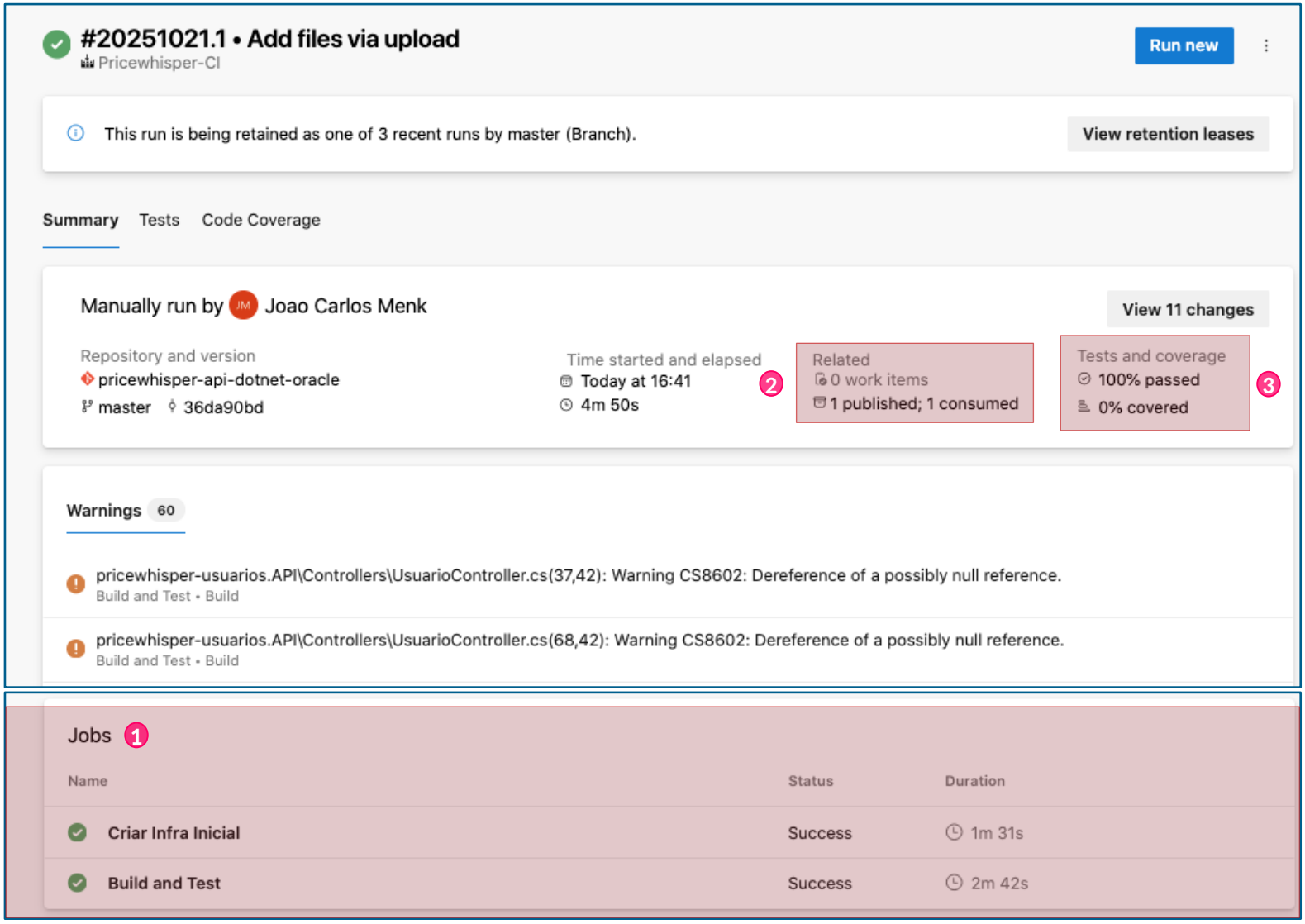
Azure Pipelines API .NET com Oracle

Salve e execute a Pipeline de CI – Verifique se existem parâmetros em Run Time



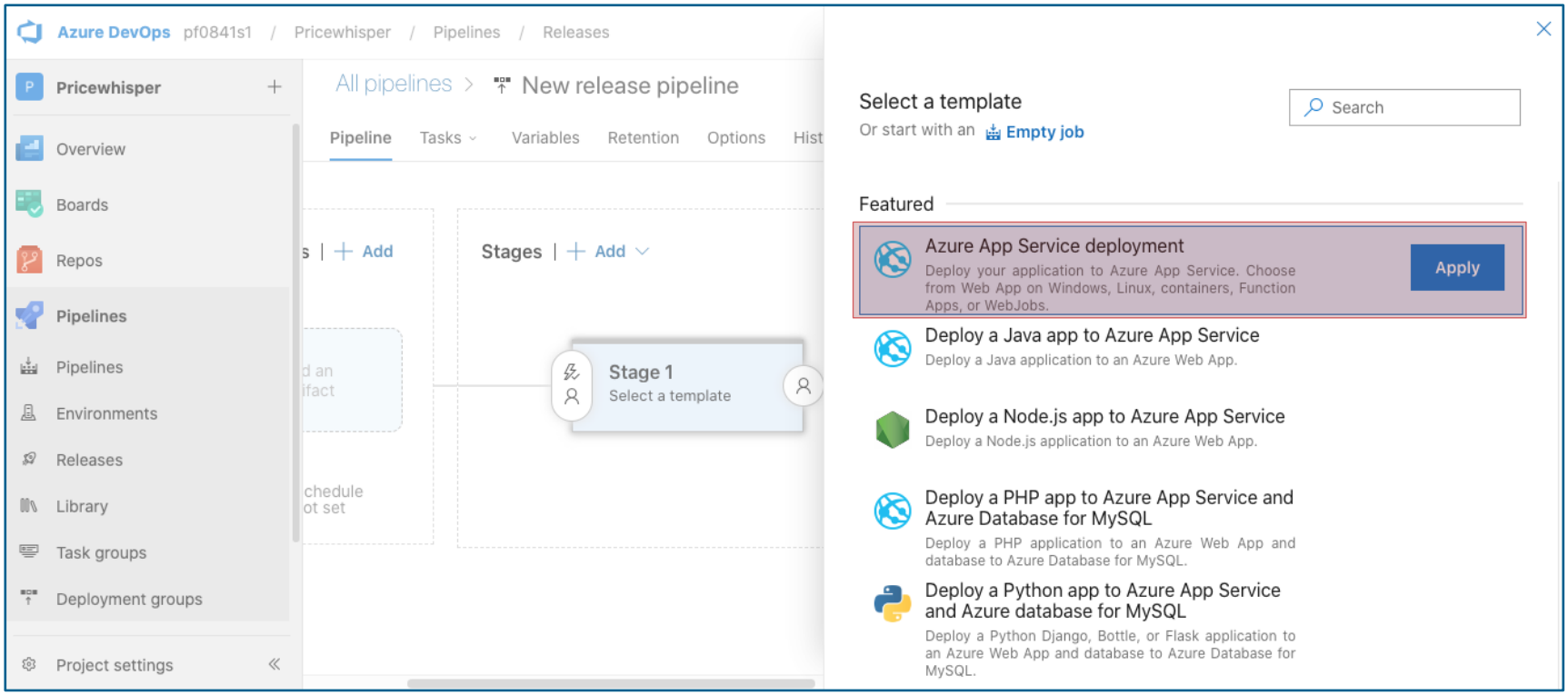
Azure Pipelines API .NET com Oracle

Analise o resultado da Pipeline de Build (CI)



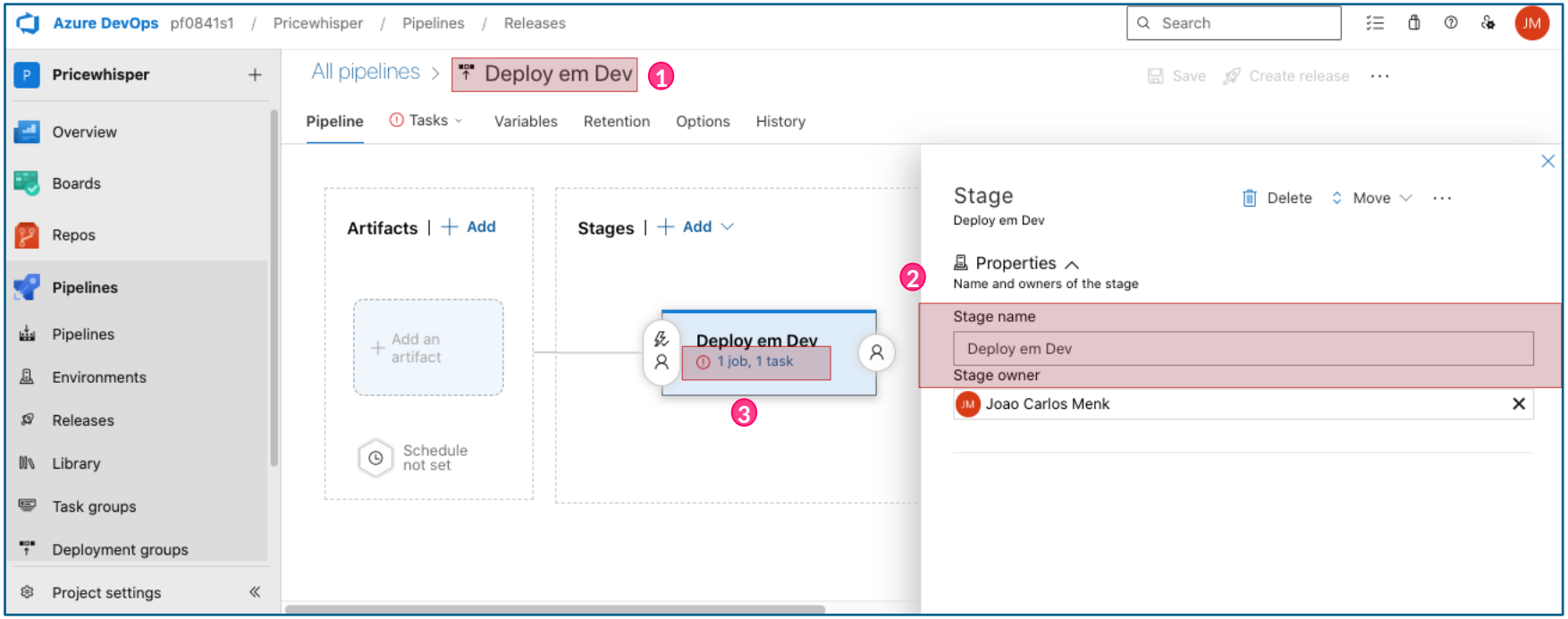
Azure Pipelines API .NET com Oracle

Crie uma nova Pipeline de Release escolhendo o Template: **Azure App Service deployment**



Azure Pipelines API .NET com Oracle

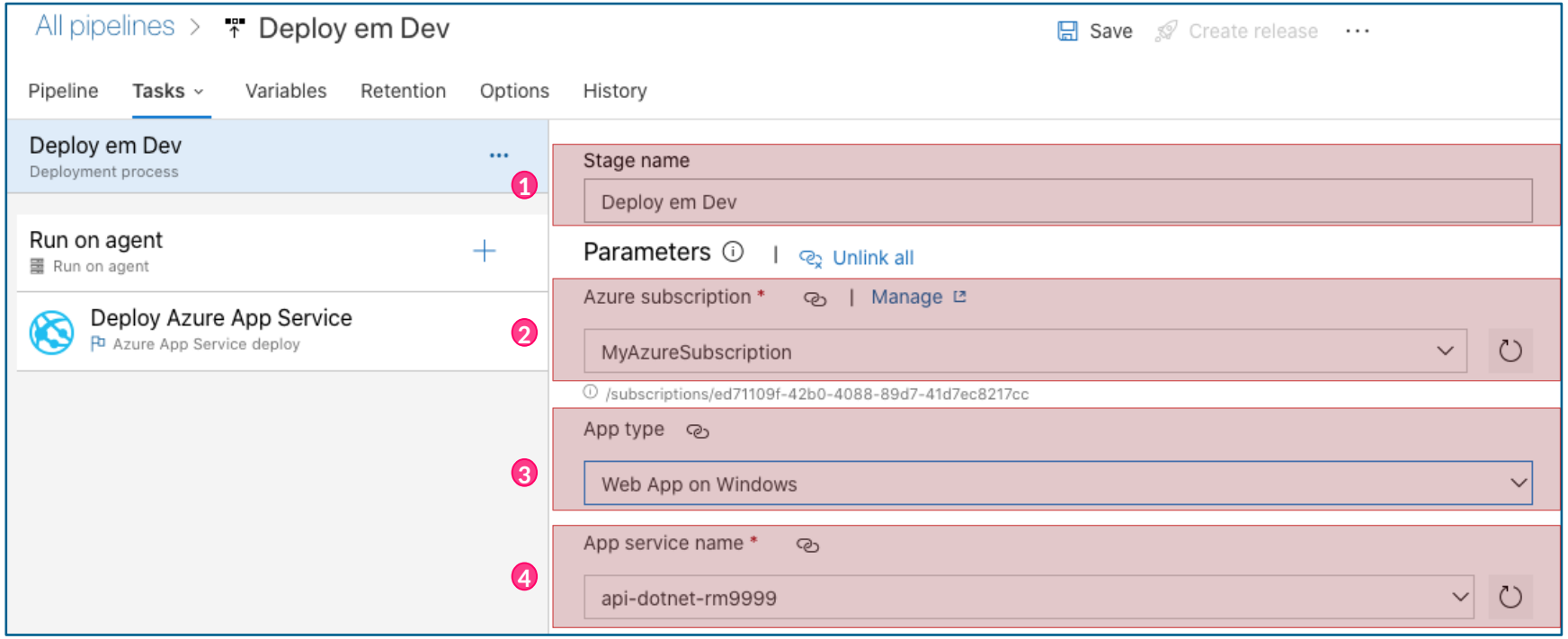
Vamos começar a definir as propriedades da Pipeline de Release começando pelo nome da Pipeline e depois alterando o nome do Stage



Após alterar os nomes, clique em: **1 job, 1 task**

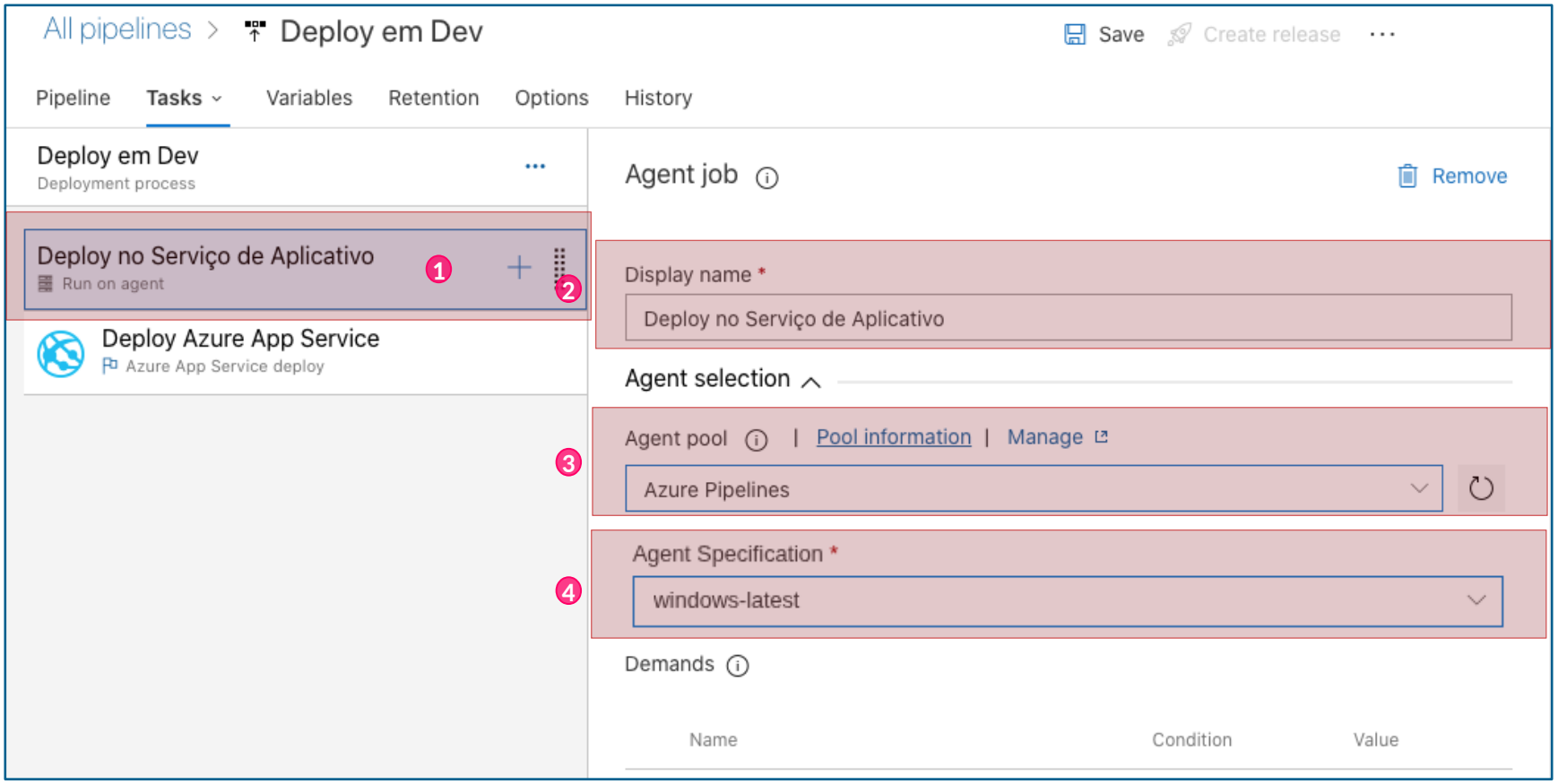
Azure Pipelines API .NET com Oracle

Deixe as propriedades conforme abaixo em nossa **Pipeline**



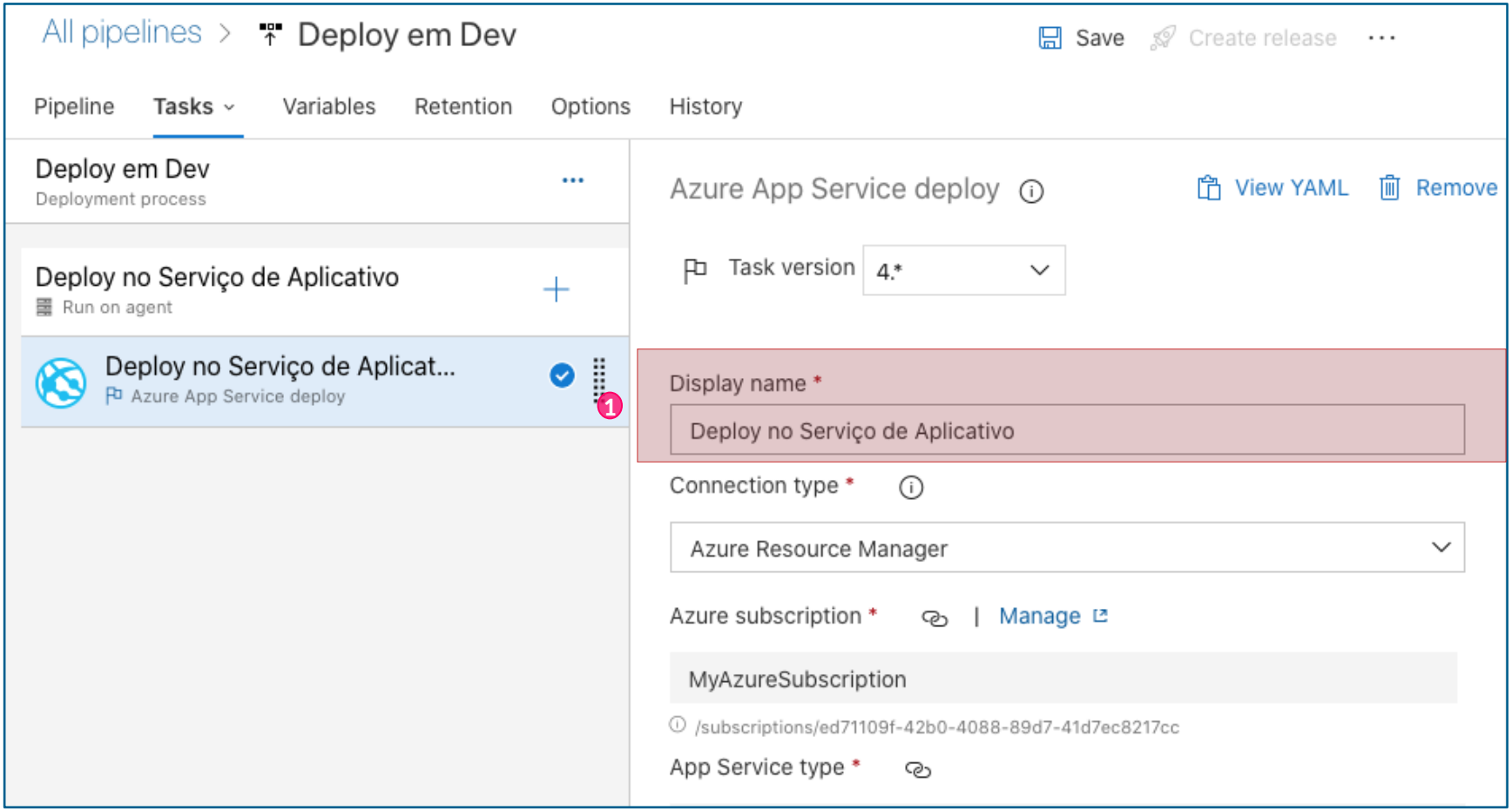
Azure Pipelines API .NET com Oracle

Deixe as propriedades conforme abaixo no **Agente**



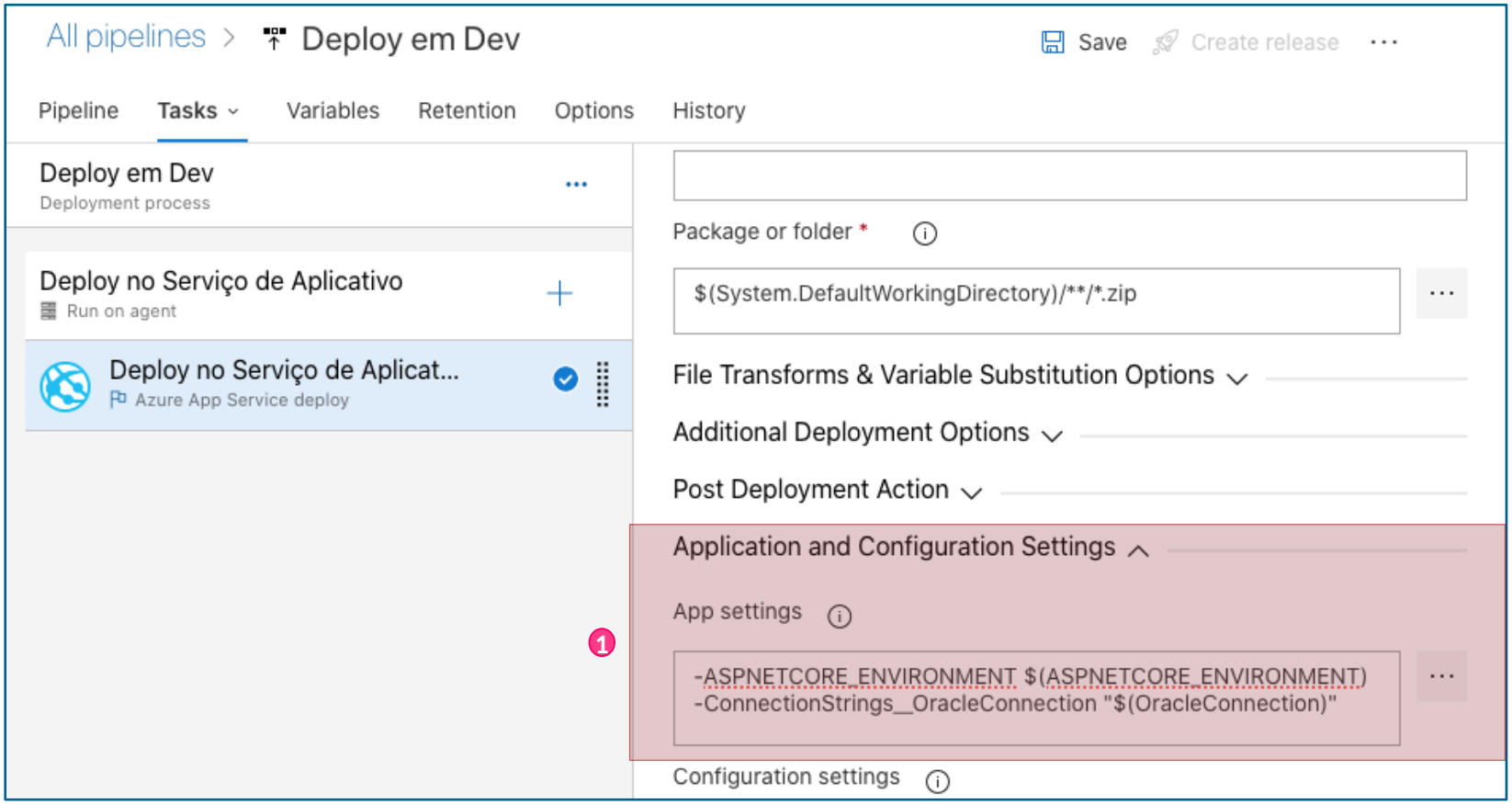
Azure Pipelines API .NET com Oracle

Deixe as propriedades conforme abaixo na Task de **Deploy**



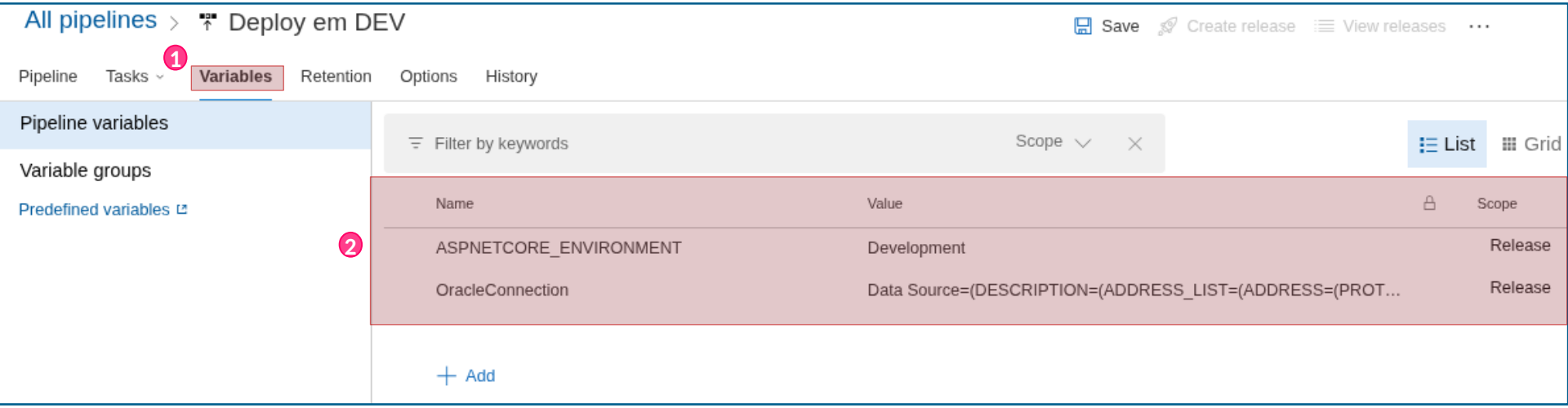
Azure Pipelines API .NET com Oracle

Deixe as propriedades conforme abaixo na Task de **Deploy**



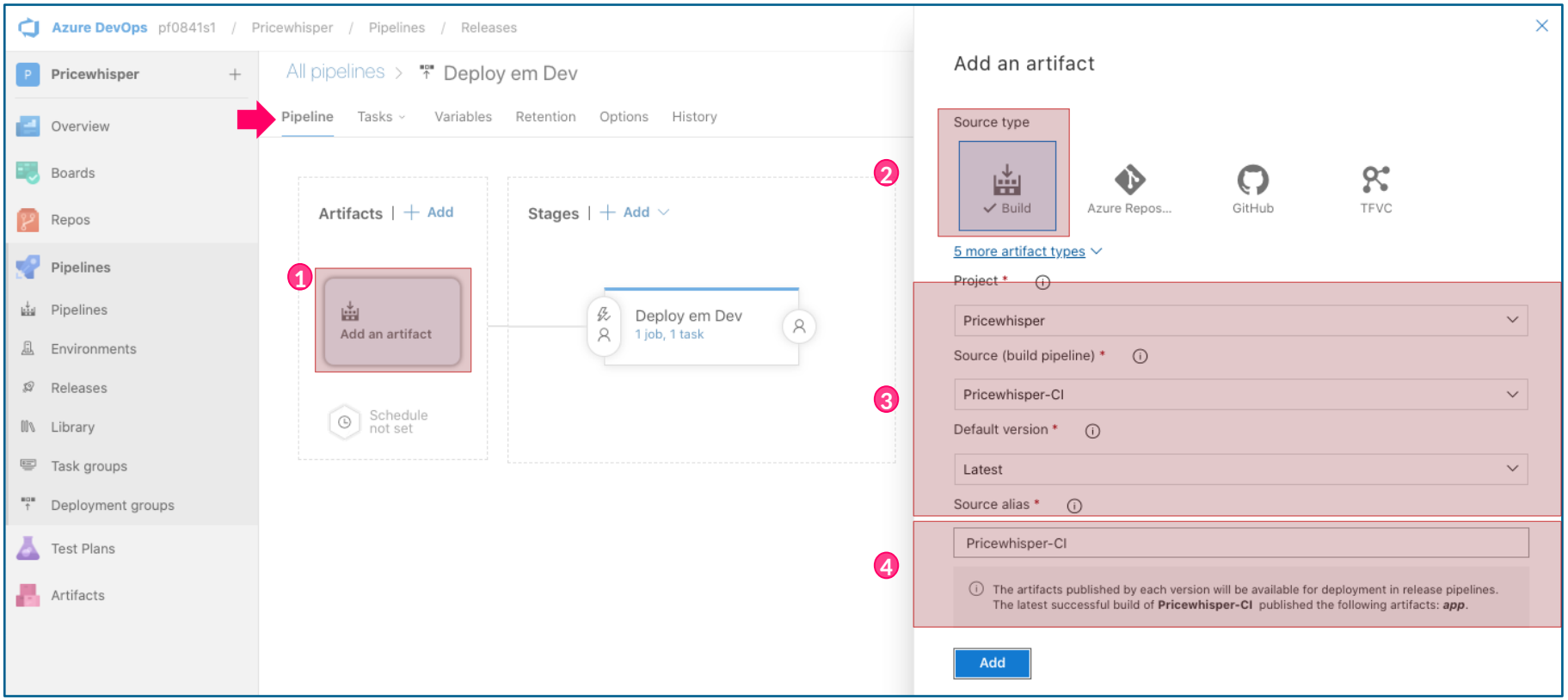
Azure Pipelines API .NET com Oracle

Agora, na parte de **Variáveis**, vamos incluir as utilizadas nessa Pipeline



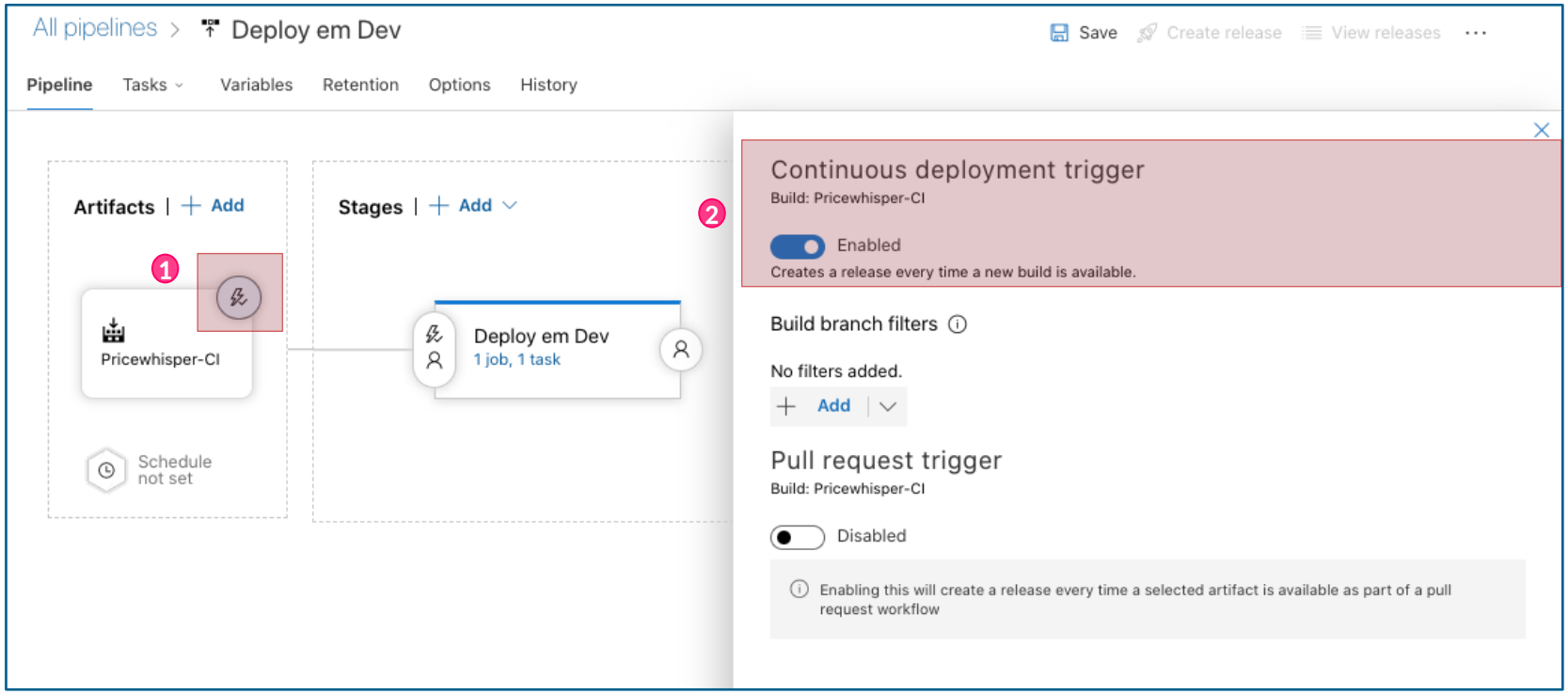
Azure Pipelines API .NET com Oracle

Com o Stage pronto clique em **Add an artifact** e escolha o Repositório do nosso Projeto



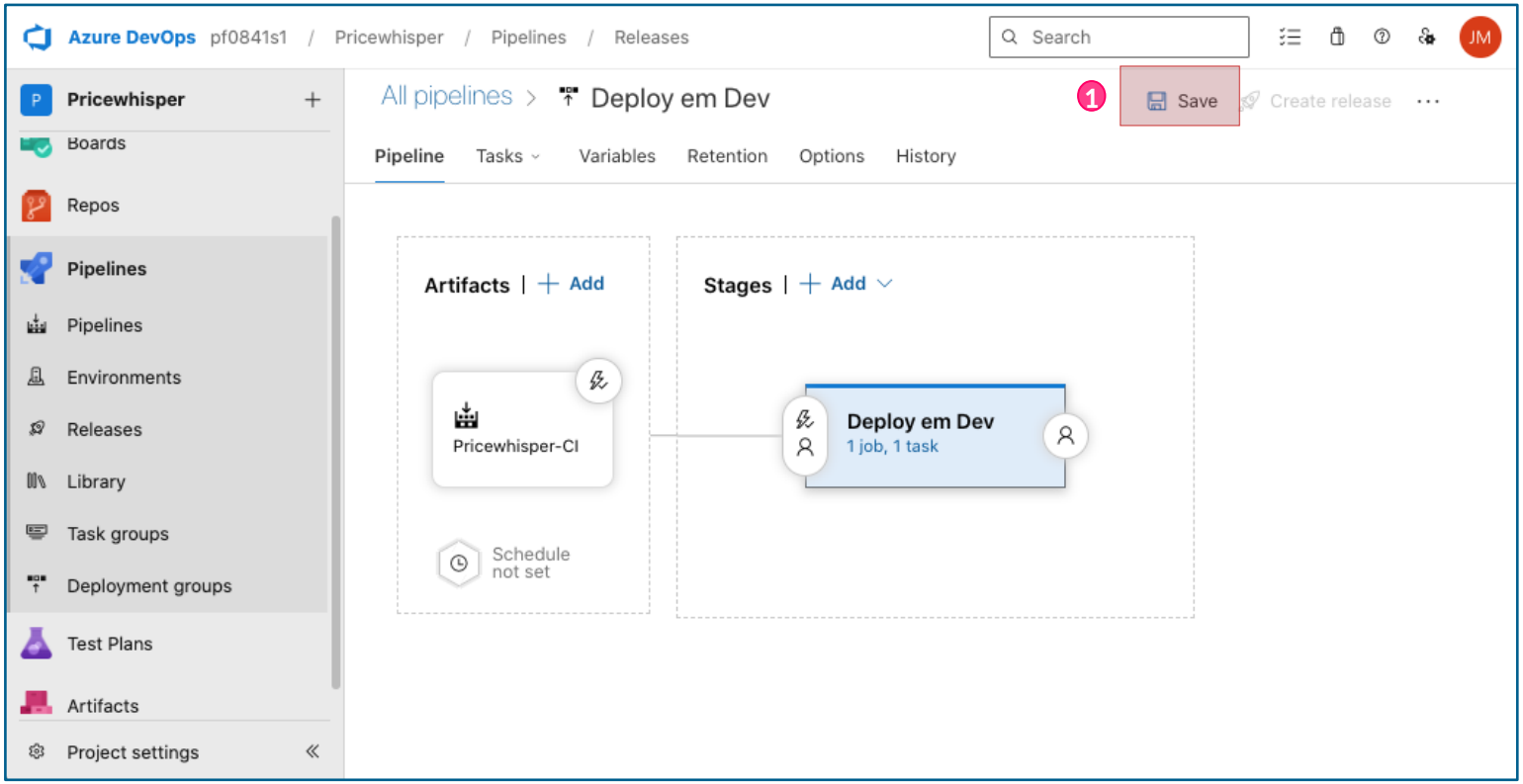
Azure Pipelines API .NET com Oracle

Habilite o **Continuous deployment**



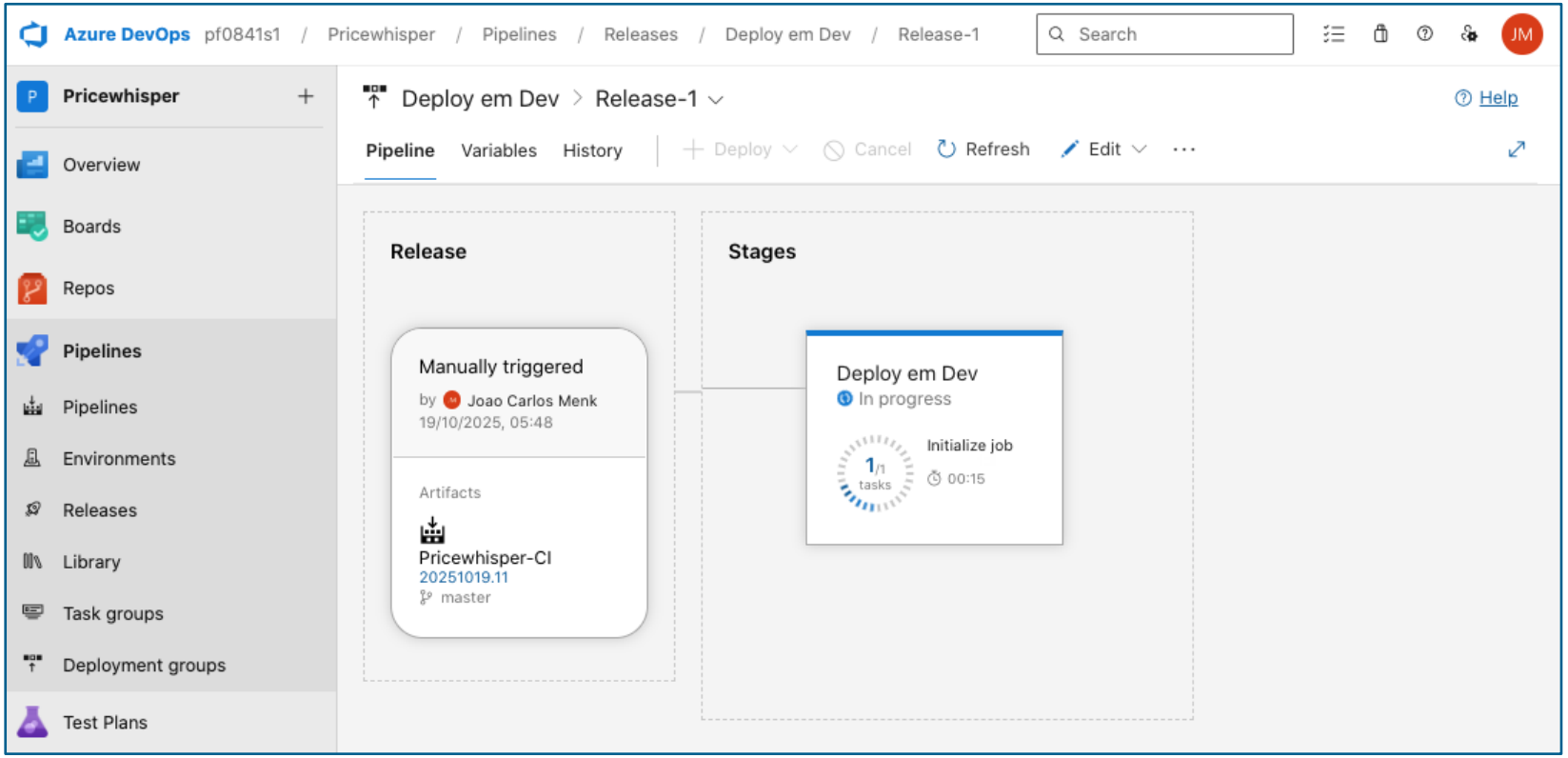
Azure Pipelines API .NET com Oracle

Salve a Pipeline de Release e depois **Crie um Release**



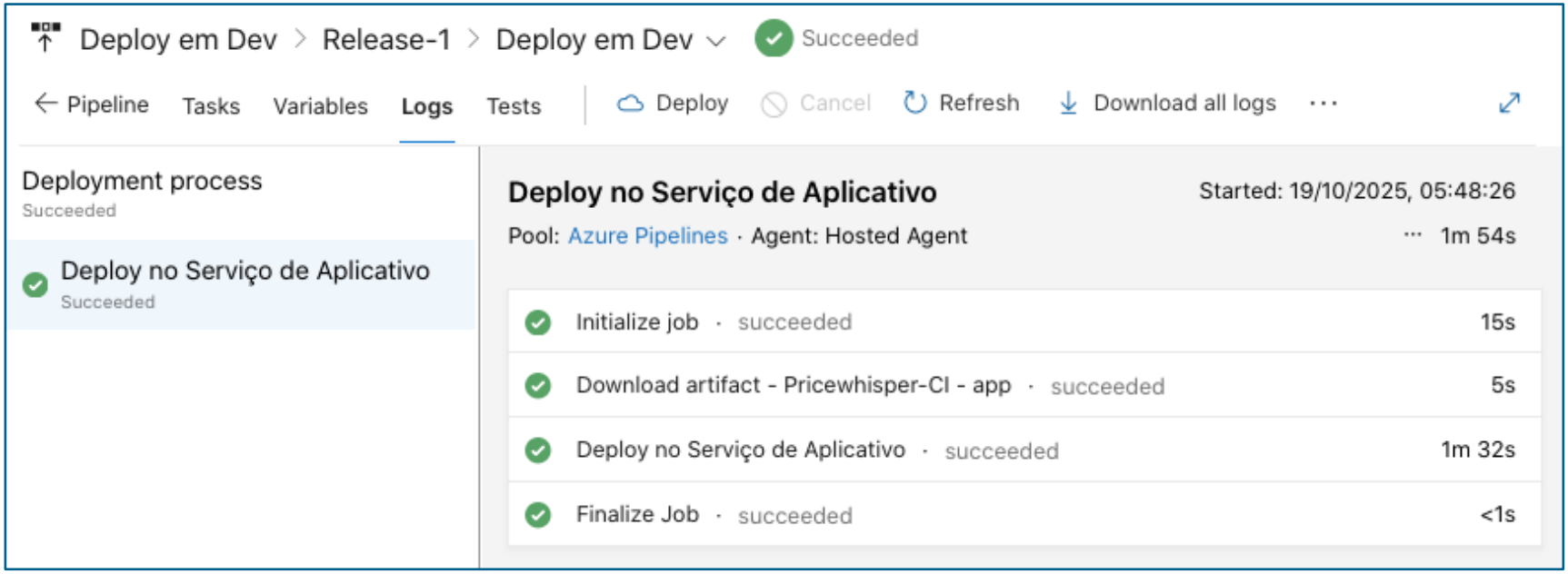
Azure Pipelines API .NET com Oracle

Aguarde a execução da Pipeline de Release



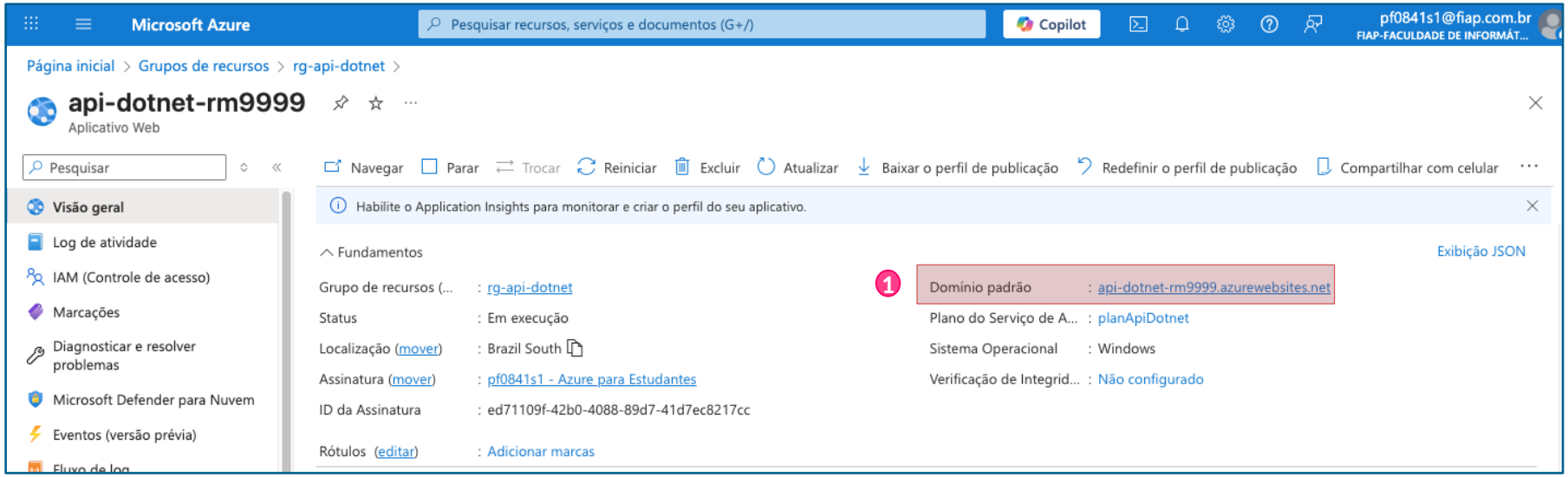
Azure Pipelines API .NET com Oracle

Analise os resultados do Logs



Azure Pipelines API .NET com Oracle

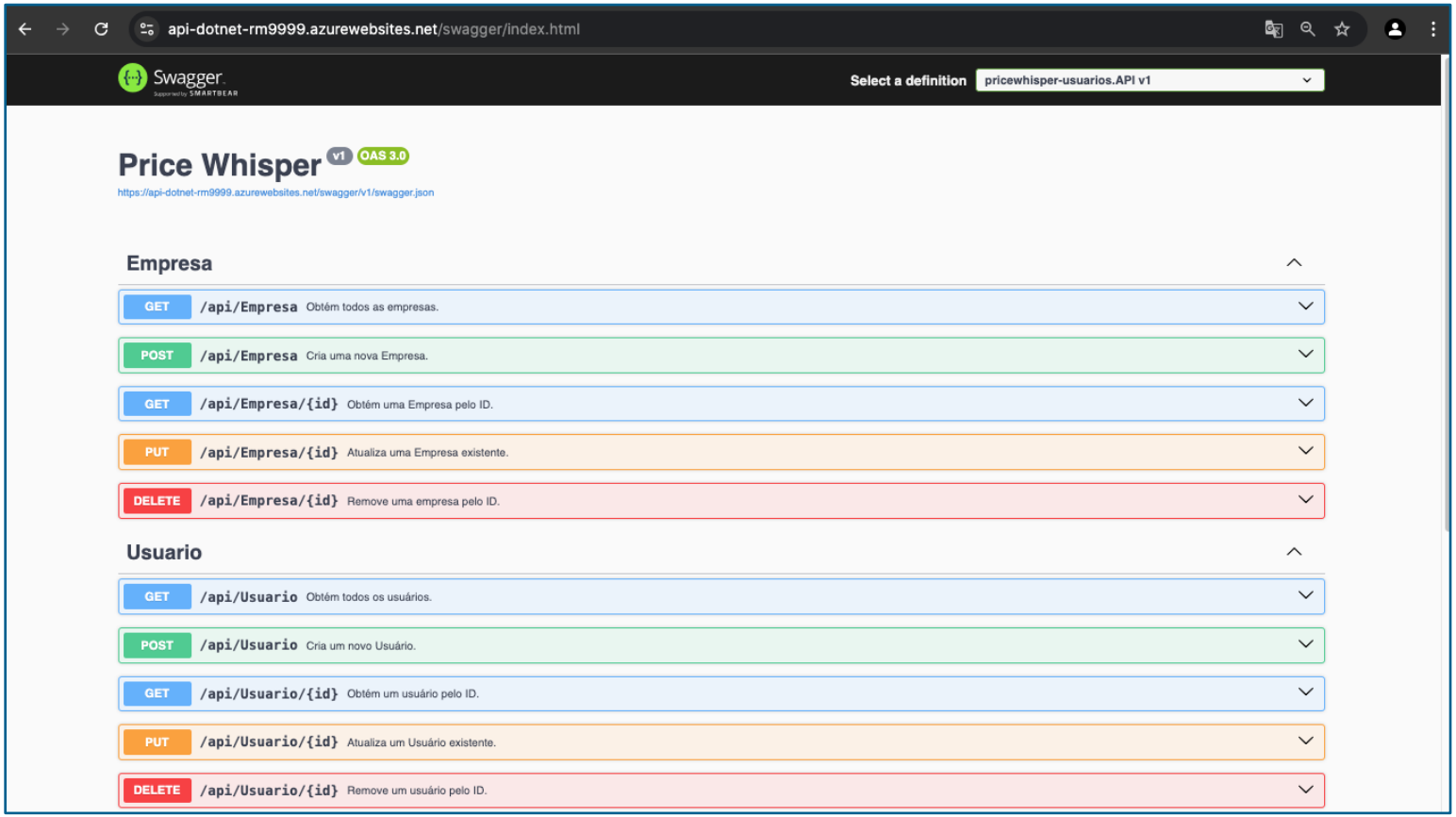
Execute a API pelo Link do Serviço de Aplicativo



Azure Pipelines API .NET com Oracle

Inclua a Rota para o Swagger e realize os testes

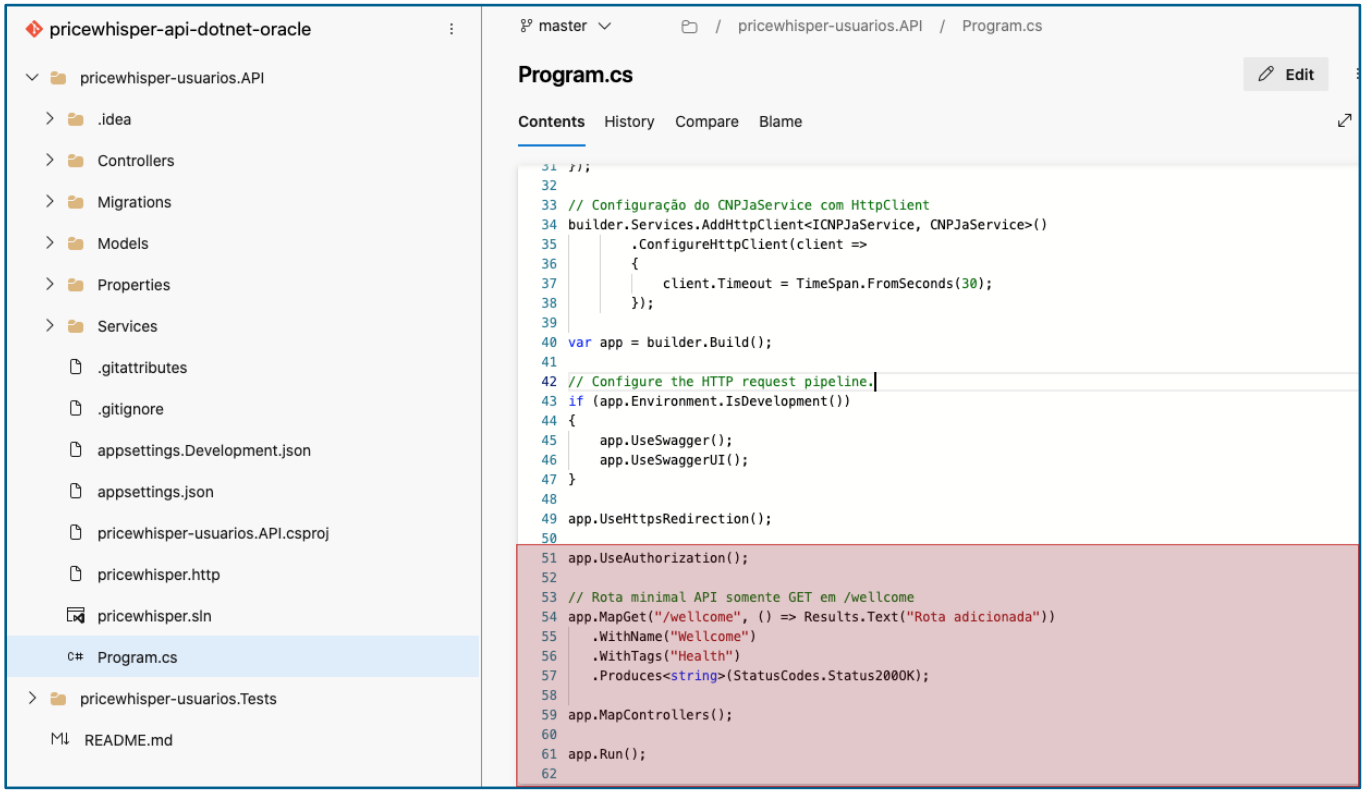
https://api-dotnet-rm9999.azurewebsites.net/swagger/index.html



Azure Pipelines API .NET com Oracle

Para testar o CI/CD completo vamos incluir uma nova rota em **Program.cs**

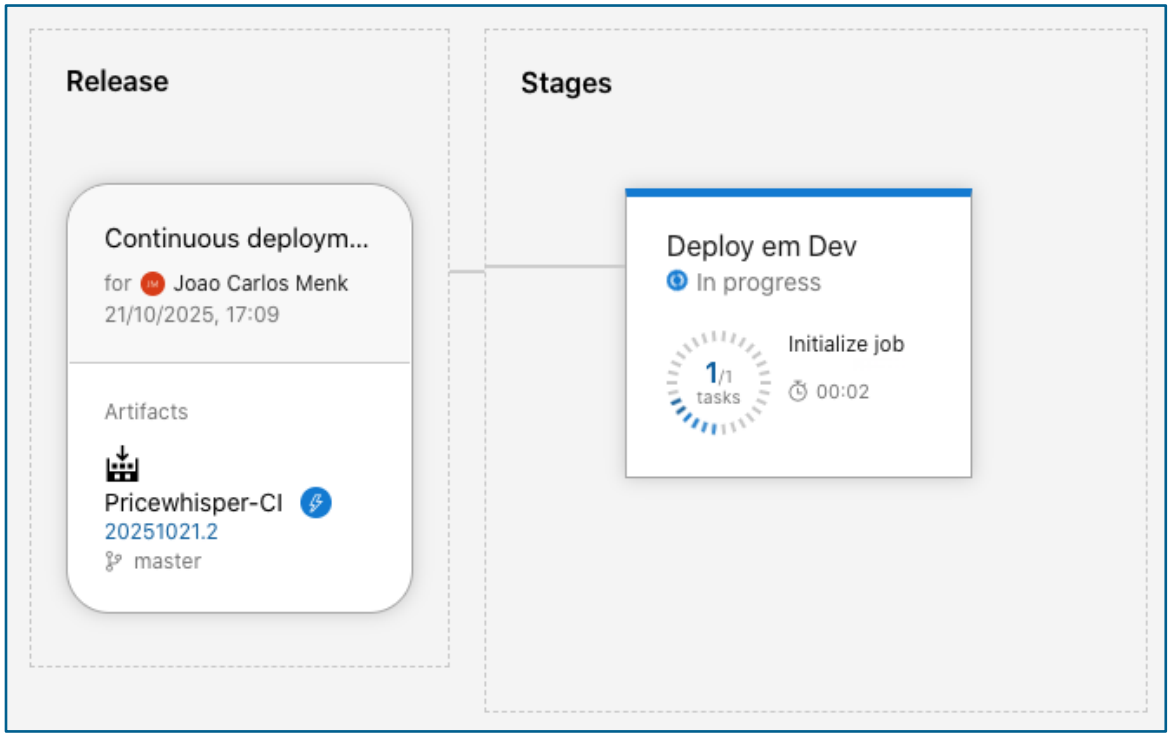
```
// Rota incluída - API somente GET em /wellcome
app.MapGet("/wellcome", () => Results.Text("Rota adicionada"))
   .WithName("Wellcome")
   .WithTags("Health")
   .Produces<string>(StatusCodes.Status200OK);
```



Azure Pipelines API .NET com Oracle

Após o commit acompanhe a Pipeline de CI e a Pipeline de CD

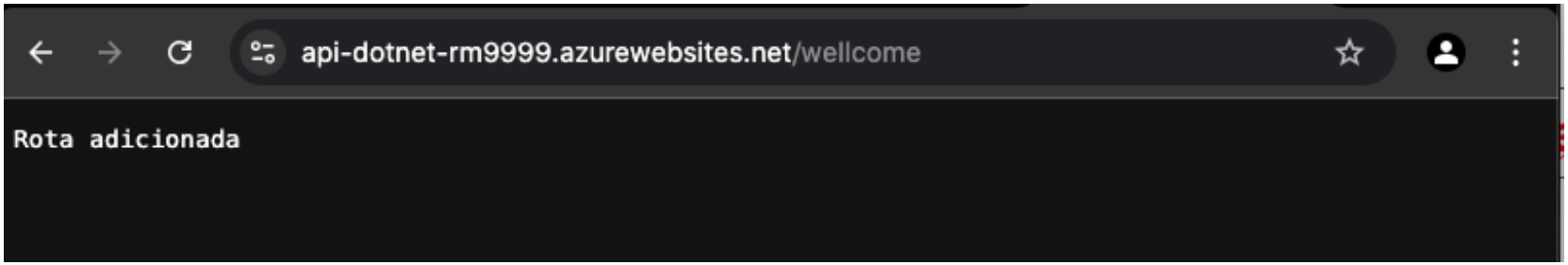




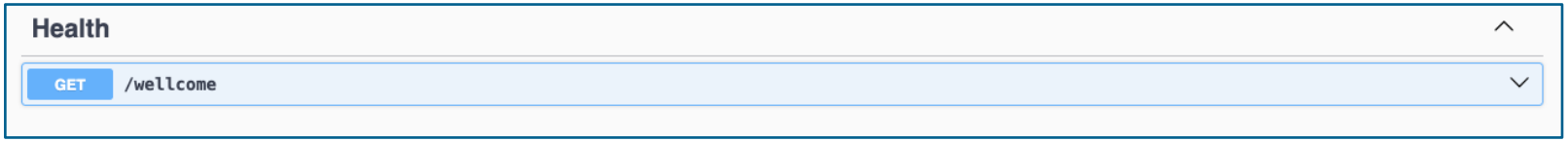
Azure Pipelines API .NET com Oracle

Inclua o Sufixo **/wellcome** na URL do seu App para testar

https://api-dotnet-rm9999.azurewebsites.net/wellcome

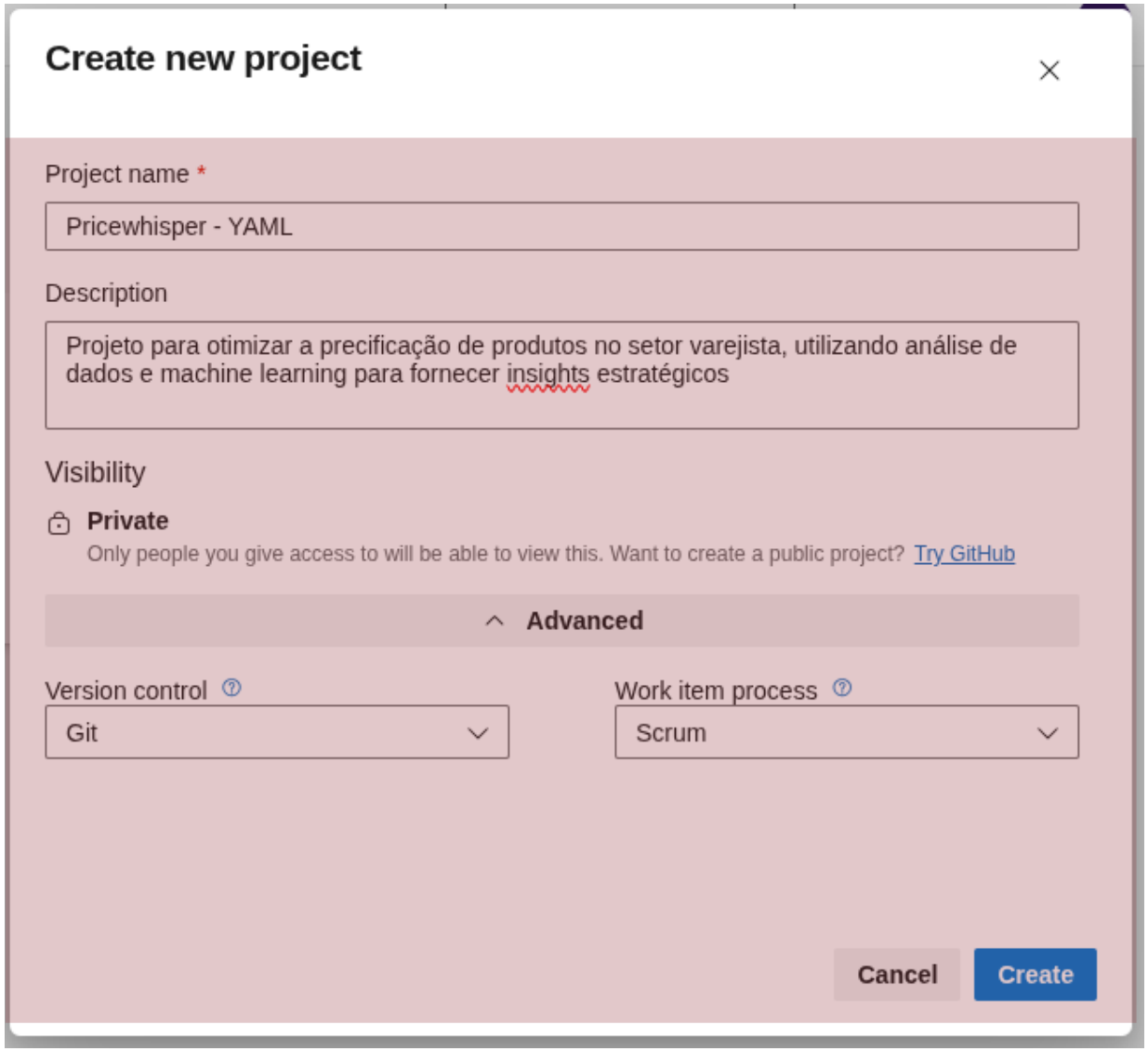


Ou utilize o Link do Swagger



Azure Pipelines API .NET com Oracle

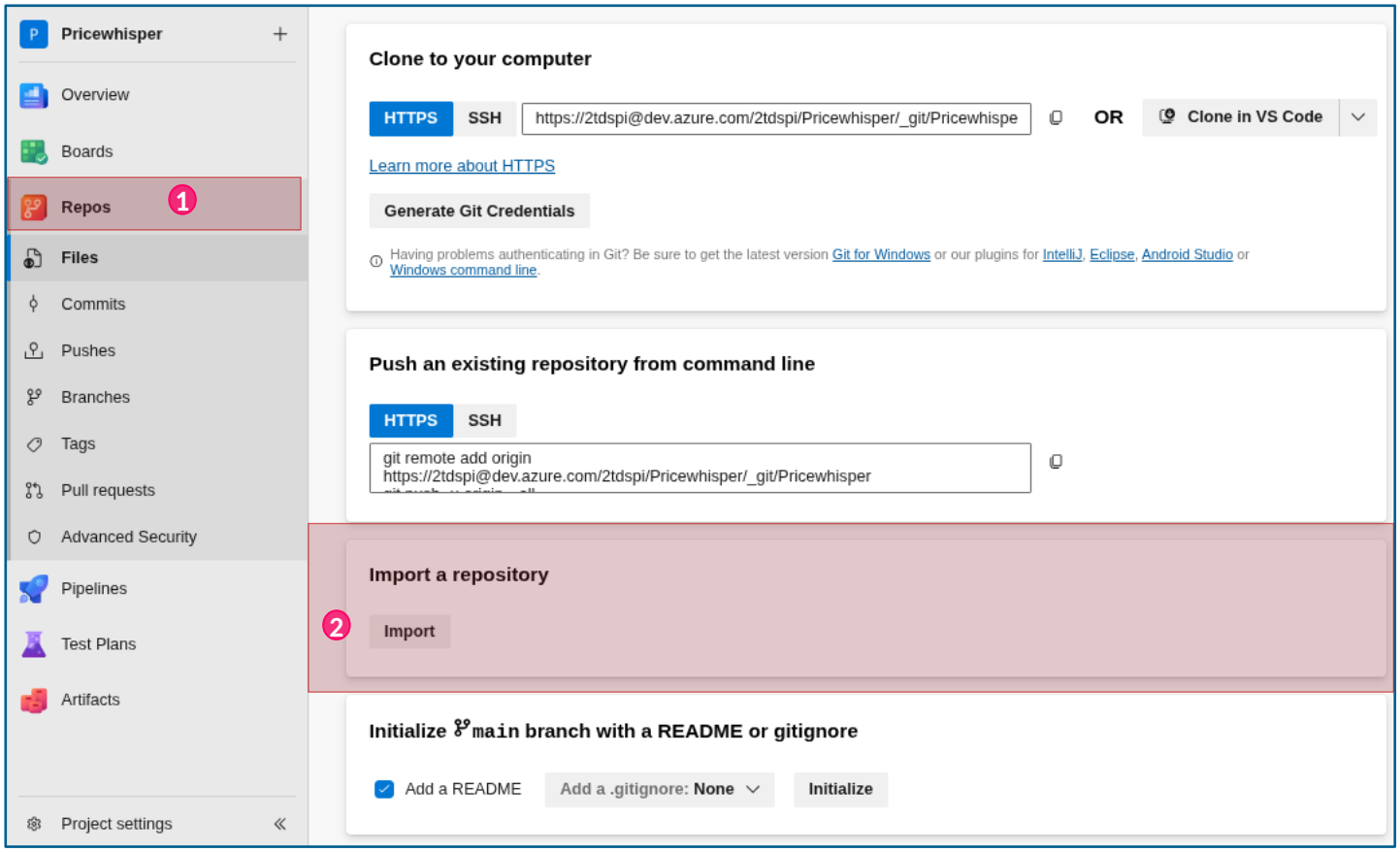
Vamos criar outro Projeto para criar uma **Pipeline CI/CD com YAML**



Azure Pipelines API .NET com Oracle

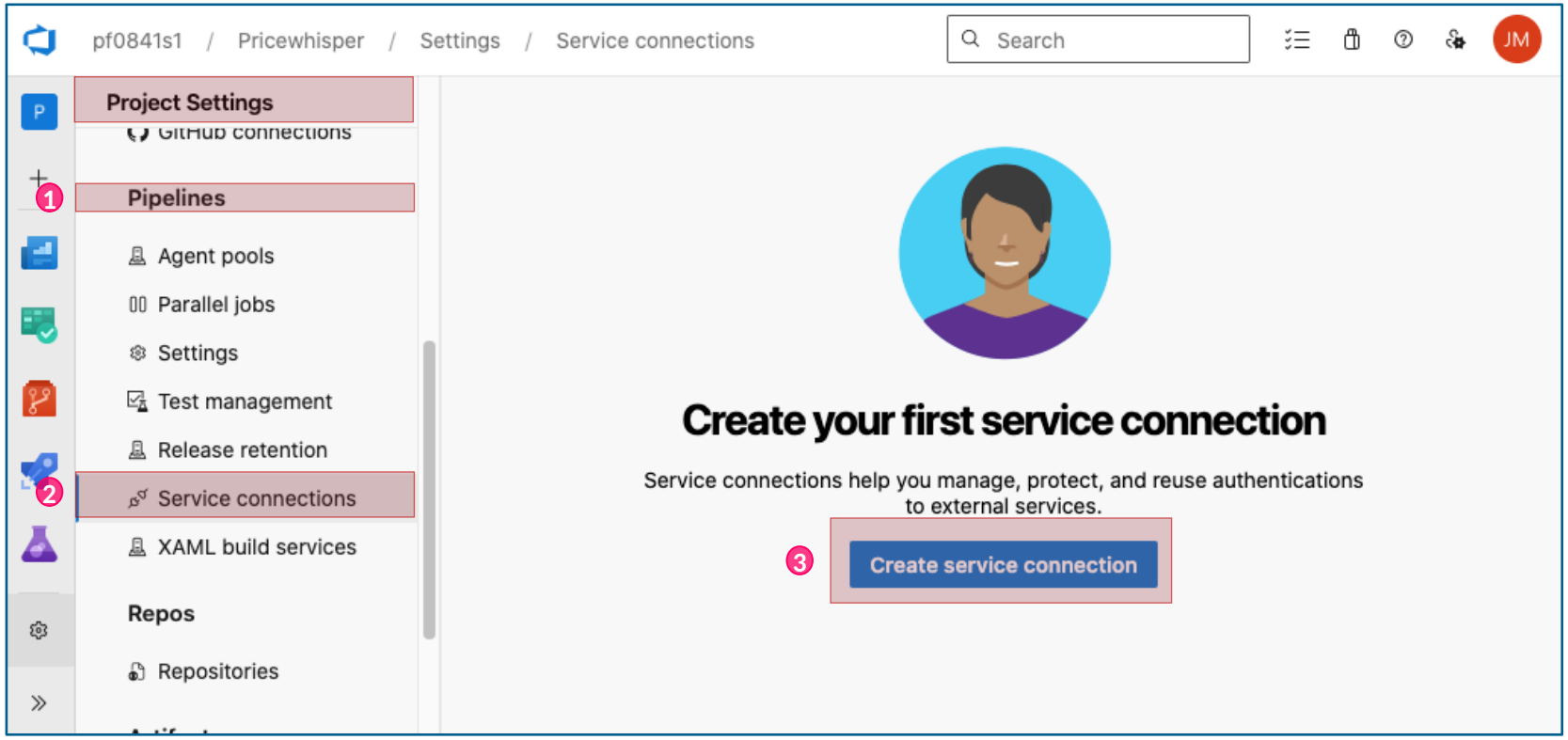
Agora vamos importar o projeto que está no GitHub para o Azure Repos

https://github.com/profjoaomenk/pricewhisper-api-dotnet-oracle.git



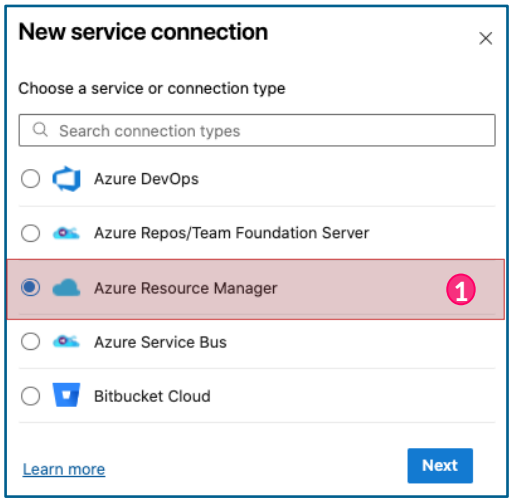
Azure Pipelines API .NET com Oracle

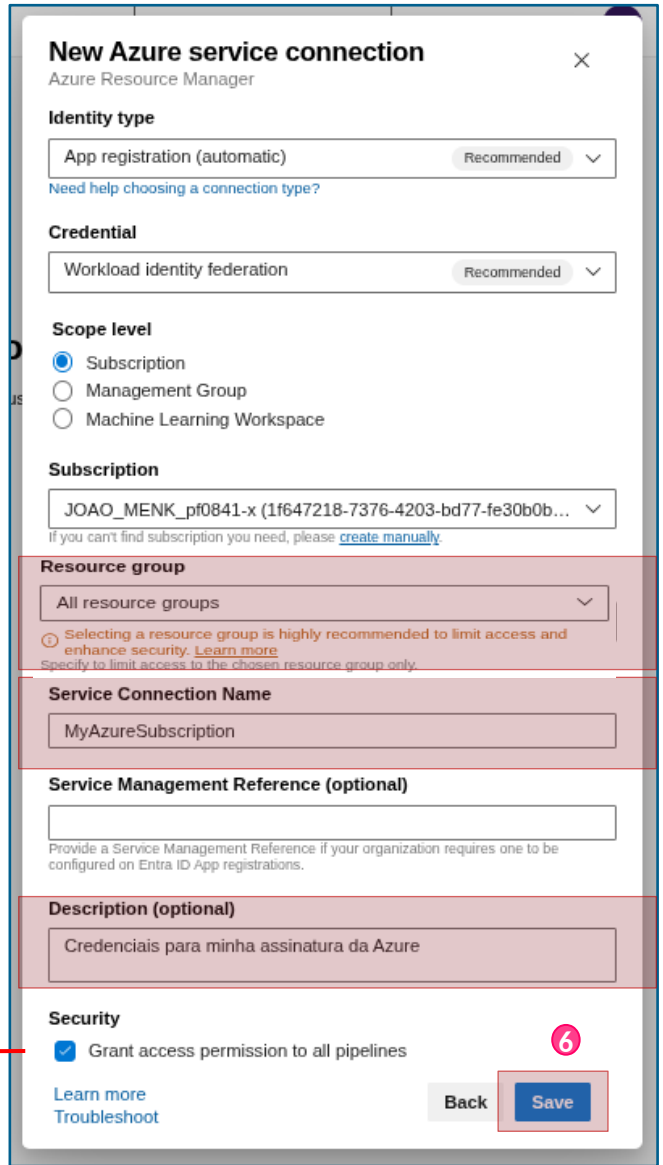
Para utilizar um aquivo YAML precisamos criar uma **Conexão de Serviço (Service Connection)** com a nossa assinatura do Azure



Azure Pipelines API .NET com Oracle

Escolha **Azure Resource Manager** como o Service connection e o configure para sua assinatura da Azure





→ **Deixe o padrão**

→ **Deixe o padrão**

→ **Nível do escopo: Subscription**

→ **Sua assinatura**

→ **Grupo de Recursos**

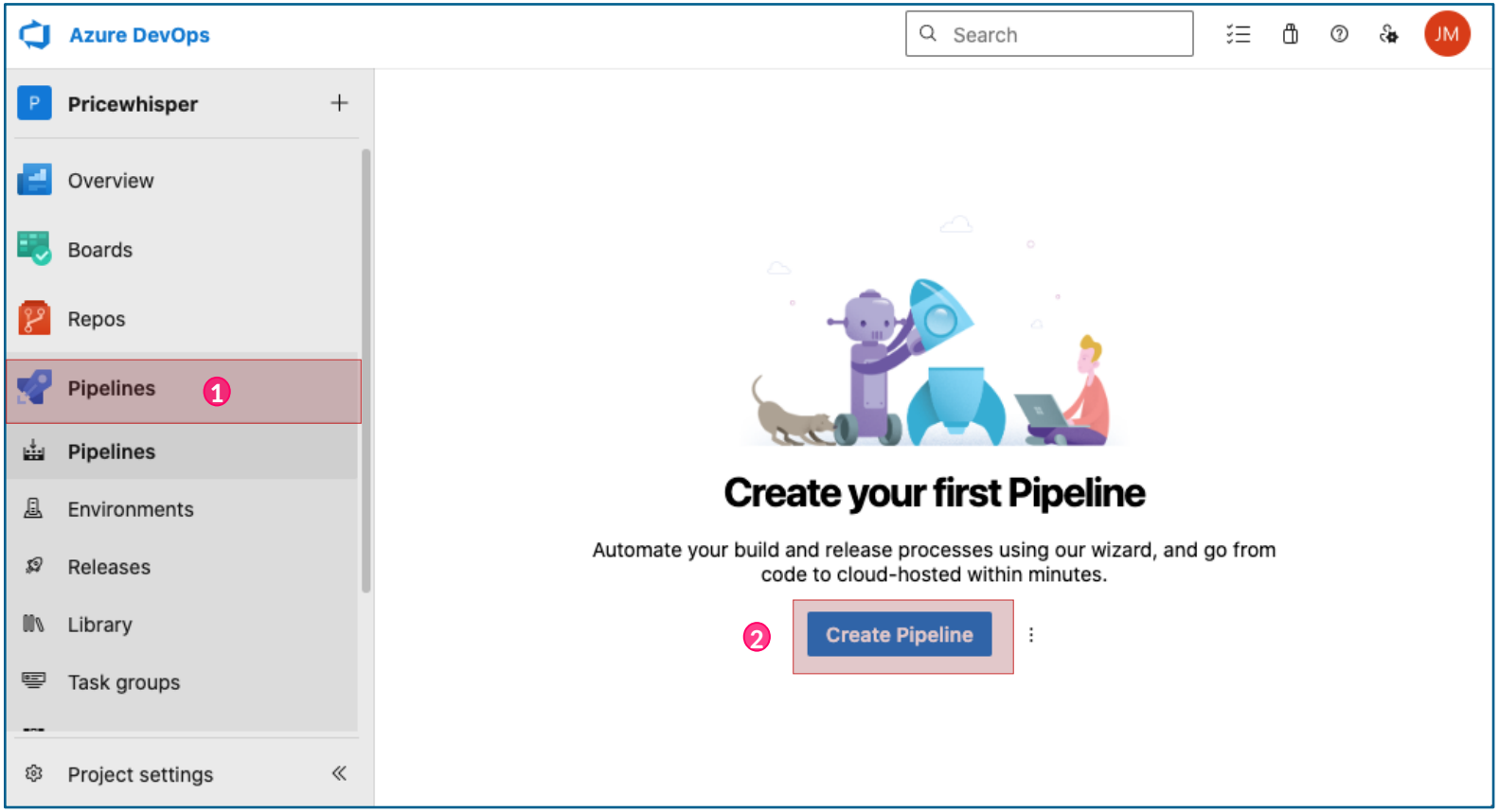
→ **Nome da Conexão de Serviço**

→ **Descrição**

Todas as Pipelines podem acessar esse Serviço ←

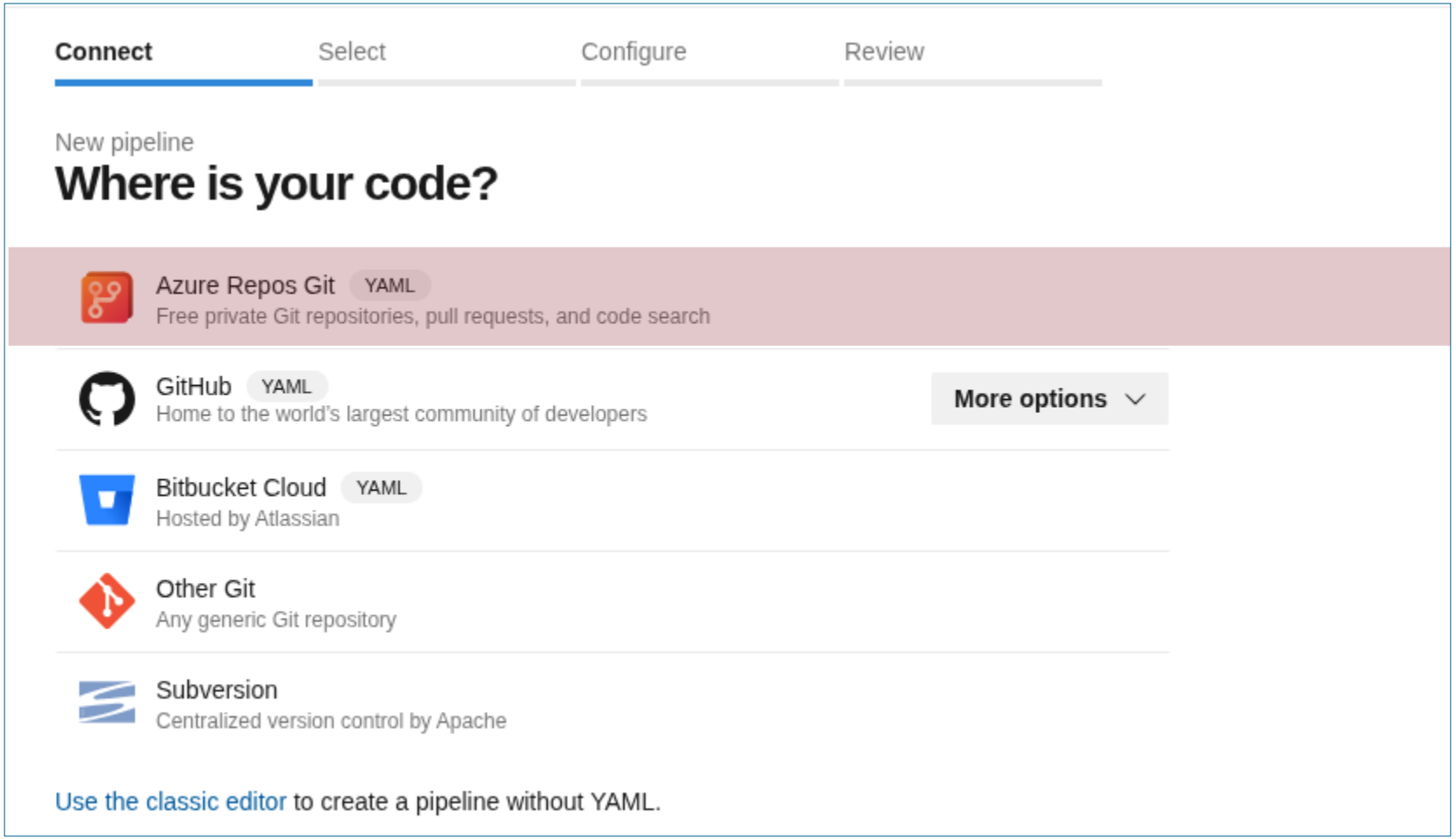
Azure Pipelines API .NET com Oracle

Crie uma nova Pipeline



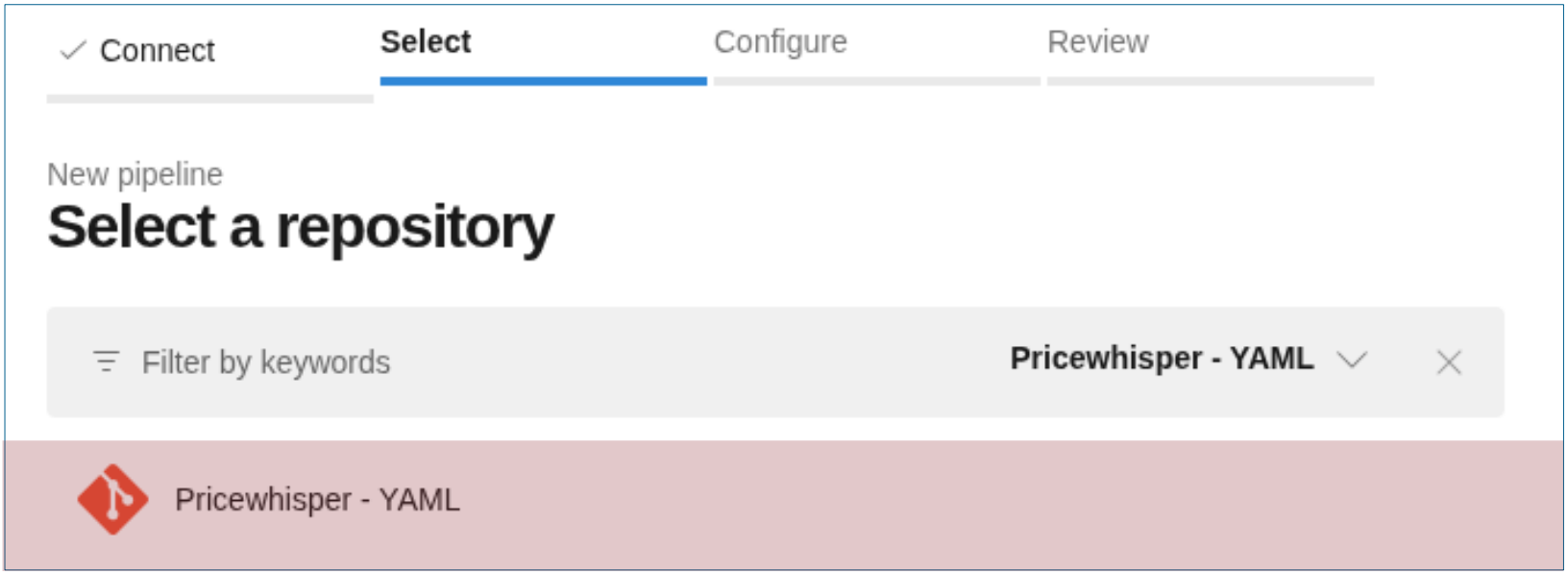
Azure Pipelines API .NET com Oracle

Escolha **Azure Repos Git YAML**



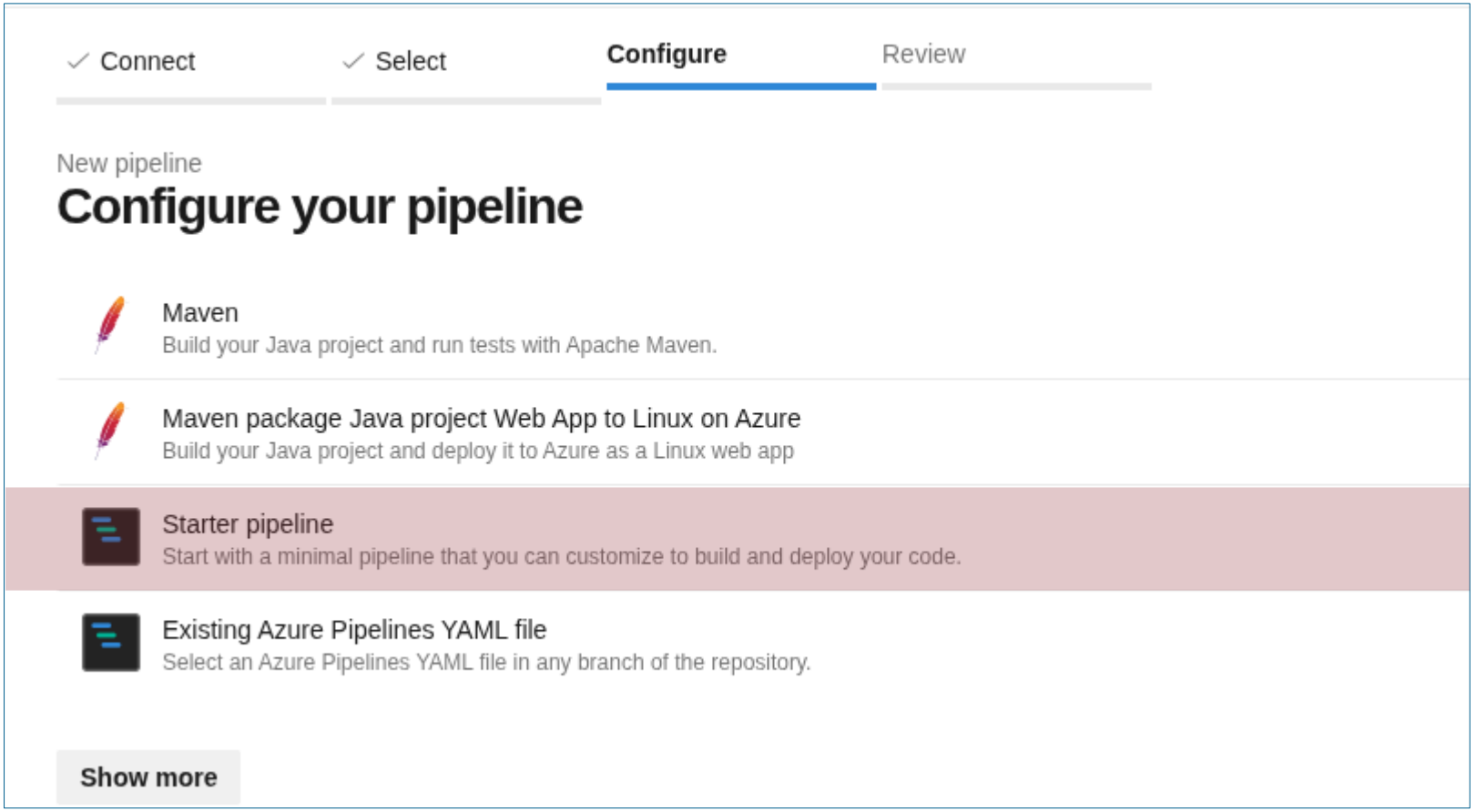
Azure Pipelines API .NET com Oracle

Selecione o Repositório



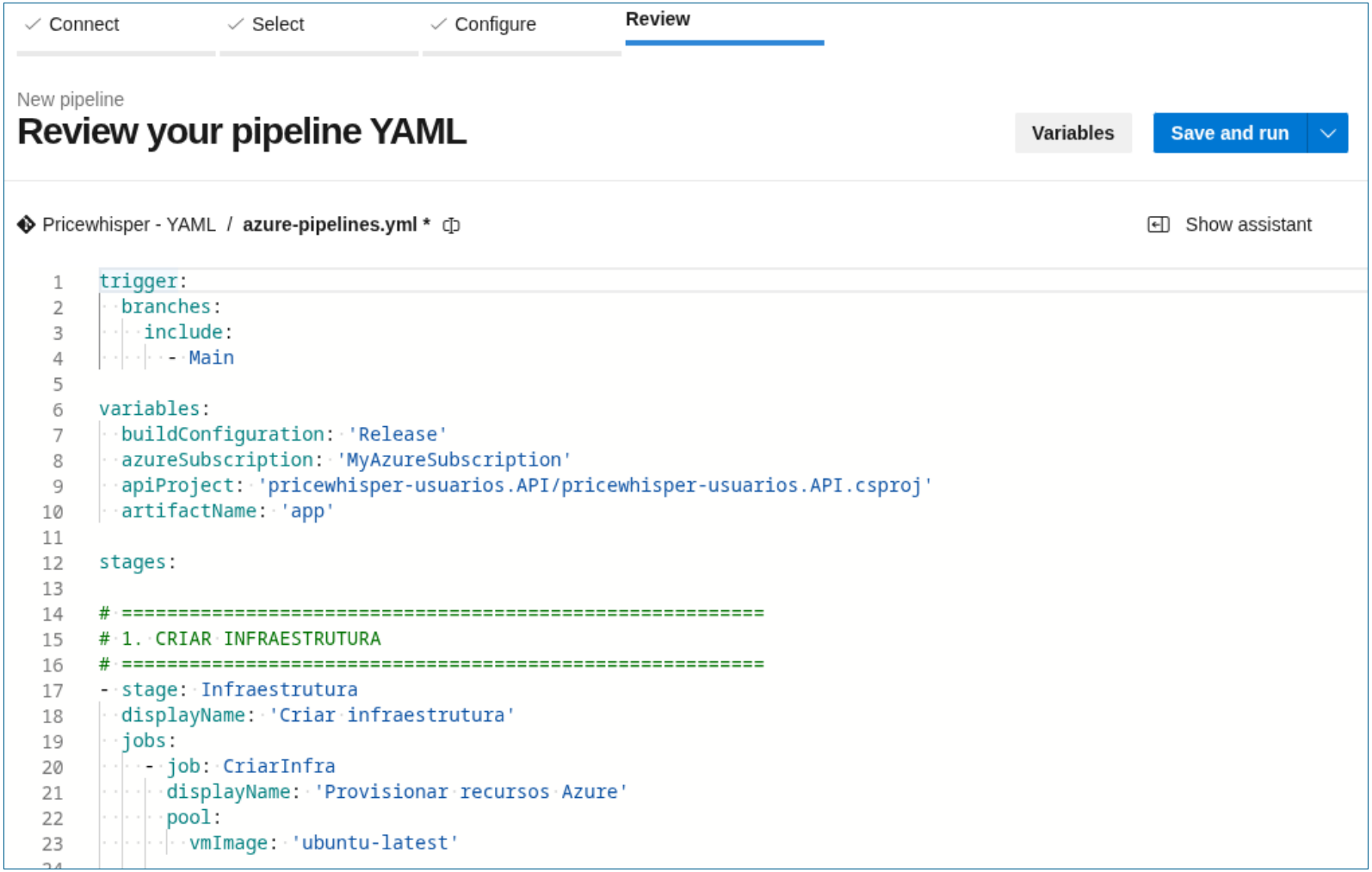
Azure Pipelines API .NET com Oracle

Escolha **Starter pipeline**



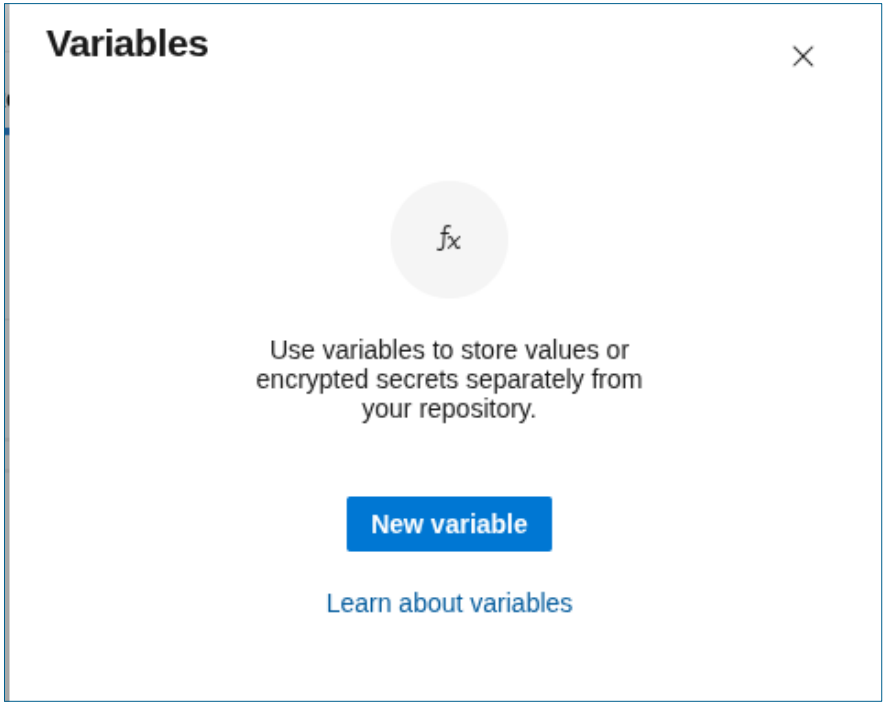
Azure Pipelines API .NET com Oracle

Copie e cola o conteúdo do arquivo **Semana_19_03_Pricewhisper-yaml.txt**



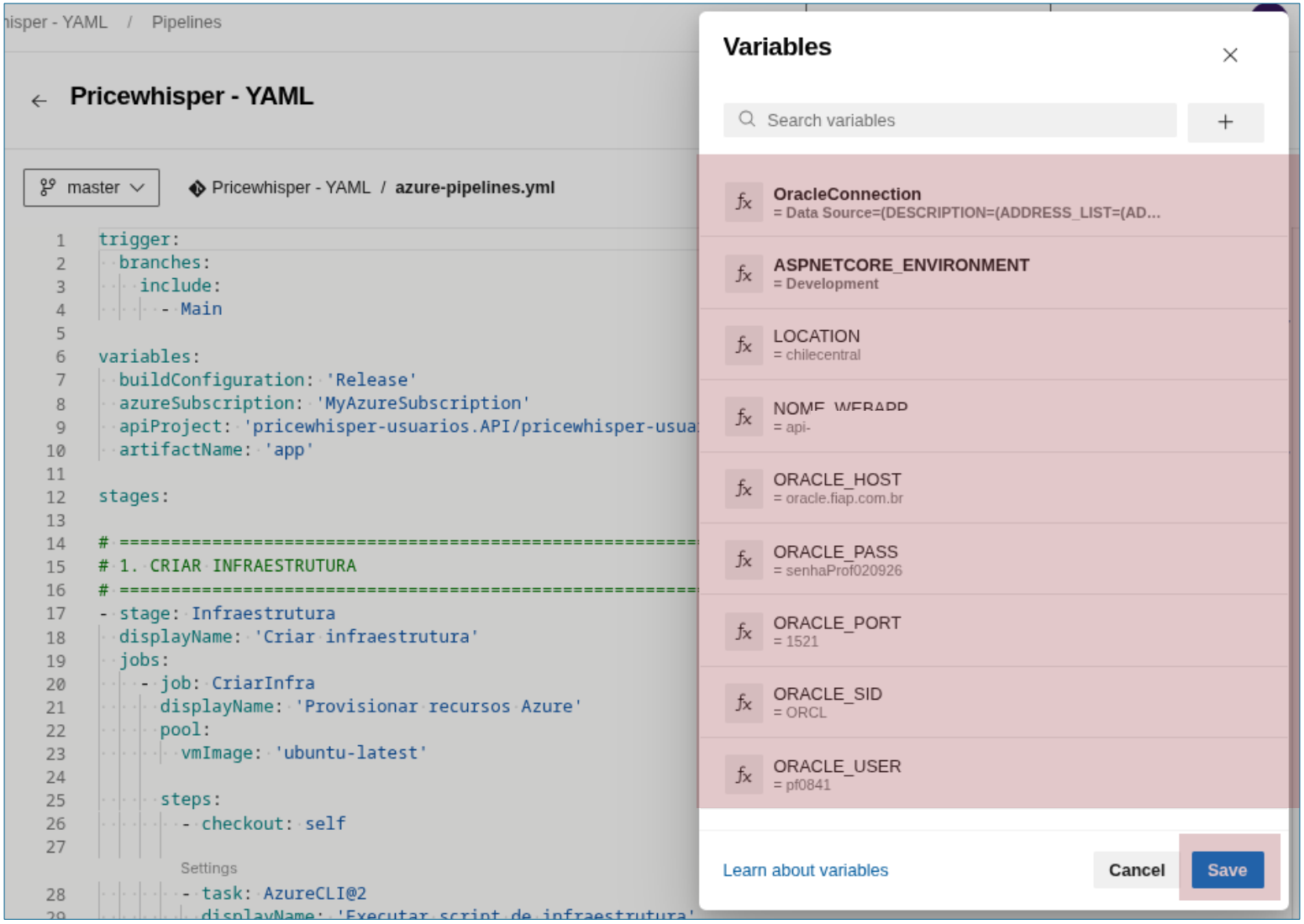
Azure Pipelines API .NET com Oracle

Clique em **Variables** e inclua as variáveis necessárias para o Projeto



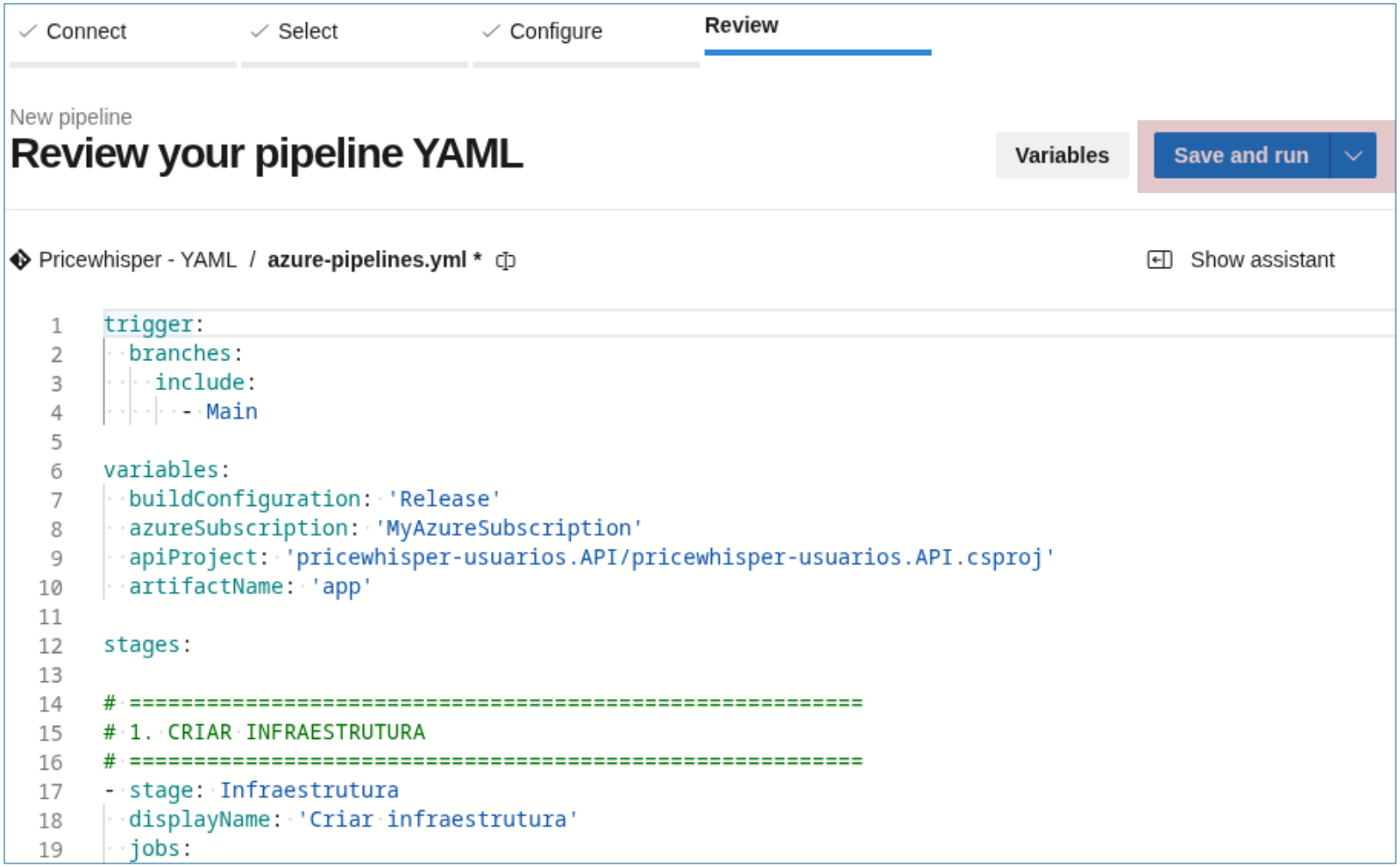
Azure Pipelines API .NET com Oracle

Variáveis incluídas, clique em **Save**



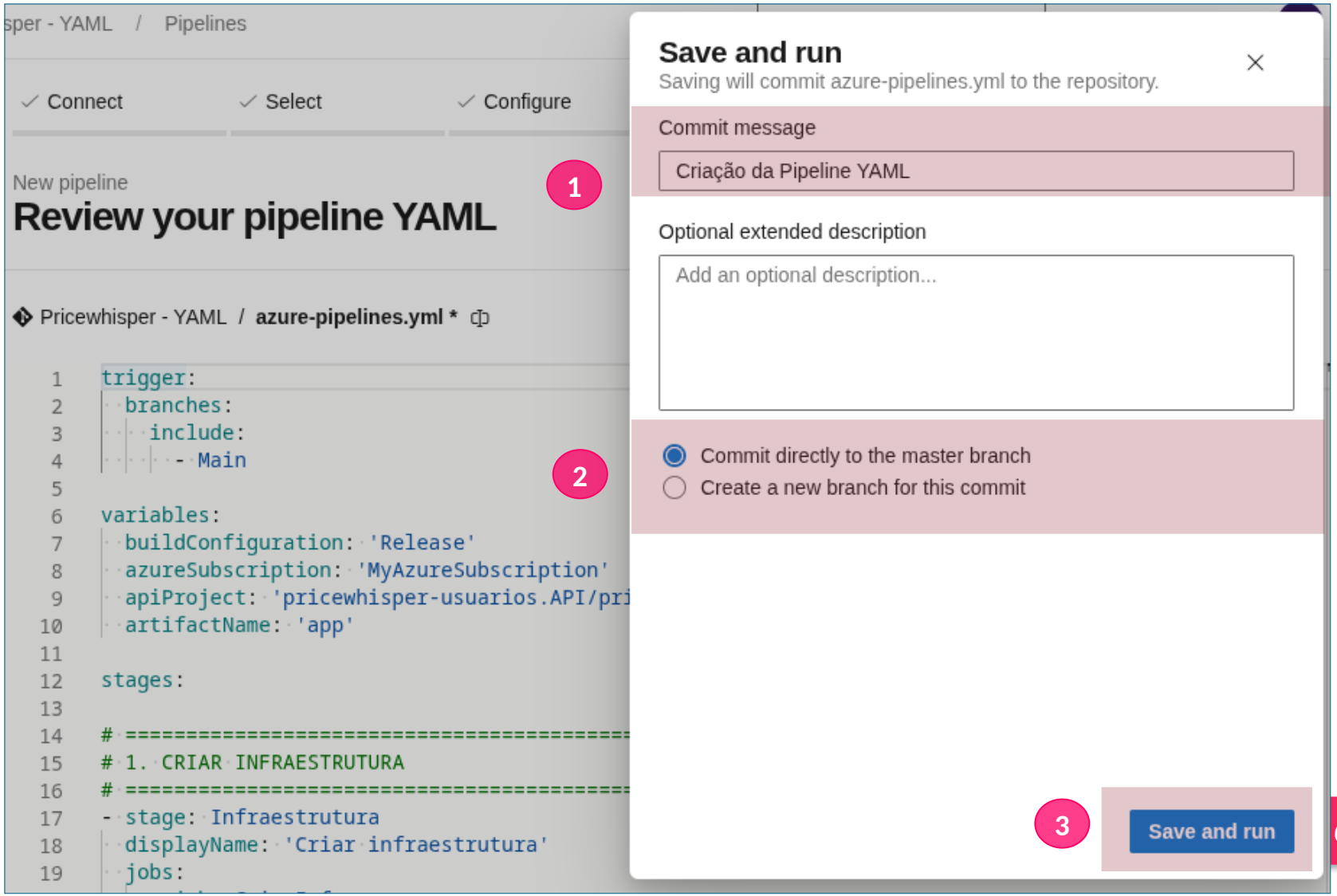
Azure Pipelines API .NET com Oracle

Agora clique em Save and run



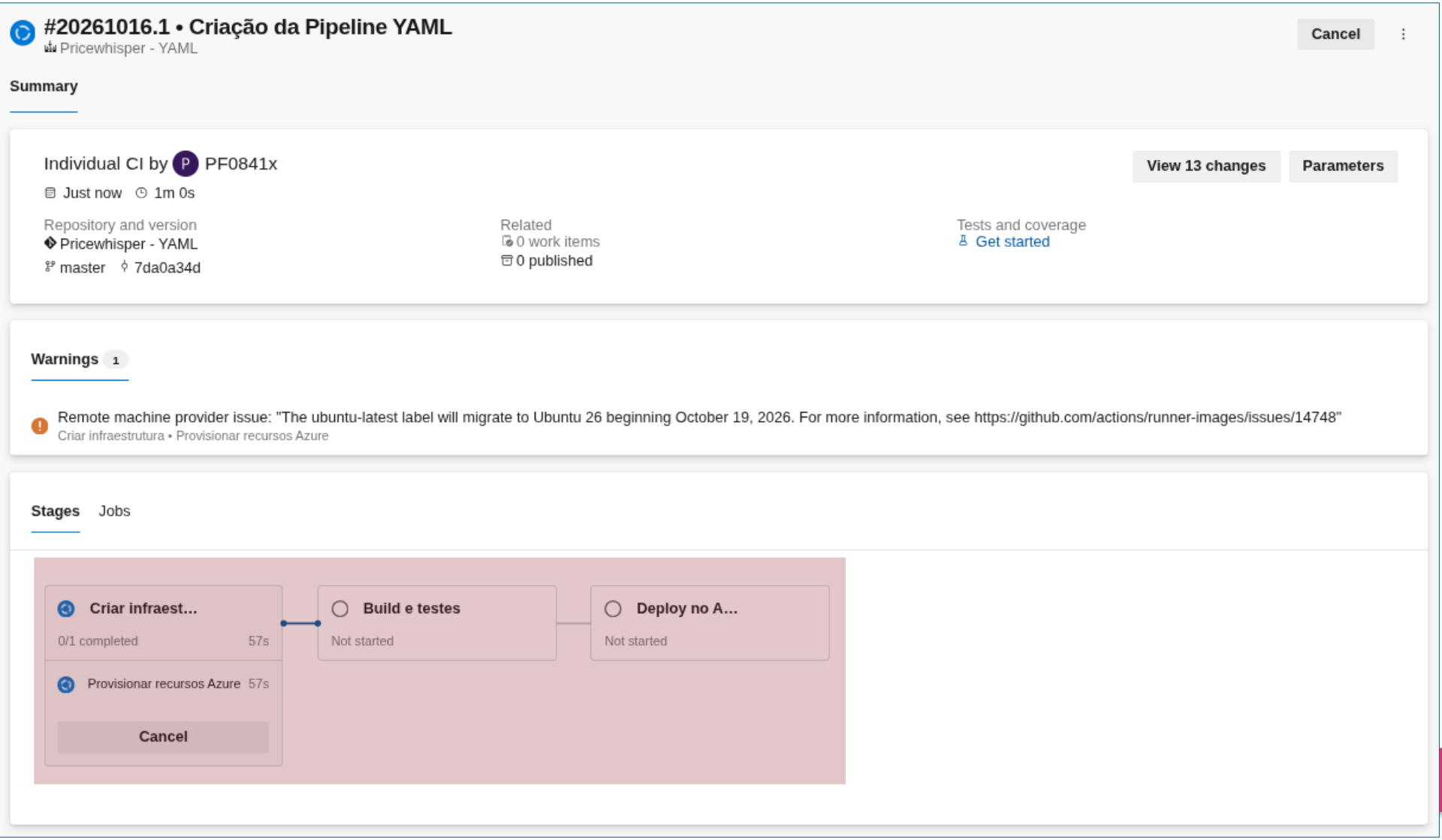
Azure Pipelines API .NET com Oracle

Informe os parâmetros de execução



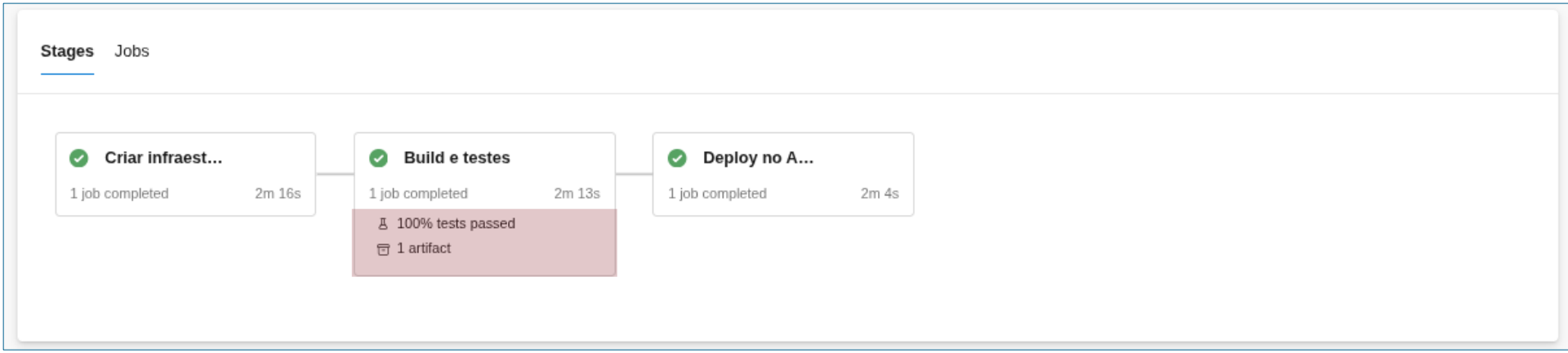
Azure Pipelines API .NET com Oracle

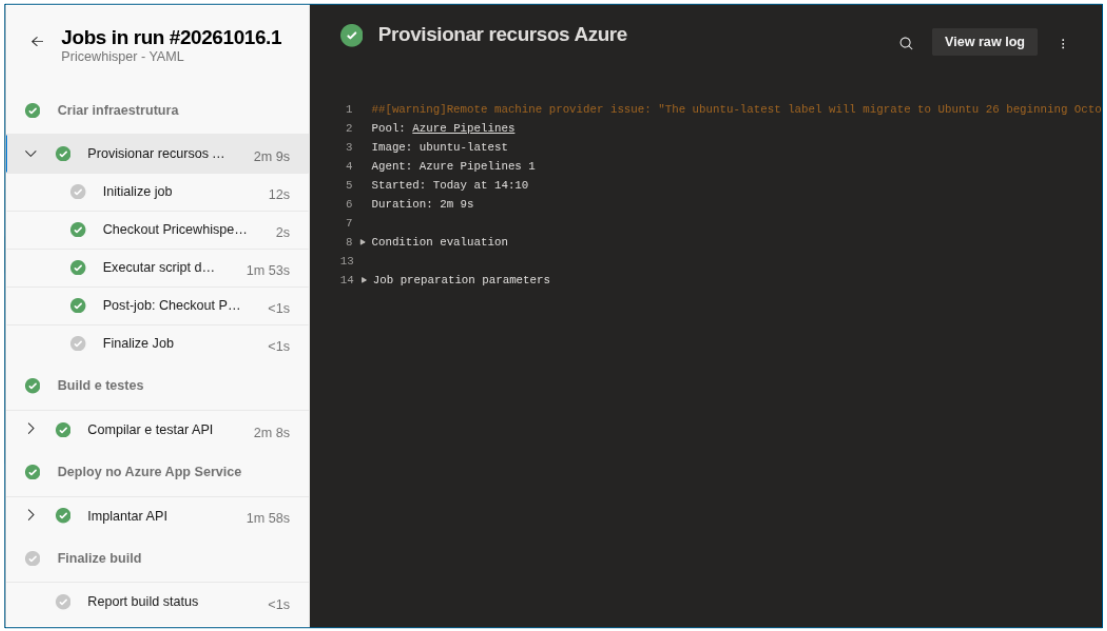
Acompanhe a execução do CI/CD



Azure Pipelines API .NET com Oracle

Ao final analise os LOGs da execução da Pipeline

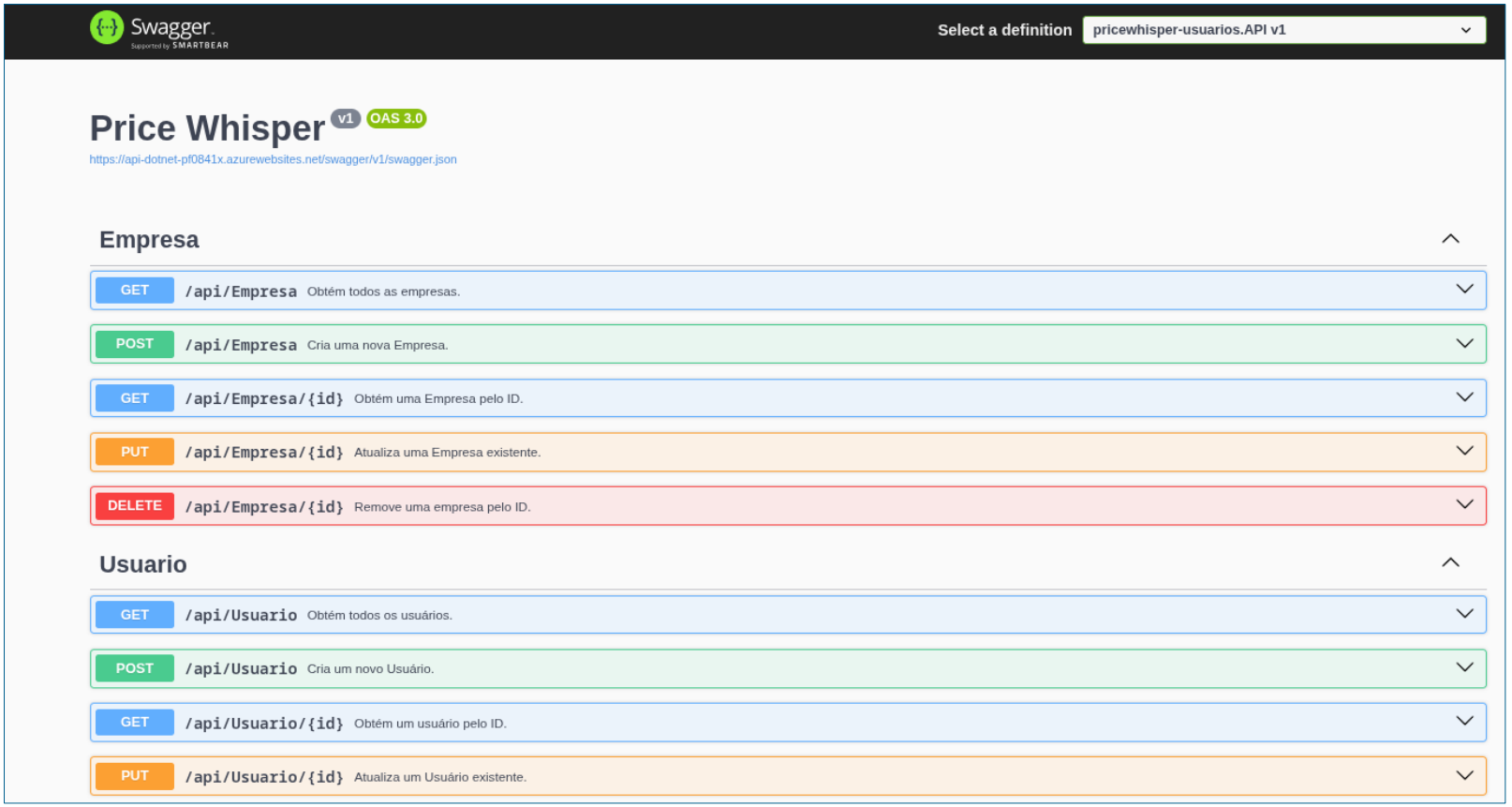




Azure Pipelines API .NET com Oracle

Verifique o App em funcionamento na nuvem

https://api-dotnet-rm9999.azurewebsites.net/swagger/index.html

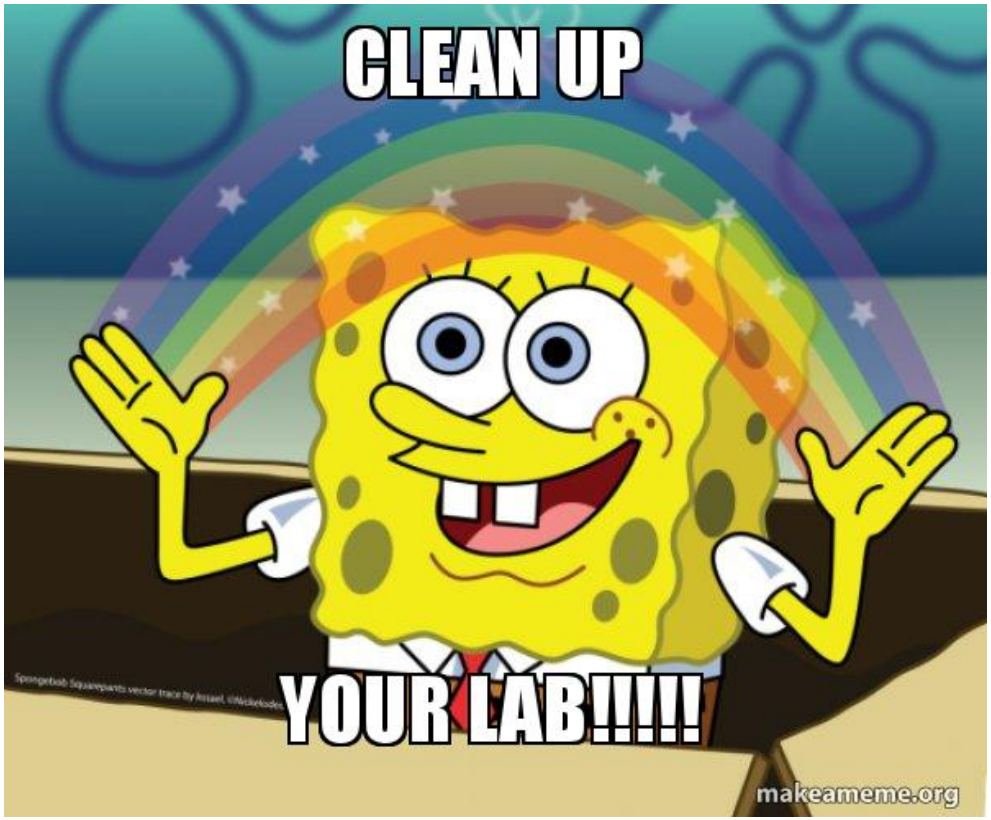


Limpar Laboratório na Azure

- ✓ Delete todos os Recursos no Portal utilizados nesse exercício
- ✓ Verifique se não sobrou nenhum Recurso dos nossos Deploys

drop table "Usuarios";

drop table "Empresas";



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